

Risky Business Cycles*

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Abstract

We propose a novel mechanism that translates risk premium fluctuations into business cycles with full macroeconomic comovement. Durable factors of production differ in their risk exposure, implying that riskier factors command a higher risk premium and deliver more output per unit invested. When risk appetite falls, firms shift towards safer factors with lower returns, leading to broad contractions in consumption, investment, employment, and output. We characterize the mechanism analytically, and evaluate its potential in a quantitative model. The model captures the empirical link that we document between elevated risk premia, recessions, and increased part-time employment.

Keywords: Business Cycles; Risk Premia; Uncertainty.

JEL Classification: E32, E24, E21, G1.

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1 Introduction

Economists have long entertained the idea that changes in investors’ willingness to undertake risky investment might be a key driver of the business cycle, a notion dating at least to [Keynes \(1936\)](#). This intuitive idea regained prominence after the Great Recession (e.g. [Bloom et al., 2018](#), [Cochrane, 2017](#)), but the project of formalizing risk fluctuations as a driver of business cycles remains incomplete. The central difficulty is that, in many models, the counterpart to a risk-induced fall in investment demand is an increase in current consumption. The countervailing forces between investment and consumption mean that such models often fail to generate the positive comovement among output, consumption, investment and labor that defines the “business cycle” ([Lucas, 1977](#)).

In this paper we propose a novel resolution to this comovement challenge, based on a real flight-to-safety mechanism. Our proposal focuses on the allocation of investment across risky inputs, rather than the overall demand for investment. The basic idea is that when aggregate risk premia increase, firms shift demand away from riskier factors of production towards safer ones. However, in equilibrium, riskier factors of production deliver higher returns per unit invested — a “production risk premium.” As a result, the reallocation towards inputs with lower returns can generate simultaneous declines in all four main macro aggregates. Embedded in a quantitative model, this mechanism produces realistic business cycles from risk premium variations, without relying on nominal rigidities or concurrent changes in technology (e.g. [Basu and Bundick, 2017](#); [Bloom et al., 2018](#)).

We begin with an empirical exercise that highlights the connection between equity risk premium fluctuations and macroeconomic *comovement* in the data. We use a vector autoregression (VAR) and a max-share procedure in the tradition of [Faust \(1998\)](#) and [Uhlig \(2004\)](#) to extract a reduced-form innovation to the equity risk premium. This empirical analysis documents that an increase in the equity risk premium is associated with substantial and persistent falls across *all* of output, consumption, investment, and employment. While more than one structural shock could drive the risk premium disturbances, the conditional comovements we estimate suggest a tight link between risk premia and the macroeconomic comovement that defines business cycles. Our results complement prior work that focuses on the cyclical relationship between risk premia and output or consumption in isolation.

Although we find that most macroeconomic quantities fall when risk premia rise, we also highlight an important exception: the number of part-time workers *increases* substantially in periods of elevated risk premia. This reallocation between different production inputs would be difficult to explain with a shock to either aggregate demand or aggregate supply, since these typically cause all inputs to move together. Other patterns in the data also

point away from these classic sources of business cycles: the risk premium disturbance we have isolated is not associated with traditionally-identified TFP shocks or large movements in the present discounted value of dividends. We also find that risk premium fluctuations are not associated with large movements in interest rates or inflation, contrary to textbook new-Keynesian propagation mechanisms.

To explain these patterns, we propose a novel theory, in which risk premium fluctuations themselves generate business cycle comovement, without a reliance on supply shocks or new-Keynesian elements. In our theory, an aggregate increase in risk premia precipitates a recession by causing a real “flight-to-safety” where firms shift their input use towards factors of production that are safer and thus endogenously offer lower returns.¹ In order to isolate how our mechanism generates business cycles and comovement without shocks to technology or endowments, we model risk premium fluctuations as caused by exogenous shocks to risk aversion. Nevertheless, if risk premia were to change in response to other economic fundamentals, our reallocation channel would function in the same way, as a propagation and amplification mechanism.

We operationalize this idea by introducing search-and-matching labor markets with two types of employment relationships: full-time and part-time. Search frictions effectively introduce labor adjustment costs, which makes hiring a risky investment. To introduce differences in riskiness across types, we assume that (i) part-time wages are more flexible and (ii) part-time employment relationships are shorter-lived. Both features are well-documented in micro data: in the Current Population Survey, separation rates for part-time workers are roughly eight times those of full-time workers, and their wages exhibit about twice the cyclical volatility (Lariau, 2017; Borowczyk-Martins and Lalé, 2019; Grigsby et al., 2021). In the model, these differences make full-time employment relationships riskier from the viewpoint of the firms, because sticky wages generate operating leverage, while the longer duration of full-time worker matches exposes the firm to aggregate risk over a longer horizon. In turn, an increase in risk aversion optimally shifts input demand towards the safer part-time positions, which is consistent with our own time series evidence, as well as a growing body of cross-sectional evidence that firms’ use of labor plays an important role in determining their risk characteristics.²

Section 3 presents a simplified analytical model that illustrates the core of our flight-to-safety mechanism. In this model, the only real investments are the vacancy costs paid to hire the two types of labor. We begin by demonstrating the comovement challenge in a

¹Cochrane (2017) similarly conjectures that a real “flight-to-safety” could lead to business cycles, but does not analyze the idea or provide a formal model of it.

²Belo et al. (2014), Favilukis and Lin (2016), Donangelo et al. (2019) all provide empirical evidence that labor relationships are a source of priced risk.

benchmark case in which both labor positions are symmetric, so that all factors of production have the same riskiness. Fluctuations in risk aversion thus change the overall demand for investment in risky labor but not the composition of labor demand. Such fluctuations in overall investment cannot generate comovement, because the resource constraint implies that they always push consumption in the opposite direction.

To introduce our mechanism, we next assume that full-time wages are rigid while part-time wages are flexible.³ This makes full-time workers riskier from the firm’s perspective, a risk that is compensated in equilibrium by a higher return of output per unit of hiring expenditure. Because of the relatively high returns of full-time labor, a pure reallocation of hiring expenditure from full-time to part-time vacancies results in lower output. Therefore, for a given amount of total investment in hiring, such a reallocation necessarily lowers consumption. In equilibrium, an increase in risk aversion causes both a change in the overall demand for investment and a reallocation towards the safer part-time positions. Our main proposition shows that the flight-to-safety mechanism can dominate the change in total savings, in which case a rise in the aggregate risk premium generates a full recession.

In Section 4, we embed the mechanism in a standard quantitative macro model. Among other things, we now allow for the full-time positions to also have lower separation rates and thus longer duration, amplifying their riskiness, and we also introduce physical capital as a third factor of production as is standard. We estimate the quantitative model with a rich set of empirical targets, encompassing both conditional moments identified in our VAR exercise and various unconditional moments, including asset pricing moments. We find that the estimated model fits the data well, generating realistic business cycle fluctuations and macroeconomic comovement in response to risk aversion shocks, without predicting excessively high or volatile risk premia or labor market fluctuations.

We use our estimated model to perform several exercises that provide additional insight into the quantitative importance of the reallocation channel. Using a counterfactual experiment, we first show that reallocation between labor types alone accounts for about 3/4 of the equilibrium fall in consumption in response to increased risk aversion. Reallocation between investment in vacancies and in physical capital also pushes consumption down substantially, while the fall in overall demand for input investment pulls consumption up. The net effect is a fall in consumption that matches the data well. We also show that the risk premium on full-time labor is essential for the model to explain business cycles, and discuss how our mechanism is different from those in several related papers.

³For simplicity, the analytical model only considers differences in the wage structure but not the duration of the two labor positions. The quantitative model introduces both channels through which part-time and full-time positions differ in their riskiness.

Related Literature

Recent work has rekindled interest in uncertainty- or risk-driven business cycles (Bloom, 2009; Gilchrist et al., 2014), but generating full macro comovement has proved difficult. These challenges have led researchers to focus on studying just a subset of macro aggregates, rather than attempting to explain full comovement (Arellano et al., 2019; Segal and Shaliastovich, 2025), or to complement risk mechanisms with concurrent first-moment shocks (Gourio, 2012; Bloom et al., 2018) or with specific assumptions about the behavior of the government sector in times of elevated risk (Bansal et al., 2021). Another solution to the comovement challenge is to use models with nominal rigidities so that output is primarily determined by final goods demand (e.g., Ilut and Schneider, 2014; Fernández-Villaverde et al., 2015; Basu and Bundick, 2017; Bayer et al., 2019; Caballero and Simsek, 2020; Christiano et al., 2014; Leduc and Liu, 2016; Challe et al., 2017).⁴ By contrast, our model generates business cycles with full comovement and it does so without relying on sticky nominal prices, suboptimal monetary policy, or markup variation.

Two recent papers, Di Tella and Hall (2021) and Ilut and Saijo (2021), also provide mechanisms that deliver business cycle comovements via a risk channel without nominal rigidities. They propose models where the marginal product of both capital and labor is uncertain – due to a labor-in-advance choice in the former, and imperfect information about productivity in the latter. In both cases, comovement requires the risk-driven fall in firms’ investment demand to be strong enough to offset households’ increased desire to save. We differ from this work in two ways. First, our mechanism is conceptually distinct: rather than relying on fluctuations in the *overall* demand for risky factors, we show that *reallocation* of input demand from riskier, higher-marginal-product factors toward safer, lower-marginal-product factors can itself generate comovement, and we demonstrate its quantitative relevance. Second, the mechanism in Di Tella and Hall (2021) relies on variation in *idiosyncratic risk*, and thus could not explain the link between aggregate risk premia and macroeconomic comovement we document and answer the long-standing comovement challenge of generating business cycles out of aggregate risk shocks.

Our paper also relates to the investment-based asset pricing literature that studies how risk affects firms’ production decisions, valuations, and firm-level dispersion in productivity. These papers model partial equilibrium firm choices of risky inputs – such as physical capital (Cochrane, 1991, 1996; David et al., 2022), inventories (Belo and Lin, 2012), or labor with adjustment costs (Belo et al., 2014) – and use exogenous stochastic discount factors specifically estimated for their respective settings. By contrast, we focus on macroeconomic

⁴Occasionally binding downward wage rigidity also amplifies the impact of uncertainty shocks on labor market variables, with or without nominal rigidities (Cacciatore and Ravenna, 2021).

comovement in general equilibrium, and our key mechanism studies firm choices across a portfolio of inputs with different risk characteristics.

Similarly, there is a literature, following [Hall \(2017\)](#), that studies how time variation in discount rates affects labor markets and the volatility of unemployment. Subsequent papers have built on this idea to provide a risk-driven explanation of the [Shimer \(2005\)](#) puzzle and other labor market phenomena (e.g. [Kilic and Wachter, 2018](#); [Mitra and Xu, 2019](#); [Kehoe et al., 2022](#)). Unlike our paper, these and other models that focus on risk-driven unemployment fluctuations largely abstract from capital accumulation or, when capital is considered, do not focus on the comovement across macro aggregates. In addition, despite their labor market focus, these models do not account for the disparate movements in part-time and full-time labor we find in the data.

2 Motivating Facts: Risk Premia and Comovement

In this section, we isolate the business cycle dynamics associated with changes in the equity risk premium. We later use these patterns to motivate our theoretical approach to capturing risk-driven recessions.

Our baseline empirical specification consists of a vector autoregression of the form

$$Y_t = B(L)Y_{t-1} + u_t. \quad (1)$$

In the above, Y_t is the vector of observed variables, $B(L)$ is the lag polynomial containing the weights on past realizations of Y_t , and u_t is the vector of residuals.

We use U.S. data, with $Y_t \equiv [gdp_t, c_t, inv_t, n_t, r_t^s, r_t^b, dp_t]'$ consisting of the logs of real per-capita output (gdp_t), consumption (c_t), investment (inv_t), employment (n_t), cum-dividend real stock log-returns (r_t^s), log real ex-post three-month treasury bill rate (r_t^b), and the aggregate dividend-price ratio (dp_t).⁵ Our sample is 1954Q1-2018Q4.⁶ The VAR is estimated in log-levels using OLS, with three lags in the polynomial $B(L)$.

2.1 Identification Approach

We seek to find a rotation matrix A that maps the econometric residual u_t to a vector of orthogonal disturbances ϵ_t , so that $u_t = A\epsilon_t$. To do so, we use a “max-share” approach

⁵Appendix [A.1](#) contains all details on data definitions and sample construction.

⁶A previous version of the paper used data starting in 1985Q1 to avoid a potential structural break at the start of the “Great Moderation.” The shorter sample results, which are very similar, are reported in Appendix [A.2](#).

similar to that introduced by Faust (1998) and Uhlig (2004). Specifically, we select A so that the disturbance associated with its first column explains the largest possible share of *expected* equity excess returns.

To target the expected excess return, we must first compute it using the VAR system in (1). The *realized* j -period cumulative excess return is given by

$$rp_{t,t+j} \equiv [r_{t+1}^s + r_{t+2}^s + \dots + r_{t+j}^s] - [r_{t+1}^b + r_{t+2}^b + \dots + r_{t+j}^b]. \quad (2)$$

We can then compute the *expectation* of excess returns using our VAR. Taking expectations over (2) and iterating backwards through the VAR system, we can express the expected excess return as a linear function of disturbances,

$$\mathbb{E}_t[rp_{t,t+j}] = (e_5 - e_6)(\tilde{B} + \tilde{B}^2 + \dots + \tilde{B}^j)(I - \tilde{B}L)^{-1}\tilde{A}\tilde{\epsilon}_t, \quad (3)$$

where $\tilde{Y}_t = \tilde{B}\tilde{Y}_{t-1} + \tilde{A}\tilde{\epsilon}_t$ is the companion form of the VAR in equation (1), with $\tilde{Y}_t$ defined as the vector of Y_t stacked along with its lags, and $\tilde{\epsilon}_t$ defined as ϵ_t stacked with zeros. In (3), e_5 and e_6 are vectors that select the stock and bond returns from $\tilde{Y}_t$, respectively.

To isolate the conditional moments associated with changes in risk premia, we then select the matrix A such that the first element $\epsilon_t^{(1)}$ explains the largest share of $\mathbb{E}_t[rp_{t,t+20}]$. To do this, we need to specify both the periodicity and horizon of the fluctuations that we wish to target. We target periodicities between 2 and 500 periods. This range effectively captures unconditional variances when variables are stationary, but allows us to perform robustness checks using VECM techniques, in which case the lag polynomial $B(L)$ has a unit root.

Regarding the horizon, we set $j = 20$ in order to focus specifically on the predictability of *long-term* (5-year) excess returns. We make this choice because influential studies like Welch and Goyal (2008) have shown that predictability of short-horizon returns is much less robust and more unstable, while finding some promise in long-horizon forecasts (as also argued by Cochrane, 2008). Moreover, by including measures of real consumption and investment (and their lags), our VAR spans information close to the types of real consumption- and investment-based variables that perform best in the exhaustive review of Welch and Goyal (2008) and Goyal et al. (2024).⁷ The return expectations implied by our VAR approach are also tightly related to forecasts made by sophisticated financial analysts included in the data constructed by Thesmar and Verner (2026). We detail this relationship in Appendix A.4.

⁷In Appendix A.3 we analyze the specific sources of the predictive power of our VAR, and indeed we find that it is mainly due to consumption, investment, and GDP. Moreover, we demonstrate that VAR specifications restricted to just these variables deliver predictive power in line with the results of the previous literature. Hence, our methodology finds predictability of similar magnitude and source to that in the previous literature.

Appendix A.5 provides additional computational details about the max-share procedure.

2.2 Empirical Results

Overall, the estimated $\epsilon_t^{(1)}$ explains 94% of the variation in the 5-year risk premium at business cycle frequencies (6-32 quarters) and 88% of the unconditional variance.⁸ Thus, this single disturbance captures most of the dynamics of the equity risk premium; hence, we will refer to $\epsilon_t^{(1)}$ as the main “risk premium disturbance.”

While our approach does not identify its deep structural origin, we observe that most recessions in our sample coincide with large positive innovations to $\epsilon_t^{(1)}$, particularly episodes associated with acute financial stress such as 1981-82 and 2008, while the relationship is weaker in the recessions of 1990 and 2001. The shock also spikes during non-recessionary episodes, including Black Monday in 1987Q4, and its largest realization occurs in 1974, amid the OPEC oil embargo and the end of the Bretton Woods system. This pattern is consistent with the interpretation that our identified shock captures periods of elevated risk aversion or reduced risk-bearing capacity in the U.S. economy. The full time series of the estimated $\epsilon_t^{(1)}$ is plotted in Appendix Figure A.5.

More formally, Figure 1 plots the impulse responses of the major business cycle variables to a one-standard-deviation change in $\epsilon_t^{(1)}$. The numbers in the panel titles represent the percent of the variance of each variable explained by our disturbance, either at business cycle frequency (first number, periodicities between 6 and 32 quarters) or essentially the unconditional variance (second number, periodicities between 2 and 500 quarters).

The first panel plots the response of the 5-year equity risk premium itself ($\mathbb{E}_t r p_{t,t+20}$), the target of our max-share procedure. We see that $\epsilon_t^{(1)}$ leads to a substantial and persistent increase: the risk premium jumps up by about 1.1% (annualized) on impact (compared to an average risk premium of 5.5% in our sample), and the impulse response is largely monotonic and persistent, with a half-life of 9 quarters. Naturally, we find that this persistent rise in the risk premium is associated with a sharp drop in stock prices on impact (bottom-right panel). This is followed by a prolonged period of higher-than-average stock returns, which underlie the elevated expected excess returns $\mathbb{E}_t(r p_{t,t+20})$.

In order to analyze the relationship with the broader business cycle and macroeconomic comovement, Figure 1 plots the responses of the four main macro aggregates: output, consumption, investment and employment. We find that all of these variables exhibit a substantial and persistent contraction, with hump-shaped dynamics. These strong dynamic

⁸For technical purposes, we compute the “unconditional” variance as the variance over periodicities between 2 and 500 quarters, which is almost identical to the unconditional variance. Using a wide-band variance, however, will also allow us to compute the same statistic for non-stationary variables.

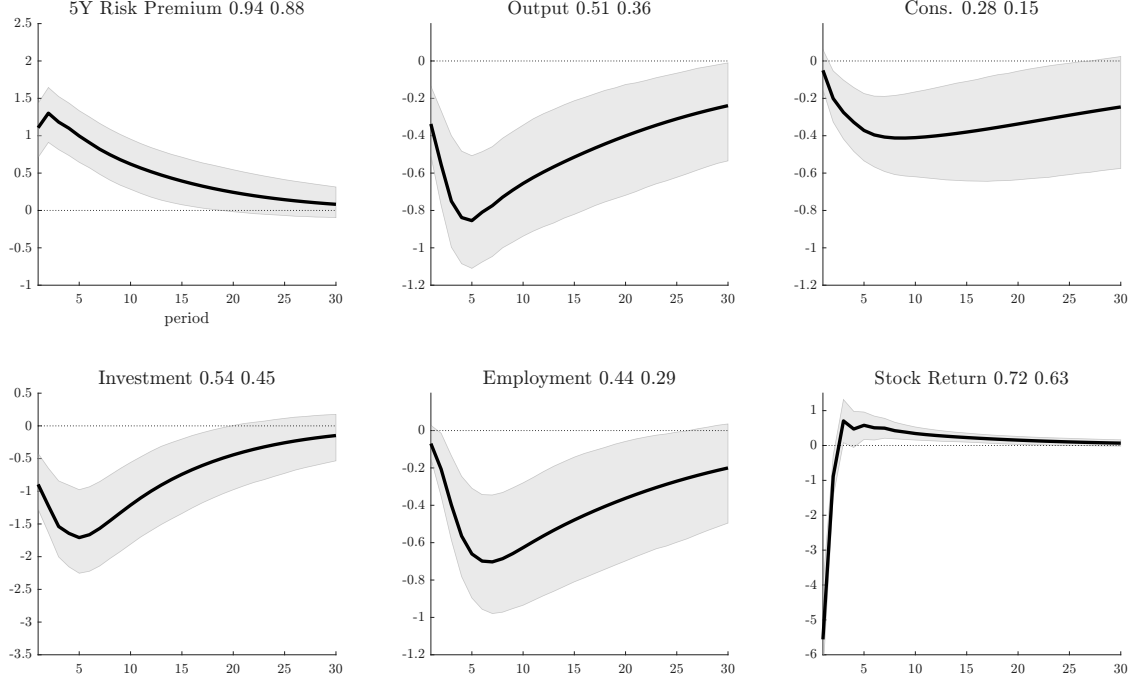


Figure 1: Impulse responses to a risk premium disturbance. Shaded areas cover the 10th- and 90th-percentiles of bootstrapped responses. Vertical axis values are in annualized percentage points for the risk premium and stock return; all other numbers are in percentage deviations. Numbers in subplot titles correspond to the percent of variance explained at the business cycle (6-32 quarters) and unconditional frequencies (2-500 quarters), respectively.

responses are also reflected in the variance decomposition statistics, which show that the risk premium disturbance accounts for roughly half of the variance in output, investment, and employment at business cycle frequencies, and about a third of the variance in consumption. The contribution to unconditional variances is slightly lower, consistent with its persistent but clearly stationary nature (as shown in the first panel).

In summary, our findings show that whatever structural shocks drive risk premia are likely also important drivers of business cycles, both in terms of generating macro fluctuations and in driving the classic observation of macroeconomic comovement. It is interesting to contrast our results with the “main business cycle” shock of [Angeletos et al. \(2020\)](#). To extract their shock, they also follow a max-share procedure but instead isolate the main driver of output (or employment, depending on the specification), and not risk premia as we do.⁹ Yet our disturbance captures a similar portion of business cycles and is correlated – with a coefficient of 0.73 – with their main business cycle shock. Taken together, these results show that

⁹Another difference is that [Angeletos et al. \(2020\)](#) target business cycle frequencies while we target the unconditional variance. However, targeting business cycle frequencies in our procedure leaves the results unchanged – in that case we obtain an $\epsilon_t^{(1)}$ series that is correlated 98.0% with our baseline series.

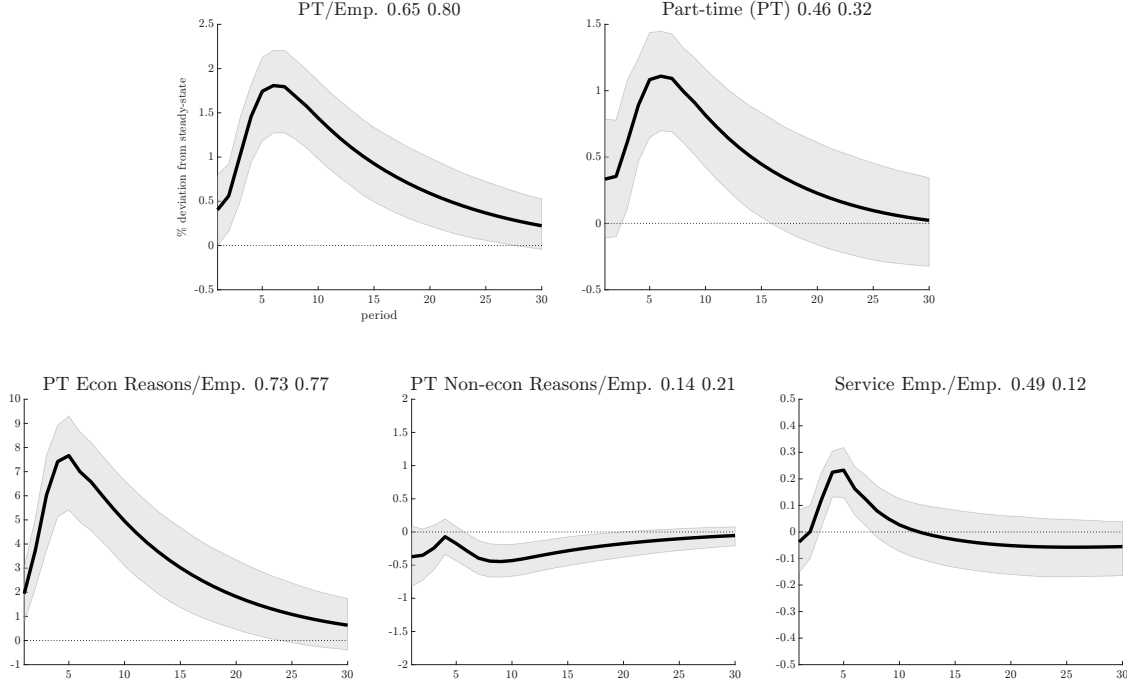


Figure 2: Part-time labor responses to a risk premium disturbance. Panels show part-time employment, the part-time share of employment, the part-time share for economic and for non-economic reasons, and the service-sector employment share. Shaded areas cover the 10th- and 90th-percentiles of bootstrapped responses. All values are in percentage deviations. Numbers in subplot titles correspond to the percent of variance explained at the business cycle (6-32 quarters) and unconditional (2-500 quarters) frequencies, respectively.

there is indeed a close connection between the drivers of business cycles and risk premium fluctuations in the data.

2.3 Additional Results

Our empirical approach is consistent with many structural interpretations for the risk premium disturbance $\epsilon_t^{(1)}$. To explore some potential interpretations, we study its implications for several additional macroeconomic variables. To save on degrees of freedom (and because not all of the additional variables are available starting in 1954Q1), we augment our VAR with a set of auxiliary variables S_t , by projecting S_t on current and past observations of Y_t :

$$S_t = \Gamma(L)Y_t + v_t. \quad (4)$$

The coefficient matrix $\Gamma(L)$, estimated via OLS, contains the same number of lags as the VAR in (1). Using the estimated values of $\Gamma(L)$, we can then compute the impulse responses for any auxiliary variable using the responses of Y_t implied by our VAR in (1).

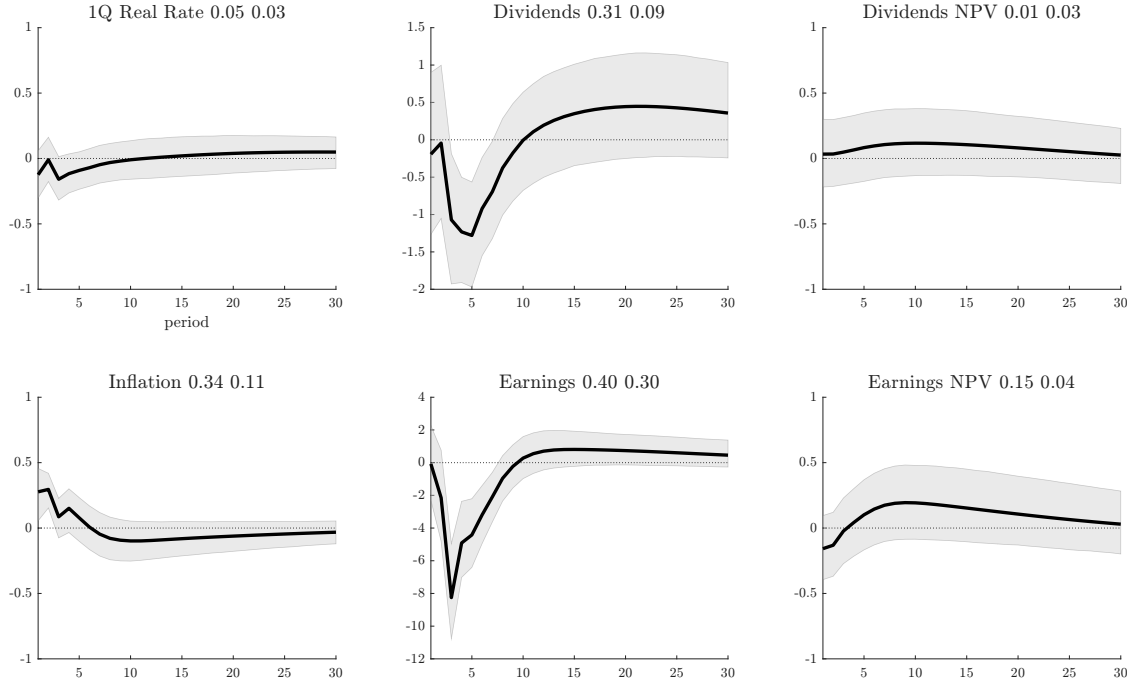


Figure 3: Additional variable responses to a risk premium disturbance. Shaded areas cover the 10th- and 90th-percentiles of bootstrapped responses. Vertical axis values are in annualized percentage points for interest rates and inflation; all other numbers are in percentage deviations. Numbers in subplot titles correspond to the percent of variance explained at the business cycle (6-32 quarters) and unconditional frequencies (2-500 quarters), respectively.

Part-time Employment

We are interested in modeling the effects of a risk-driven reallocation across investment in different factors. As intuitively argued by [Cochrane \(2017\)](#), such a flight-to-safety may generate recessions out of any disturbance driving higher risk premia. Given the existing evidence that labor relationships are a source of priced risk for firms (e.g. [Belo et al., 2014](#)), our conjecture is that such reallocation might occur between part-time and full-time labor demand, since full-time arrangements have less flexible wages and a duration that is 8 times longer than part-time (e.g. [Lariau, 2017](#)). Both features should make the value of a full-time worker more cyclical and thus riskier to firms.

The top panels of Figure 2 show that part-time employment does indeed rise significantly in response to our disturbance. The top-left panel shows that the share of part-time employment increases persistently, increasing by 1.8% at the peak. This rise in the share of part-time employment happens at the same time as the economy experiences an overall employment *fall* (0.7% at its trough) and a significant rise in the aggregate risk premium. The right panel further shows that part-time employment in fact increases in levels. Thus, the higher part-time share is not merely the result of a smaller decline in part-time relative to

full-time employment; it reflects an absolute increase in part-time positions. Because part-time employment rises even as total employment falls, our disturbance drives different labor inputs in *opposite* directions, a reallocation pattern that a conventional aggregate supply or demand shock would not easily produce.

This response of aggregate part-time employment could reflect sectoral reallocation or changes in worker composition rather than the firm-side adjustment that we emphasize. We explore this possibility in two ways. First, we examine whether the shock induces a shift toward sectors that are on average more part-time labor intensive. The bottom-right panel of Figure 2 shows that the share of employment in the service sector does increase following the risk premium disturbance, but the response is quantitatively small, peaking at about 0.23%, or 0.17 percentage points above its sample average of 76%. Since the average part-time employment share in services is roughly 18% compared to about 7% in goods-producing industries, this sectoral shift can account for an increase in the aggregate part-time employment share of only about $0.17 \times 0.11 = 0.019$ percentage points. Meanwhile, the aggregate part-time employment share rises by 0.36 percentage points in response to our shock (i.e., 1.8% above its average of 20%). Thus, sectoral reallocation can account only for a small fraction of the response of the aggregate part-time share.

Second, we distinguish between part-time employment for economic and non-economic reasons. This distinction is informative because part-time work for economic reasons reflects workers who would prefer but are unable to find full-time employment, which goes to the heart of the firm-demand mechanism we have in mind. On the other hand, part-time work for non-economic reasons is more naturally associated with voluntary labor-supply choices (e.g., home production) or may reflect a composition effect across demographic groups with different preferences for part-time work. Empirically, the bottom-left panels of Figure 2 document that the increase in aggregate part-time employment is entirely driven by workers employed part-time for economic reasons, while part-time employment for non-economic reasons actually falls. Moreover, at business cycle frequencies, our shock explains about three quarters of the variance of part-time work for economic reasons, while it has little relation to changes in part-time work for non-economic reasons. These patterns are difficult to reconcile with a purely demographic or voluntary labor-supply explanation but hint at a shift in labor demand, which is the type of mechanism we formally model below.¹⁰

Before moving to theory, we document additional empirical patterns that point away from some common alternative business cycle explanations.

¹⁰Appendix A.6 examines the demographic margin directly, and shows that the conditional responses of employment shares by sex and by age group are too small to account for the estimated increase in the part-time share; the age composition shift in fact works against it.

Safe Interest Rates and Firm Profitability

A common finding in the finance literature is that asset price fluctuations are largely driven by changes in risk premia (i.e. risky discount rates), rather than changes in safe interest rates or expected firm cash flows (e.g. [Cochrane, 2011](#)). The first row of Figure 3 shows that the conditional response to our risk premium disturbance is consistent with this view. In particular, the first panel shows that short-term interest rates do not change substantially in response to the risk premium disturbance, while the top-right panel shows that the present value of dividends (discounted using the average risky interest rate) moves little in response to the shock. Underlying this insignificant response of the present value of dividends, the top-middle panel shows that dividends themselves exhibit a short-lived initial decline followed by a subsequent persistent increase. The bottom-right panels of the same figure show that earnings display a larger response, but also in this case these fluctuations do not translate into a significant response of their present discounted value.¹¹ This further suggests that the root cause is unlikely to be a classic first-moment shock.

Inflation

In many models, high risk premia are associated with higher desired savings and a lower natural rate of interest. With new-Keynesian frictions, a lower natural interest rate could depress aggregate demand, opening an output gap that would precipitate a broad recession across all four macro aggregates ([Basu and Bundick, 2017](#)). If this channel were driving the recession in response to our shock, however, we would expect the increasing output gap to drive a corresponding fall in inflation. However, the bottom-left panel of Figure 3 shows that, if anything, an elevated risk premium is associated with a modest *increase* in inflation. This pattern could still be consistent with the shock raising desired precautionary saving, but it points away from the propagation mechanism of a textbook new-Keynesian aggregate demand shortfall.

Disentangling Supply Shocks

One concern with the max-share approach is that identified shocks could reflect a combination of underlying shocks, rather than a single structural disturbance. This is the main reason we refrain from calling our estimated disturbance a “structural shock.” Of particular

¹¹For all NPV calculations, we discount future log-deviations of each variable from its deterministic path using a constant discount factor based on the average real risky return, adjusted for trend growth: $\frac{1+\bar{g}}{1+\bar{r}}$. We set $\bar{g}$ and $\bar{r}$ equal to the average growth rate of GDP per capita and the average equity return in our sample, respectively. These quarterly rates are 0.42% and 1.76%. Present values are computed by iterating on the VAR and equation (4) for $N = 60$ periods.

concern given our findings above, the small estimated effects on inflation might result from the commingling of supply and demand disturbances that have a similar effect on real quantities but opposed effects on inflation.¹² We try to address this concern by first identifying productivity or supply shocks and controlling for these shocks when estimating our risk premium disturbance. If the baseline estimation approach commingles risk with a first-moment supply shock then this cleaning step might qualitatively change our results.

Our general approach is to include TFP in our set of auxiliary variables, and then use TFP to identify supply shocks using a number of different procedures for robustness. In each case, we first extract an identified supply shock and then identify the risk premium disturbance using the max-share approach after controlling for the supply disturbances. The full methodological details of these approaches are outlined in Appendix A.7.

No matter how we control for TFP shocks, however, the time series of risk premium disturbance remains at least 97% correlated with our baseline and impulse responses are nearly identical. While we cannot rule out the possibility that our identified disturbances conflate multiple structural shocks, we think this evidence helps to address the concern that our estimated shock simply reflects a combination of standard supply and demand shocks.¹³

Taking stock, our empirical analysis shows that, following a risk premium disturbance, the U.S. economy experiences a protracted recession across all macro aggregates, a shift towards part-time employment, little movement in safe interest rates or in the present discounted value of firms' dividends, and a modest, short-lived increase in inflation. These empirical patterns motivate us to seek an explanation that is based on a risk-driven reallocation towards safer but lower-return production inputs.

3 Analytical Model and Intuition

The central challenge our paper addresses is how an increase in risk aversion could generate macroeconomic comovement. In standard macroeconomic models, risk premia affect the overall desire to save or invest, but a change in aggregate investment typically causes consumption to move in the opposite direction.

Our theoretical contribution is to show that when durable factors of production have different risk-return profiles, factor reallocation can overcome this comovement barrier through

¹²Dieppe et al. (2021); Moorjani (2024); Dou et al. (2025) all raise versions of this concern regarding the main business cycle shock of Angeletos et al. (2020).

¹³Appendix A.8 also addresses the general concern that our predictability results are driven by over-fitting. In it, we use a Monte Carlo exercise to argue that overfitting is unlikely to explain our findings regarding excess return predictability.

a *composition* effect. From the firm’s perspective, factor investment becomes a portfolio choice across durable factors with different risk-return profiles. When risk aversion rises, it disproportionately reduces demand for the riskier factor, causing a flight-to-safety reallocation toward safer factors. But, since risky factors offer a higher return as compensation for bearing higher risk, this reallocation lowers the effective return on aggregate factor investment and pushes both output and consumption down.

In this section, we present a minimal, two-period analytically tractable model of the reallocation mechanism we propose. The model features labor market search frictions and durable employment relationships, which effectively makes hiring an investment decision subject to risk considerations. We allow for two types of labor positions – part-time and full-time – motivated by our prior empirical result that elevated risk premia are associated with a significant reallocation towards part-time labor.

After describing the model environment below, we first study a baseline case in which both durable labor inputs have the same risk profile and show that macroeconomic comovement is impossible. We then present our main result for the case in which the two labor inputs have different risk profiles. We use this case to show that changes in risk tolerance can drive macroeconomic comovement, and to clarify the intuition for why this case differs from the symmetric input baseline.

3.1 Model Environment

The setting is a two-period, general-equilibrium economy, in which the representative household has recursive Epstein-Zin preferences and there are two labor markets, labelled “full-time” (f) and “part-time” (p). Labor markets are subject to a standard search friction and employment relationships are durable, meaning that labor hired today produces both immediately and in the future period. This makes the labor demand choice of the firm a risky *investment* choice, as firms weigh the returns on labor both from production today and from production in the (uncertain) future. We are interested in studying the effect of a change in risk aversion, which Epstein-Zin preferences allow us to manipulate in isolation. Later, in Section 4, we extend this model to include an infinite horizon and physical investment with capital as a factor of production.

Households consume and supply labor inelastically in both markets, while firms make vacancy posting decisions. Firms are the main decision makers in the economy: they hire labor by posting vacancies in the two separate labor search markets, V_f and V_p , at a cost per vacancy of φ . These search markets are only open in the initial period, and successfully hired workers produce in both periods $t \in \{0, 1\}$. Matches are formed according to a Cobb-

Douglas matching function so that vacancies are filled with a probability $P_i \equiv M(V_i, S)/V_i$ for $i \in \{f, p\}$, where $M(V, S) = V^\varepsilon S^{1-\varepsilon}$ is the matching technology.

For simplicity, in this analytical version of the model we consider only one difference between the two labor markets: the wages paid to part-time workers are more flexible than the wages paid to full-time workers, as is true in the data. The rest of the structure of the two labor markets is symmetric – the vacancy posting cost φ and the matching function elasticity ε are the same, and there is simply a unit mass, $S = 1$, of searchers in each market. The simplicity is intentional: it isolates the reallocation channel in the cleanest possible environment. In the quantitative model in Section 4, we introduce significantly more richness, where among other things we introduce empirically disciplined differences in separation rates and effective labor input across types alongside endogenous household search across markets. Such extensions improve both the empirical realism of the model and its quantitative performance, but they are not needed for our key qualitative result.

Firms combine labor types using a CES aggregator to produce output according to:

$$Y_t = Z_t \tilde{N} = Z_t \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}, \quad (5)$$

where $\theta \geq 1$ is the elasticity of substitution across labor types and $\tilde{N}$ is the resulting labor aggregate. We reserve $N \equiv N_f + N_p$ for total employment (the head count), which differs from the aggregate $\tilde{N}$ whenever $\theta < \infty$. We do not put a time subscript on the labor quantities since here labor is hired once and for all. We normalize $Z_0 = 1$, and assume Z_1 is log-normal with unit mean, $\log(Z_1) \sim N(-\frac{1}{2}\sigma_z^2, \sigma_z^2)$. Period-one productivity, Z_1 , is the only source of risk in the economy.

Since vacancies V_i can be expressed as $V_i = N_i/P_i$ and firms take the equilibrium matching probabilities P_i as given, we can express the firm's problem as a choice over labor inputs, N_f and N_p . The representative firm maximizes the sum of its expected profits discounted by the household's stochastic discount factor $M_{0,1}$ (defined below):

$$\max_{N_f, N_p} \pi_0 + E_0[M_{0,1}\pi_1], \quad (6)$$

subject to $N_f \geq 0, N_p \geq 0$, and the expressions for firm profits

$$\pi_0 = \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}} - W_{f,0}N_f - W_{p,0}N_p - \varphi \frac{N_f}{P_f} - \varphi \frac{N_p}{P_p}, \quad (7)$$

$$\pi_1 = Z_1 \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}} - W_{f,1}N_f - W_{p,1}N_p. \quad (8)$$

The stochastic discount factor, $M_{0,1}$, in (6) is that of the representative household. The household has Epstein-Zin preferences,

$$\mathbb{V}_0 = \left[(1 - \beta)C_0^{1-1/\psi} + \beta(\mathbb{E}_0 C_1^{1-\gamma})^{\frac{1-1/\psi}{1-\gamma}} \right]^{\frac{1}{1-1/\psi}}, \quad (9)$$

owns the representative firm, supplies labor inelastically and consumes subject to

$$\begin{aligned} C_0 &= \pi_0 + W_{f,0}N_f + W_{p,0}N_p, \\ C_1 &= \pi_1 + W_{f,1}N_f + W_{p,1}N_p, \end{aligned}$$

where $W_{f,t}N_f + W_{p,t}N_p$ is the household's time- t labor income. The household's stochastic discount factor is given by

$$M_{0,1} \equiv \left(\frac{\partial \mathbb{V}_0 / \partial C_1}{\partial \mathbb{V}_0 / \partial C_0} \right) = \beta \left(\frac{C_1}{C_0} \right)^{-1/\psi} \left(\frac{C_1}{(\mathbb{E}_0 C_1^{1-\gamma})^{\frac{1}{1-\gamma}}} \right)^{1/\psi - \gamma}. \quad (10)$$

The intertemporal elasticity of substitution is denoted by ψ , and risk aversion by γ . In order to isolate how risk premia propagate to the real economy, we focus directly on the effects of changes to the risk-aversion coefficient γ . However, changes in risk premia that originate from other sources (e.g., changes in fundamental volatility) would propagate in the same way.

To close the model, we need to specify wages. We consider two options. The first is a flexible wage that is renegotiated each period via Nash bargaining. We assume a household bargaining weight $\eta = 1 - \varepsilon$, which eliminates congestion externalities in the labor search market. Workers' outside option is zero, so the wage is η times the marginal product of labor:

$$W_{i,t}^{Nash} = \eta Z_t \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{1}{\theta-1}} N_i^{-\frac{1}{\theta}}. \quad (11)$$

The second alternative is a wage that does not adjust to the productivity shock in the second period, but is otherwise similarly proportional to the marginal product of labor

$$W_{i,t} = \eta \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{1}{\theta-1}} N_i^{-\frac{1}{\theta}}. \quad (12)$$

In particular, equation (12) implies $W_{i,t} = E_0[W_{i,1}^{Nash}]$. Thus, *at the time of hiring*, the wage takes into account current conditions and how they affect the expected marginal product of labor going forward. But the wage is *rigid* to the actual realizations of future shocks to the marginal product of labor. This generates operational leverage and makes labor positions

riskier from the viewpoint of the firm (e.g., Favilukis and Lin, 2016).

Wage rigidity also introduces the possibility that, in some states of the world, the future payments committed to labor could exceed labor's future product. In the remainder of this section, we assume that such states cannot dominate the future return to labor:

Assumption 1. *The present value of the firm's future $t = 1$ surplus, $E_0[M_{0,1}(MPN_{i,1} - W_{i,1})]$, is positive.*

Under the flexible wage in (11), Assumption 1 always holds. Under the rigid wage in (12), Assumption 1 also holds trivially when $\gamma = 0$. However, as risk aversion increases, the household discount factor puts larger weight on those future periods with negative realized returns to labor. Thus, Assumption 1 imposes an upper bound on γ which rules out pathological equilibria where firms perceive *negative* risk-adjusted second period returns to labor and would hence ex-ante prefer to separate at the end of the first period.

3.2 No-comovement benchmark

Proposition 1 crystallizes the classic comovement challenge in the context of our model: when both labor types have identical wage structures, they are equally risky and no reallocation occurs. In this case, an increase in risk aversion only changes total labor demand, and consumption-labor comovement is impossible.

Proposition 1. *Suppose that either (i) wages in both sectors are given by (11); or (ii) both wages are given by (12) and Assumption 1 holds. Then:*

1. *The risk premia on both types of labor, N_f and N_p , are identical.*
2. *(C_0, Y_0, N) cannot all move in the same direction in response to a change in risk aversion.*

The proof of Proposition 1 is given in Appendix B, but we provide key steps of the argument here. The first period resource constraint,

$$C_0 = Y_0 - \varphi N_f^{\frac{1}{\varepsilon}} - \varphi N_p^{\frac{1}{\varepsilon}}, \quad (13)$$

is key for understanding the contemporaneous comovements that occur in response to higher risk aversion. Equation (13) is reminiscent of the standard $Y = C + I$ equilibrium condition, since in a model of labor frictions vacancy posting is a form of real investment. Defining $MPN_{f,0} = \partial Y_0 / \partial N_f$ and $MPN_{p,0} = \partial Y_0 / \partial N_p$, and differentiating the resource constraint:

$$\frac{\partial C_0}{\partial \gamma} = \left(MPN_{f,0} - \frac{\varphi}{\varepsilon} N_f^{\frac{1-\varepsilon}{\varepsilon}} \right) \frac{\partial N_f}{\partial \gamma} + \left(MPN_{p,0} - \frac{\varphi}{\varepsilon} N_p^{\frac{1-\varepsilon}{\varepsilon}} \right) \frac{\partial N_p}{\partial \gamma}. \quad (14)$$

For the next step of the argument, it is important to note that each of the terms in parentheses above is negative: marginal hiring costs exceed the current-period marginal product of labor because matches deliver the marginal product today plus an expected surplus in the future.

Under identical wage setting, both labor types are symmetric from the viewpoint of the firm: they have the same wages, the same impact on production, and the same risk premium. Because of this symmetry, $N_f = N_p \equiv \bar{N}$, so both marginal products ($MPN_{f,0} = MPN_{p,0} \equiv MPN_0$) and the effects of risk aversion on labor demand ($\frac{\partial N_f}{\partial \gamma} = \frac{\partial N_p}{\partial \gamma} \equiv \frac{\partial \bar{N}}{\partial \gamma}$) are equalized. Using this symmetry, and defining $H = \varphi N_f^{\frac{1}{\varepsilon}} + \varphi N_p^{\frac{1}{\varepsilon}}$ as the aggregate expenditure on vacancy costs, we can simplify equation (14) down to

$$\frac{\partial C_0}{\partial \gamma} = \left(\frac{MPN_0}{\varphi \bar{N}^{\frac{1-\varepsilon}{\varepsilon}} / \varepsilon} - 1 \right) \frac{\partial H}{\partial \gamma}. \quad (15)$$

Since the parenthetical term in (15) is negative, consumption and labor always move in opposite directions. If an increase in risk aversion depresses aggregate labor demand (so H and by extension $\bar{N}$ fall), the fall in vacancy expenditures pushes consumption up in equilibrium, and vice versa.¹⁴ Thus, obtaining comovement — making consumption *fall* alongside employment — requires a mechanism beyond a reduction in aggregate labor demand.

3.3 Reallocation channel leads to comovement

Our theoretical contribution is to show that when durable factors of production have different risk-return profiles, factor reallocation can overcome this comovement barrier through an input *reallocation* effect. When risk aversion rises, it disproportionately reduces demand for the riskier factor, causing a flight-to-safety toward the safer factor. Because the riskier factor earns a higher return per unit of investment — this is the equilibrium compensation for bearing higher risk — this reallocation lowers the effective return on aggregate investment and pushes consumption down. When this reallocation effect is strong enough, it can overwhelm the mechanism captured by (15) to deliver full comovement. We now formalize this argument.

We introduce a wedge in the relative riskiness of different inputs by assuming that the full-time wage is sticky, while the part-time wage is flexible. This makes full-time labor matches relatively risky for the firm, since its surplus is exposed to productivity fluctuations. In this

¹⁴In the appendix, we show that under flexible wages, $\psi > 1$ implies that an increase in γ causes N to fall and C_0 to rise; the opposite holds for $\psi < 1$. Rigid wages amplify the effects of the risk premium, so N falls for a wider range of ψ . Our goal is to highlight the impossibility of getting C and N to move in the same direction, regardless of which direction labor moves.

case, firms' optimal labor demand satisfies the following Euler-type asset pricing relations,

$$\frac{\varphi}{\varepsilon} N_p^{\frac{1-\varepsilon}{\varepsilon}} = MPN_{p,0} \left[1 + \frac{1}{R} + Cov(M_{0,1}, Z_1) \right] \quad (16)$$

$$\frac{\varphi}{\varepsilon} N_f^{\frac{1-\varepsilon}{\varepsilon}} = MPN_{f,0} \left[1 + \frac{1}{R} + \frac{1}{\varepsilon} Cov(M_{0,1}, Z_1) \right]. \quad (17)$$

Equations (16) and (17) equate the cost of hiring on the left-hand side with the expected present-discounted benefit of an extra worker of a given type. Comparing these equations, it is clear that rigid wages amplify the risk premium on full-time labor, since the (negative) covariance term $Cov(M_{0,1}, Z_1)$ is divided by $\varepsilon < 1$ in (17). This wedge in relative riskiness is the key ingredient that makes risk-driven comovement possible.

Proposition 2. *Suppose that wages in the part-time sector are given by (11) and wages in the full-time sector are given by (12) and Assumption 1 holds. Then,*

1. *The risk premium on full-time labor is larger than the risk premium on part-time labor.*
2. *An increase in risk aversion causes reallocation of hiring expenditure towards part-time labor.*
3. *When $\psi = 1$ and $\theta > \frac{1+\beta}{\beta(1-\varepsilon)}$, there exists a threshold $\tilde{\gamma}$ such that, for $\gamma > \tilde{\gamma}$, (C_0, Y_0, N) are all falling in risk aversion.*

The key result of Proposition 2 is the macroeconomic comovement that can obtain from risk aversion shocks when there are differences in the relative riskiness of the two labor types (point 3). The result follows from points 1 and 2, and we now explain the intuition.

Starting with Point 1, in asset pricing terms the higher risk premium on full-time labor translates into a higher equilibrium rate of return (or dividend-to-price ratio). Defining the “price” of labor investment as the marginal cost of hiring, $\frac{\varphi}{\varepsilon} N_i^{\frac{1-\varepsilon}{\varepsilon}}$ for $i \in \{p, f\}$, we can use (16) and (17) to derive the following relationship between returns of the two labor types:

$$r_f \equiv \frac{MPN_{f,0}}{\varphi N_f^{\frac{1-\varepsilon}{\varepsilon}} / \varepsilon} > \frac{MPN_{p,0}}{\varphi N_p^{\frac{1-\varepsilon}{\varepsilon}} / \varepsilon} \equiv r_p. \quad (18)$$

Intuitively, in order to compensate for its higher riskiness, full-time labor generates more output per unit of hiring expenditure than part-time labor.

Point 2 of the Proposition states that the risk premium difference also implies that an increase in risk aversion reallocates hiring expenditure from the riskier full-time sector to the safer part-time sector. This is true for any parameterization of the economy with positive risk aversion. To see why, note that equations (16) and (17) imply that a marginal increase

in γ lowers the present discounted value of full-time labor by more than that of part-time labor, since the covariance term enters with coefficient $\frac{1}{\varepsilon} > 1$ in (17). Firms therefore find full-time vacancies relatively less attractive at any positive level of risk aversion, and shift expenditure toward part-time hiring as risk aversion increases. The difference in output per dollar invested highlighted in Point 1 directly implies that, holding total hiring expenditure fixed, this reallocation of hiring expenditure towards the part-time sector lowers output and, by the resource constraint (13), consumption.

A fundamental question is whether the reallocation channel is ever strong enough to overwhelm the aggregate labor demand channel that we highlighted in Proposition 1. To see how these two forces operate on equilibrium consumption, we can now consider what the derivative of the resource constraint (i.e. equation (14)) looks like in this version of the model. Defining $\omega_p \equiv \varphi N_p^{\frac{1}{\varepsilon}}/H$ as the share of total hiring expenditure dedicated to part-time workers, and letting $\bar{r} \equiv (1 - \omega_p)r_f + \omega_p r_p$ be the (share-weighted) average return ratio, we can express the differentiated resource constraint as:

$$\frac{\partial C_0}{\partial \gamma} = \underbrace{(\bar{r} - 1)}_{<0} \frac{\partial H}{\partial \gamma} + H \underbrace{(r_p - r_f)}_{<0} \underbrace{\frac{\partial \omega_p}{\partial \gamma}}_{>0}. \quad (19)$$

The first term is the same aggregate labor demand channel we already analyzed in the symmetric model and Proposition 1. The second term on the right-hand side of (19) captures the new reallocation channel that is at the center of this paper. Because an increase in risk aversion always shifts expenditure from full-time to part-time hiring ($\frac{\partial \omega_p}{\partial \gamma} > 0$), and because full-time labor delivers more output per unit of hiring expenditure, this reallocation unambiguously depresses consumption. The term $H(r_p - r_f) < 0$ measures the consumption cost of this shift: for every dollar reallocated from full-time to part-time vacancy expenditure, the economy loses $(r_f - r_p)$ units of current output per dollar invested, which is then scaled by total hiring investment H .

Clearly, without this novel reallocation channel — a change in ω_p — comovement is not possible in this model. Furthermore, point 3 of Proposition 2 establishes the specific condition under which the reallocation channel dominates the aggregate labor demand channel, so that an increase in risk aversion indeed generates a simultaneous decline in consumption, output, and employment. Setting $\psi = 1$ not only simplifies the stochastic discount factor, but also serves as a useful benchmark since this is the value of ψ at which labor demand does not change with risk aversion when all wages are flexible. The condition that θ is sufficiently large helps to ensure that enough reallocation toward part-time workers occurs in response to the change in risk aversion. This parameter is among those that we estimate in the

subsequent quantitative analysis.

4 Quantifying the Mechanism

In this section, we ask whether *pure* shocks to risk aversion could account quantitatively for the patterns we see in the data. We estimate an extended version of the model to match the impulse responses identified in Section 2, as well as several unconditional moments in the data related to labor markets and risk premia. We then use the estimated model to perform several counterfactual exercises that elucidate the importance of reallocation in driving macroeconomic comovement.

4.1 Extended Model

The model consists of a representative household and a representative firm. The household consumes, supplies labor inelastically, and invests in firm equity as well as corporate and government debt instruments. The firm produces final goods and invests in capital and in two types of labor (via labor search markets) in order to maximize shareholder value.

Households

There is a unit mass of identical households. In period t , the household chooses aggregate consumption (C_t), government bond holdings (B_{t+1}), corporate bond holdings (B_{t+1}^c), and holdings of equity shares in the firms (X_{t+1}) to maximize lifetime utility,

$$\mathbb{V}_t = \max \left[(1 - \beta) C_t^{1-1/\psi} + \beta (\mathbb{E}_t \mathbb{V}_{t+1}^{1-\gamma_t})^{\frac{1-1/\psi}{1-\gamma_t}} \right]^{\frac{1}{1-1/\psi}}, \quad (20)$$

subject to the budget constraint denoted in terms of the consumption numeraire,

$$C_t + P_t^e X_{t+1} + Q_t^c (B_{t+1}^c - d B_t^c) + \frac{1}{R_t} B_{t+1} \leq (D_t^e + P_t^e) X_t + B_t^c + B_t + E_t^l + T_t.$$

In the above, Q_t^c is the price of a multi-period corporate bond, R_t is the one-period safe real interest rate, P_t^e is the price of a share of the representative firm that pays a real dividend D_t^e , and E_t^l is the household's total labor earnings (detailed below). T_t denotes lump-sum transfers. We model corporate bonds following [Gourio \(2012\)](#), and assume they repay a constant fraction $1 - d$ of the principal each period. These bonds are only needed to create an empirically relevant amount of financial leverage in firms, since we will eventually match the average equity risk premium in the data. Government bonds are in zero net supply, and

serve to define the safe real rate.

The Epstein-Zin preferences in equation (20) give the stochastic discount factor:

$$M_{t,t+1} \equiv \left(\frac{\partial \mathbb{V}_t / \partial C_{t+1}}{\partial \mathbb{V}_t / \partial C_t} \right) = \beta \left(\frac{C_{t+1}}{C_t} \right)^{1-1/\psi} \left(\frac{C_t}{C_{t+1}} \right) \left(\frac{\mathbb{V}_{t+1}}{(\mathbb{E}_t \mathbb{V}_{t+1}^{1-\gamma_t})^{\frac{1}{1-\gamma_t}}} \right)^{1/\psi-\gamma_t}. \quad (21)$$

In parallel with the analytical model, we are interested in shocks to risk aversion, so γ_t is allowed to vary exogenously over time.

Labor markets are subject to search and matching frictions. The structure of the labor market generalizes the one in Section 3 in several ways. First, while we continue to assume that total labor supplied by the representative household is fixed, we now model an optimal choice between search in either of two labor markets. Second, we allow for differences in separation rates between the two position types, a realistic assumption that amplifies the firms' perceived riskiness of longer-lived full-time employment relationships. Third, we allow for a parameter Ω that differentiates the labor input contributed by full- and part-time workers. Finally, we allow for vacancy posting costs and matching functions to differ across labor markets. We discuss each of these asymmetries in more detail below.

Household labor search works as follows. At the end of each period, call it $t-1$, employed workers separate from their previous job with probability ρ_f or ρ_p , depending on which sector they were employed in.¹⁵ Normalizing total labor supply to unity, this leaves a mass of $1 - (1 - \rho_f)N_{f,t-1} - (1 - \rho_p)N_{p,t-1}$ non-employed workers who must choose which market to search in. The household allocates the search effort of these potential workers to whichever market brings a higher contribution to household utility, dividing the available search effort according to

$$\tilde{S}_{f,t} + S_{p,t} = 1 - (1 - \rho_f)N_{f,t-1} - (1 - \rho_p)N_{p,t-1}. \quad (22)$$

In addition to the search effort from non-employed workers, we assume that workers in the part-time sector who do not experience an exogenous separation have the opportunity to do “on-the-job” search in the full-time sector. Hence, total search effort in the full-time sector is given by $S_{f,t} = \tilde{S}_{f,t} + (1 - \rho_p)N_{p,t-1}$.¹⁶

Let $\mathbb{S}_{i,t}$, $i \in \{f, p\}$, be the utility of providing a unit of search effort in market i . If search effort is to be positive in both markets, the household must be indifferent between searching in either market, so that in each period $\mathbb{S}_{f,t} = \mathbb{S}_{p,t}$. We provide explicit derivations of the $\mathbb{S}_{i,t}$

¹⁵There is some countercyclical in the rate of layoffs but quit rates are procyclical. Together, total separation rates seem to be fairly acyclical, which is why we model them as constant.

¹⁶This assumption captures the empirical facts that workers frequently move from part-time to full-time employment without an intervening unemployment spell (Borowczyk-Martins and Lalé, 2020), and that on-the-job search is pervasive and most intense at the lower rungs of the job ladder (Faberman et al., 2022).

in Appendix C.3.

Workers who find no employment in either market are unemployed, and receive a benefit b_t that corresponds to monetary unemployment benefits as well as any other time-use benefits they might accrue from not working. Workers employed in the part-time sector receive a wage as well as a flow κ_t that corresponds to the benefits (e.g., of home production) from the additional time made available by only working part-time. Both b_t and κ_t are time-varying only because they are cointegrated with the stochastic trend in our economy. Aggregate household earnings are given by:

$$E_t^l = W_{f,t}N_{f,t} + (W_{p,t} + \kappa_t)N_{p,t} + b_t(1 - N_{f,t} - N_{p,t}). \quad (23)$$

Firms

The representative firm has cash flows,

$$D_t = Y_t - W_{f,t}N_{f,t} - W_{p,t}N_{p,t} - I_t - \varphi_{f,t}V_{f,t} - \varphi_{p,t}V_{p,t}. \quad (24)$$

It maximizes profits by choosing employment for the two types of contracts, $N_{f,t}$ and $N_{p,t}$, the two types of vacancies, $V_{f,t}$ and $V_{p,t}$, physical capital, K_{t+1} , and investment, I_t . The variables $W_{i,t}$ and $\varphi_{i,t}$ denote the real wage and the vacancy posting cost for the labor contract of type $i \in \{f, p\}$, all of which the firm takes as given.

The firm's objective is to maximize $\tilde{P}_t^e \equiv \mathbb{E}_t \sum_{s=0}^{\infty} M_{t,t+s} D_{t+s}$ subject to a production function with labor-augmenting technology Z_t

$$Y_t \leq K_t^\alpha (Z_t \tilde{N}_t)^{1-\alpha}, \quad (25)$$

a CES labor aggregator that combines the inputs of the full-time and part-time workers into the labor aggregate $\tilde{N}_t$ (which we distinguish from total employment, the head count $N_t \equiv N_{f,t} + N_{p,t}$),

$$\tilde{N}_t = \left((1 - \Omega) N_{f,t}^{\frac{\theta-1}{\theta}} + \Omega N_{p,t}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}, \quad (26)$$

where the parameter $\Omega \in (0, 1)$ allows for differences in the effective labor input of part-time relative to full-time workers, allowing us to match the empirical share of part-time employment. The firm also faces a capital accumulation equation with quadratic capital adjustment costs,

$$K_{t+1} = \left(1 - \delta - \frac{\phi_K}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 \right) K_t + I_t, \quad (27)$$

and the laws of motion for employment as perceived by the firm,

$$N_{f,t} = (1 - \rho_f)N_{f,t-1} + \Theta_{f,t}V_{f,t}, \quad (28)$$

$$N_{p,t} = (1 - \rho_{p,t})N_{p,t-1} + \Theta_{p,t}V_{p,t}, \quad (29)$$

where $\Theta_{i,t}$ is the probability of filling a type- i vacancy and again $\rho_{p,t} \equiv \rho_p + (1 - \rho_p)P_{f,t}$ takes into account part-timers that experience a job-to-job transition from a part-time position to a full-time position.

Equations (28)-(29) imply that workers engage in production as soon as they are hired. We adopt this timing assumption following [Christiano et al. \(2016\)](#) because a period in our model corresponds to one quarter and it would be implausible to assume a whole quarter of delay between a worker-firm match and the start of employment.

We assume that the representative firm can raise capital by issuing equity shares and debt. Specifically, we follow [Jermann \(1998\)](#) by assuming the representative firm finances a percentage of its physical capital stock each period through debt. As in [Gourio \(2012\)](#), this financing occurs with multi-period riskless bonds. Firm debt evolves according to $B_{t+1}^c = dB_t^c + L_t$, where the parameter $d \in [0, 1]$ is the portion of outstanding debt that does not mature in the current period, and hence determines the effective duration of a bond as $\frac{1}{1-d}$ quarters. The net amount of new borrowing each period, $Q_t^c L_t = \xi K_{t+1}$, is proportional to the quantity of capital owned by the firm. Under these assumptions, the steady-state leverage ratio of the firm is given by $B^c/K = \xi/(1 - d) \equiv \nu$. This is a parameter we will estimate. The price of the multi-period bond (Q_t^c) is determined by the pricing equation $Q_t^c = \mathbb{E}_t [M_{t,t+1}(dQ_{t+1}^c + 1)]$. Total firm cash flows are divided between payments to bond holders and equity holders according to: $D_t^E = D_t - B_t^c + \xi K_{t+1}$.

Since the [Modigliani and Miller \(1958\)](#) theorem holds in our model, financial policies such as leverage decisions do not affect firm value or optimal firm decisions. Leverage does, however, affect the volatility of cash flows to shareholders and, therefore, the price of equity and its risk premium. The introduction of leverage allows us to map equity returns from the model to the data, where firms carry significant financial leverage.

Wage-setting

We make a set of assumptions about wage determination that simplify our equilibrium computations and serve as a realistic baseline for examining the importance of our mechanism.

The initial values of the wages in both sectors are the Nash-bargained wages that would emerge in a non-stochastic steady-state with $Z = 1$. Denoting with η_f and η_p the bargaining

power of full-time and part-time workers, Nash bargaining implies:

$$\mathbb{W}_i - \mathbb{U} = \frac{\eta_i}{1 - \eta_i} \frac{\varphi_i}{\Theta_i}. \quad (30)$$

Here, $\mathbb{W}_i - \mathbb{U}$ represents the value of a match to the worker net of the value of unemployment. Nash bargaining implies that this is proportional to the value of the match to the firm. Given free entry, the latter equals the expected cost of filling a vacancy, $\frac{\varphi_i}{\Theta_i}$, where Θ_i is the job-filling probability. We provide additional details in Appendix C.3.

We assume that wages in the part-time sector are set by Nash bargaining each period, while wages in the full-time sector are sticky, and equal each period to their previous value plus an adjustment for the change in the level of productivity (detailed below). The wage in the part-time sector is flexibly Nash-bargained. These assumptions qualitatively match the observation of [Lariau \(2017\)](#) that part-time positions have more flexible wages than full-time positions.¹⁷

Government

The government finances an exogenous stream of expenditures that slowly adjusts to the productivity level of the economy. (The process for G_t is described below.) The initial value of the government expenditure in a non-stochastic steady-state with $Z = 1$ is $G = \bar{g}Y$. The government finances a balanced budget via lump-sum taxes, and as a result government bonds are always in zero net supply: $B_t = 0$, for all t .

Market Clearing

At the aggregate level, the labor workforce at time t in the two sectors is:

$$N_{f,t} = (1 - \rho_f)N_{f,t-1} + \mathcal{M}_{f,t}, \quad (31)$$

$$N_{p,t} = (1 - \rho_{p,t})N_{p,t-1} + \mathcal{M}_{p,t}, \quad (32)$$

where $\mathcal{M}_{f,t}$ and $\mathcal{M}_{p,t}$ are the matches from the Cobb-Douglas matching functions of the full-time and part-time sectors, respectively. These matching functions are given by

$$\mathcal{M}_{i,t} = \chi_i V_i^{\epsilon_i} S_i^{1-\epsilon_i}, \quad (33)$$

¹⁷As a separate observation, part-time workers are predominantly paid by the hour, while the majority of full-time workers are salaried workers. The empirical evidence indicates that salaries are stickier than hourly wages [Barattieri et al. \(2014\)](#).

for $i \in \{f, p\}$. The corresponding job-finding and vacancy-filling probabilities as a function of labor market tightness $\theta_{i,t} = \frac{V_{i,t}}{S_{i,t}}$ are respectively: $P_{i,t}^m = \chi_i \theta_{i,t}^{\epsilon_i}$ and $\Theta_{i,t} = \chi_i \theta_{i,t}^{\epsilon_i - 1}$.

Finally, the aggregate resource constraint in the economy is given by

$$Y_t = C_t + I_t + \varphi_{f,t} V_{f,t} + \varphi_{p,t} V_{p,t} + G_t. \quad (34)$$

In order to ensure our model satisfies the national accounting identity, we follow [den Haan and Kaltenbrunner \(2009\)](#) by including job posting costs in defining our model analogue to measured investment, i.e., $\tilde{I}_t \equiv I_t + \varphi_{f,t} V_{f,t} + \varphi_{p,t} V_{p,t}$.

Exogenous Processes

The economy is perturbed by two disturbances. Technology, Z_t , follows a random walk, consistent with the utilization-adjusted U.S. TFP data of [Fernald \(2014\)](#), while risk aversion, γ_t , follows an AR(1) process in logs:

$$\log(Z_t) = \log(Z_{t-1}) + \sigma_z \epsilon_t^z \quad (35)$$

$$\log(\gamma_t / \gamma_{ss}) = \rho_\gamma \log(\gamma_{t-1} / \gamma_{ss}) + \sigma_\gamma \epsilon_t^\gamma. \quad (36)$$

Direct shocks to risk aversion have also been used in other contexts, for example to explain the dynamics of the term structure of interest rates ([Dew-Becker, 2014](#)).

Because our economy has a unit root in productivity, we impose additional assumptions to ensure that the model has a balanced growth path. In particular, we assume that the cost of vacancy posting, the workers' outside options, the sticky full-time wage, and government expenditure are all cointegrated with technology, with a common error-correction rate of ω . Specifically, for each variable $X \in \{\varphi_{f,t}, \varphi_{p,t}, \kappa_t, b_t, W_{f,t}, G_t\}$, we assume that $X_t = \Gamma_t \bar{X}$ where $\bar{X}$ is the deterministic steady-state value, and $\Gamma_{t+1} = \Gamma_t^\omega Z_t^{1-\omega}$, with $\omega \in [0, 1)$. Combining these relationships for (e.g.) the case of the wage, we have

$$W_{f,t} = \left(\frac{Z_{t-1}}{\Gamma_{t-1}} \right)^{1-\omega} W_{f,t-1}. \quad (37)$$

To the extent that $\omega > 0$, the full-time wage only gradually “catches up” with productivity.

We solve the model using a third-order perturbation technique, as implemented in the Dynare code package. Dynare computes a third-order approximation to the equilibrium conditions around the non-stochastic steady-state of the economy. Impulse responses are computed by comparing the path of the economy over an extended period in which the realizations of all shocks are identically zero to the counterfactual path in which a single

Table 1: Calibrated Parameters

Name	Description	Value
β	Discount rate	0.994
ϕ_k	Capital Adj. Cost	10.000
ψ	Intertemporal elasticity of substitution	2.500
α	Capital share	0.300
δ	Capital depreciation rate	0.025
$\bar{g}$	Steady-state G/GDP	0.200
d	Corporate bond duration	0.975
Labor Markets		
ρ_f	Separation Rate - full-time	0.042
ρ_p	Separation Rate - part-time	0.335
η_f	HH's bargaining power - full-time	0.500
η_p	HH's bargaining power - part-time	0.500
Exogenous Processes		
σ_z	Std. dev. of tech shock	0.008

one-standard-deviation shock to γ_t is realized. For a simplified version of our empirical model, we have also compared Dynare's third-order approximation results to those from a global solution computed using projection techniques. In these cases, we find the third-order approximation works well. We present the full set of conditions that describe the equilibrium in Appendix C.

4.2 Calibrated Parameters and Steady-State Targets

To begin, we calibrate a set of standard parameters to values that are consistent with the literature and summarized in Table 1. We set $\beta = 0.994$ as in Basu and Bundick (2017). We fix the depreciation rate to $\delta = 0.025$ and the capital share parameter to $\alpha = 0.3$. Because the estimated model includes risk, this will imply an unconditional capital income share that is slightly less than 0.3. We fix the long-run share of government expenditures to GDP to 20% and the bond duration parameter $d = 0.975$ as in Gourio (2012), which implies corporate debt has a 10-year maturity.

Estimates of capital adjustment costs vary considerably in the literature, and range from values around 2 in macro contexts (e.g., Basu and Bundick, 2017) to values of 18 or higher in micro studies (e.g., Galeotti and Schiantarelli, 1991). We set our adjustment cost parameter $\phi_k = 10$. This value lies in the middle of this range and is in line with the value obtained by Belo et al. (2022), which estimates neoclassical investment models on a rich dataset of

market value data of U.S. publicly-traded firms.

We set the elasticity of intertemporal substitution to $\psi = 2.5$, which is in line with the macro-finance practice of picking IES higher than one (Schorfheide et al., 2018). This value is relatively high compared to the macro literature that focuses on quantities only, but overall the quantitative fit of the model does not rely on any particular restriction on ψ . To illustrate this, in Appendix F.2 we reestimate the model assuming $\psi = 0.5$, and find that the overall difference in fit with our benchmark is modest.

We calibrate the separation rates of full-time and part-time workers, ρ_f and ρ_p , to satisfy two features of the data. First, we fix $\rho_p/\rho_f = 8$, matching recent estimates of the relative difference in separation rates of part-timers to full-timers from the longitudinal dimension of the U.S. Current Population Survey (CPS) (Lariau, 2017; Borowczyk-Martins and Lalé, 2019). Second, we fix the level of separations in the full-time sector (ρ_f) so that the average separation rate across both labor sectors equals the aggregate quarterly rate in the U.S. economy of 10% (Yashiv, 2008). We also fix the Nash bargaining parameters to $\eta_f = \eta_p = 0.5$, the midpoint of empirical estimates (e.g., Flinn, 2006) but have found that alternative choices for these parameters make very little difference.

Finally, we use the Basu et al. (2006) utilization adjustments to U.S. TFP, as implemented in quarterly data by Fernald (2014), to calibrate the process for productivity. Over our sample period, productivity is an almost perfect random walk with standard deviation in growth rates of $\sigma_z = 0.008$.

The remaining parameters are estimated by matching the impulse responses in the model to a risk-aversion shock ϵ_t^γ to the empirical responses from Section 2.1 along with the eight additional unconditional moments reported in Table 2. Our approach is to place extremely high weight on the unconditional moment targets in the estimation procedure to force the model to match the unconditional moments perfectly, and then see how the model does in terms of conditional dynamics.

Among these unconditional moments, the average equity premium, the share of part-time workers, and the average unemployment rate are directly observed in the data, and we match their average values over our sample period. The targeted average vacancy rate of 3.5% comes from the full-sample average of the JOLTS dataset (which starts in 2000). In line with Blanchard and Galí (2010), we target a ratio of hiring costs to GDP of 1%.

We also target the standard deviations of (HP-filtered) employment and vacancies (using the series created by Barnichon, 2010), in order to ensure that the model delivers a Beveridge curve in line with the data. We also note that, since it successfully matches both of these moments, our model is not subject to the Shimer critique – this is because in our model the risk aversion shocks generate large fluctuations in employment and vacancies.

Table 2: Unconditional Target Moments

Description	Target	Model Implied
Equity risk premium	0.055	0.055
Share of part-time	0.198	0.196
LR unemployment	0.059	0.059
Vacancy Rate	0.035	0.035
Hiring cost/GDP	0.010	0.010
Part-time earn./Full-time earn.	0.375	0.375
Std. HP log(Emp/Pop)	0.013	0.013
Std. HP log(vacan.)	0.138	0.138

Finally, we target a ratio of part-time to full-time earnings of 0.375. We arrive at this ratio by assuming that part-time workers work one-half the number of hours of full-time workers (in line with CPS averages), and earn an hourly wage that is 25% lower than similar full-time workers ($0.375 = 0.5 \times 0.75$), in line with studies on this wage penalty ([Aaronson and French, 2004](#); [Bick et al., 2022](#)).¹⁸ Our results change very little if we make substantially different assumptions about the part-time wage penalty.

4.3 Estimation Procedure

Aside from the long-run targets in Table 2, our impulse response matching exercise is standard. We target the impulse responses of output, consumption, investment, total employment, part-time employment, equity returns, and the real interest rate estimated using our “Method 1” for controlling for the effects of supply shocks. The set of estimated parameters, denoted by Π , includes the steady-state risk aversion parameter γ , the aggregate leverage ratio ν , the vacancy posting costs, φ_f and φ_p , the value of outside options κ and b , the production share of part-time labor Ω , the elasticity of substitution between the two types of labor θ , the four parameters governing the aggregate matching technologies, the cointegration parameter ω , and the parameters of the risk aversion shock, ρ_γ and σ_γ .

Let $\hat{\Psi}$ denote the column vector stacking point estimates of each impulse response at each horizon along with our unconditional target moments and let $\Psi(\Pi)$ denote the corresponding theoretical responses and moments. The estimated parameter vector is

$$\hat{\Pi} \equiv \arg \min_{\Pi} (\hat{\Psi} - \Psi(\Pi))' W (\hat{\Psi} - \Psi(\Pi)). \quad (38)$$

¹⁸See also: <https://www.epi.org/publication/part-time-pay-penalty>.

Table 3: Estimated Parameters

Name	Description	Point Est.	Std Err.
γ_{ss}	Steady-state risk aversion	36.236	1.807
ν	Leverage Ratio	0.767	0.014
Labor Markets			
φ_f	Vacancy posting cost - full-time	0.955	0.088
φ_p	Vacancy posting cost - part-time	0.234	0.027
κ	Value if no perm posit.	0.725	0.008
b	Value if unemployed	1.194	0.007
Ω	Labor contrib. of part-time	0.228	0.003
θ	Elas. between full- & part-time	5.079	0.278
ϵ_f	Matching elasticity - full-time	0.392	0.020
ϵ_p	Matching elasticity - part-time	0.025	0.003
χ_f	Matching technology - full-time	0.449	0.019
χ_p	Matching technology - part-time	1.594	0.223
ω	Gradual wage adj.	0.976	0.002
Risk Aversion Process			
ρ_γ	AR(1) risk av. shock	0.934	0.010
σ_γ	Std. dev. of risk av. shock	0.438	0.049

Note: Standard errors computed via bootstrap, by reestimating model parameters targeting $N=100$ different (bias-corrected) impulse responses drawn from the VAR bootstrap procedure. All parameter estimates except for ϵ_p are interior.

The matrix W is a diagonal weighting matrix consisting of the inverse of the bootstrapped variances of each impulse response in $\hat{\Psi}$, plus very large weights for our unconditional target moments. Given the extreme weights on our eight unconditional targets, we are essentially targeting $7 \times 30 = 210$ impulse response moments with just 7 degrees of freedom.

4.4 Estimation Results and Model Fit

The estimation procedure searches for a global optimum, and Table 3 reports the estimated parameters $\hat{\Pi}$ along with their corresponding standard errors.¹⁹

Our estimate of $\gamma \approx 36.2$ is similar to or lower than the values used by other quantitative papers focused on matching risk premium facts in business cycle models (e.g., Piazzesi and Schneider, 2006; Rudebusch and Swanson, 2012; Basu and Bundick, 2017; Caggiano et al., 2021). That this estimate remains “high” relative to microeconomic estimates of risk aversion is a manifestation of the well-known “equity premium” puzzle, the resolution of which remains debated.²⁰ Like the papers cited above, we seek to match the observed risk premium

¹⁹We impose wide parameter bounds on the search space, and only ϵ_p obtains a (lower) bound.

²⁰Potential solutions include habit formation (Campbell and Cochrane, 1999), long-run risk (Bansal and Yaron, 2004), rare disasters (Barro, 2006), or parameter/model uncertainty (Weitzman, 2007; Barillas et al.,

without taking a strong stand on this debate. The estimated risk-aversion process implies that, along the response to a one-standard-deviation shock, the coefficient of risk aversion rises from its steady-state value to about 56.2.²¹

The labor market parameters that can be compared to the literature are those of the full-time sector. Our estimates imply a replacement ratio, measured as unemployment surplus flow divided by the full-time wage, of 0.89. This value is somewhat higher than in calibrations intended to capture only the narrow monetary size of unemployment benefits but is lower than in [Hagedorn and Manovskii \(2008\)](#) who argue that this parameter should also reflect disutility of providing labor. The elasticity of full-time matches with respect to searchers is 0.39, close to [Gertler and Trigari \(2009\)](#) and in the middle of the values used in the literature.

The point estimate of the leverage ratio parameter ν is 0.77, which implies a liability-to-equity ratio of 3.3. This ratio is above the Flow of Funds average of 2.2 for aggregate non-financial corporations during our sample, but we view this as reasonable given that our empirical target for the equity price corresponds to the aggregate of S&P 500 firms, which are more levered and include large financial firms. In [Appendix F.3](#), we reestimate the model imposing $\nu \leq 0.65$, which caps the liability-to-equity ratio at approximately 2, and discuss the implications of such a restriction.

Moving to the main results, [Figure 4](#) shows that the impulse responses implied by the estimated model (blue-dotted lines) match the data well, including the aggregate comovement patterns that traditionally define the business cycle. On the macroeconomic side, the changes in output, consumption and employment track the data quite closely. Investment undershoots modestly, but the model-implied responses are substantial and remain for the most part within the standard error bands of the VAR.

The estimated model also captures two central conditional features of asset prices. First, the model closely matches the persistent increase in the 5-year equity risk premium. Second, it matches the steep fall in stock returns on impact and the subsequent long period of above-average returns. Working against this fall in asset prices is a concurrent fall in safe interest rates, which the model somewhat overstates relative to the data. Nevertheless, the increase in the risk premium in the model is both larger (by a factor of approximately 2.6) and more persistent than the real rate decline. The pattern of much larger fluctuations in risk premia relative to safe interest rates is consistent with the data.

To further assess the external validity of our estimated model, we ask how well it reproduces some *untargeted* macroeconomic and asset pricing moments. The left panel of

[2009](#)).

²¹Because our solution procedure is local and we target impulse responses, this is the most relevant statistic for assessing the size of risk aversion fluctuations.

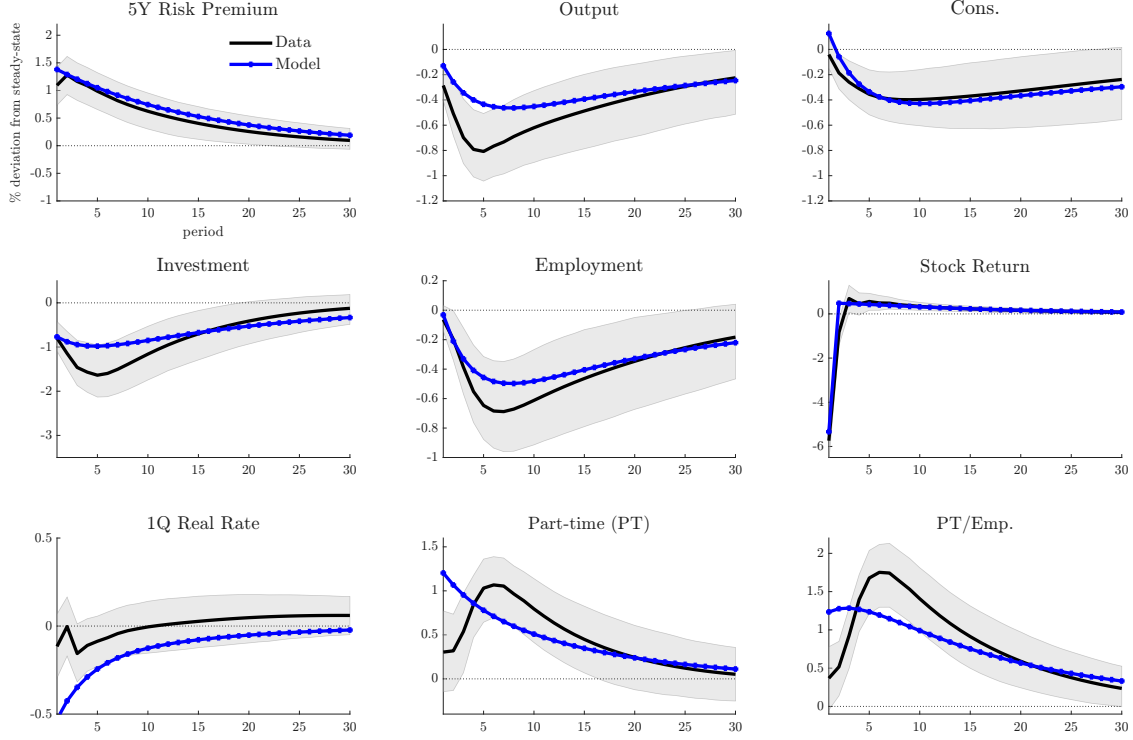


Figure 4: Impulse responses to a risk premium disturbance along with model-implied responses to a risk aversion shock. Shaded areas cover the 10th- and 90th-percentiles of bootstrapped responses. Vertical axis values are in annualized percentage points for the risk premium and stock return; all other numbers are in percentage deviations.

Table 4 shows that our model matches the unconditional volatility of output, investment, employment and the part-time employment share. The model overpredicts to some extent the volatility of consumption but still implies that consumption is less volatile than output.

The right panel of the table shows that the model also succeeds in replicating several untargeted unconditional asset pricing moments. Despite the somewhat large interest rate response to risk aversion shocks, our model predicts an unconditionally low and stable risk-free rate in line with the data. Besides reproducing the unconditional levels of risk premia that we targeted in estimation, our model also implies realistic unconditional variability of excess returns, which was not targeted. Indeed, the model's unconditional standard deviation of the (annualized) 1-quarter excess return of 31.4% is quite close to the empirical counterpart of 29.8%. Moreover, the annualized Sharpe ratio, calculated using quarterly returns as $SR = \frac{E[\log(R_{t+1}^E/R_t^b)]}{std[\log(R_{t+1}^E/R_t^b)]}$, implied by our model is 0.34, which is quite close to the empirical value of 0.35 in our sample.

Table 4: Unconditional and Asset Pricing Moments

Real Variables			Asset Prices		
Moment	Data	Model	Moment	Data	Model
Std(Y)	1.49	1.57	$E[r_t^b]$	1.54	1.86
Std(C)	0.82	1.33	Std(r_t^b)	2.53	1.36
Std(I)	3.20	2.97	$E[rp_{t,t+1}^1]$	5.50	5.50
Std(N)	1.31	1.34	Std($rp_{t,t+1}^1$)	29.76	31.39
Std(N_p/N)	2.21	2.18	Ann. Sharpe Ratio	0.35	0.34

Note: Unconditional model moments based on a single simulation of 5000 periods. Data moments based on data from our empirical sample period. All real variables are logged and then HP-filtered with a penalty parameter $\lambda = 1600$. Asset pricing moments are annualized percentages (i.e., quarterly percentages multiplied by 4) and are not filtered. Standard deviations are in percent. Sharpe ratio is for annualized equity returns.

4.5 Unpacking the Mechanism

Our analytical results in Section 3 show that reallocation between durable factors of production with different riskiness and marginal products is central for achieving macroeconomic comovement, while fluctuations in the overall demand for risky inputs naturally cause consumption to move in the opposite direction of output in equilibrium. In this section, we quantify these different channels and their effects on consumption and GDP.

We start by rewriting the aggregate resource constraint $Y_t = C_t + H_t + G_t$ where

$$H_t \equiv I_t + \varphi_{f,t}V_{f,t} + \varphi_{p,t}V_{p,t} \quad (39)$$

captures the combined investment in all three factors of production, and hence the total amount of “savings” in the economy. We then decompose H_t as

$$H_t = (\omega_{I,t} + \omega_{f,t} + \omega_{p,t})H_t \quad (40)$$

where the ω ’s are the shares of total savings dedicated to investment in the three different factors of production, e.g., $\omega_{I,t} \equiv \frac{I_t}{H_t}$. Movements in these shares, holding H_t fixed, capture changes in the relative demand for the three different types of risky inputs.

We next implement decompositions of the impulse responses of C_t and Y_t to a risk aversion shock by considering alternative paths for the three share parameters as well as the total level of savings in the economy. We use these counterfactual exercises to isolate the effects of three different equilibrium forces: (1) the reallocation from full-time to part-time labor only, (2) the reallocation from labor to physical capital, and (3) the overall change in the demand for risky inputs. The sum of these three separate effects adds up to the equilibrium responses of C_t and Y_t , as plotted in Figure 4.

In our first counterfactual, we fix the level of total savings H_t and capital investment I_t to their steady states, and using the actual impulse response of the shares of savings going to

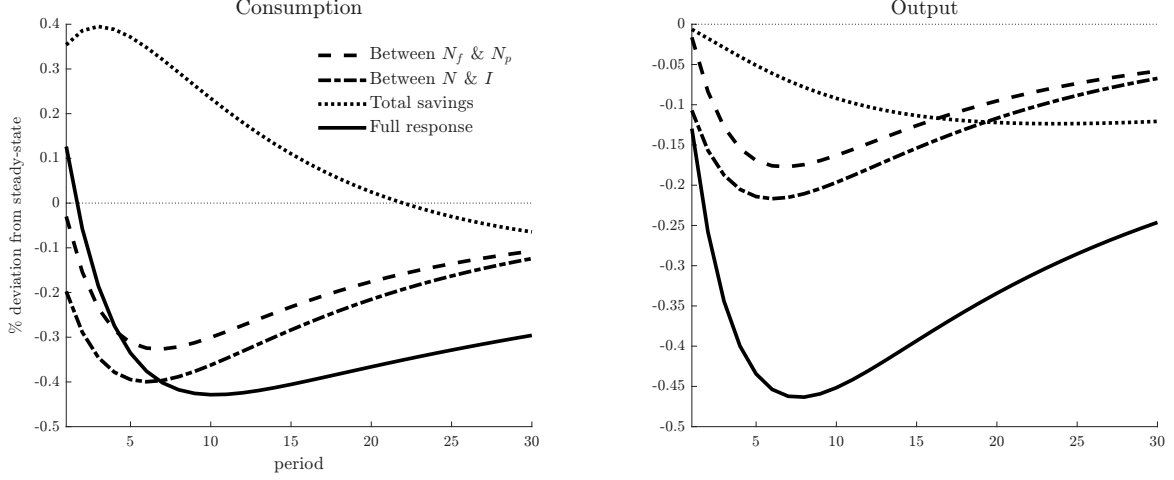


Figure 5: Impulse response decomposition. The figure depicts the paths of consumption and output due to reallocation between labor types, between labor and capital, and due to changes in total savings. The three components sum up to the equilibrium responses of consumption and output, plotted as the solid “Full response” line in each panel and identical to the corresponding responses in Figure 4.

full-time and part-time vacancies, $\omega_{f,t}$ and $\omega_{p,t}$, we construct counterfactual vacancy postings that are influenced only by the change in the *relative demand* for different labor types:

$$\tilde{V}_{f,t}^1 = \frac{(\bar{H} - \bar{I})}{\varphi_{f,t}} \frac{\omega_{f,t}}{\omega_{f,t} + \omega_{p,t}}; \tilde{V}_{p,t}^1 = \frac{(\bar{H} - \bar{I})}{\varphi_{p,t}} \frac{\omega_{p,t}}{\omega_{f,t} + \omega_{p,t}}.$$

We then use these counterfactual vacancy postings to isolate the implications of this reallocation of labor demand on the economy.²² In Figure 5, the dashed line shows the resulting counterfactual paths of consumption (left panel) and GDP (right panel) after a risk aversion shock, given that only the relative demand for full-time and part-time labor changes, holding physical capital demand and the total demand for risky inputs fixed.

Consistent with our analytical model, this “labor reallocation only” experiment causes a substantial drop in both consumption and output. To give a sense of magnitude, the first column of Table 5 cumulates the counterfactual IRFs over three years following the shock and divides them by the respective cumulative impact in the estimated model (with all channels active). We find that 87% of the equilibrium fall in consumption and about one-third of the fall in output can be accounted for by the labor reallocation margin alone.

Next, in the second counterfactual, we also allow the share of capital investment $\omega_{I,t}$ to follow its equilibrium path, along with the adjustment between the investment in the two types of labor, but still keep total savings fixed at its steady-state value $H_t = \bar{H}$.

²²Because labor supply is exogenous in our economy, the paths of the $\omega_{i,t}$ and H_t can be combined with matching functions and resource constraints to fully determine counterfactual output and consumption.

Table 5: Reallocation effects on C and GDP

	Between N_f & N_p	Between N & I	Fall in total savings	Equilibrium
Consumption	-0.87	-1.16	1.03	-1.00
Output	-0.36	-0.48	-0.15	-1.00

Note: Contribution of each channel, as share of equilibrium consumption/output change. Consumption responses cumulated over 12 quarters, without discounting. Negative numbers reflect decreasing cumulated consumption. Each column reflects the marginal contribution, beyond previous channels, so that the first three columns sum to minus one.

We then use equation (40) to back out counterfactual impulse response paths for physical investment ($\tilde{I}_t^2 = \omega_{I,t}\bar{H}$), and full-time and part-time labor demand ($\tilde{V}_{f,t}^2 = \omega_{f,t}/\varphi_{f,t}\bar{H}$ and $\tilde{V}_{p,t}^2 = \omega_{p,t}/\varphi_{p,t}\bar{H}$), and trace the impact on the economy keeping all other equilibrium conditions the same. This counterfactual thus keeps the total demand for risky inputs fixed, but now allows for reallocation of demand from risky full-time labor to both part-time labor and capital.

We plot the marginal effect of also allowing the investment share to adjust with the dash-dot lines in Figure 5. This additional input demand reallocation also pushes consumption and output down – contributing to a further fall in output and consumption equal to 48% and 116% of their equilibrium responses, respectively. We want to emphasize that this does not constitute a conceptually different channel – this is simply a different instance of the input demand reallocation mechanism illustrated in the analytical model. The sum of the dashed and the dash-dot lines gives the full effect of shifting the input composition, while keeping the overall demand for risky inputs fixed.

Finally, the dotted lines in Figure 5 reflect the contribution of the remaining margin, allowing total savings H_t to follow its equilibrium path. The fall in the overall demand for risky inputs — captured by this line — pushes consumption and output in opposite directions. Following the economic logic in Proposition 1, a fall in the overall demand for risky inputs lowers output; however, the resources saved from this lower investment necessarily show up in higher consumption, thus causing the model to imply a *negative* comovement between consumption and output. Cumulating the responses over three years, the third column of Table 5 shows that the fall in total savings, H_t , contributes an *increase* in consumption equal to 103% of the equilibrium consumption fall and also contributes 15% to the cumulative fall in output. This result emphasizes our analytical intuition and showcases that the model’s success in generating business cycle comovement is entirely due to the reallocation of demand from riskier to safer factors of production.

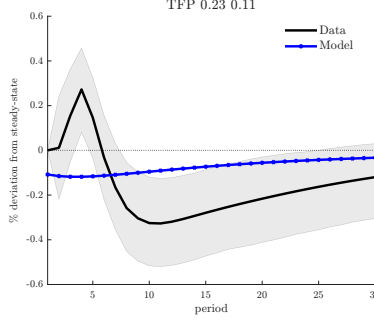


Figure 6: Model and data responses of measured TFP.

Implications for measured TFP

The reallocation from full-time to part-time labor also has implications for measured TFP because the empirical measurement of productivity may not properly capture the heterogeneity between these two types of labor.

As shown in Appendix E, there are different ways of measuring TFP depending on the data available to the econometrician. The baseline case we analyze there is one where the econometrician does not separately observe full-time and part-time labor and works with total employment.²³ In that case, our model implies the following change in *measured TFP* following a risk aversion shock

$$\Delta \log(\tilde{Z}_t^e) = (1 - \alpha)\Delta \log(\tilde{N}_t) - \omega_N \Delta \log(N_t).$$

Above, ω_N is the labor share, $\tilde{N}_t$ is the CES aggregate of full-time and part-time labor in the production function (eq. (26)), and $N_t = N_{f,t} + N_{p,t}$ is total employment.

Figure 6 shows the impulse response of measured TFP in the model against the response of TFP in our baseline treatment of the data. In the model, TFP falls by about 0.1% at the trough, which corresponds to approximately 26% of the trough in output. Because measured TFP takes into account total employment but not how full-time and part-time workers affect output separately, shifting a worker from full-time to part-time changes output (and the labor aggregate $\tilde{N}_t$), but not the total number of employed workers ($N_t = N_{f,t} + N_{p,t}$). Thus, the resulting change in output is interpreted as a shift in measured TFP.

The figure also shows that empirical TFP falls by more than implied by our quantitative model, but still the model's response remains within the confidence bands of the data

²³We consider TFP measurement based on total employment, not hours, because our model abstracts from the intensive margin of labor and it is the number of employed agents (in full-time and part-time jobs) that enters the production function. In the Appendix, however, we also present the case of using total hours and results are unchanged.

at most horizons. A possible reason for the discrepancy is that we model only one source of heterogeneity in input riskiness whereas other dimensions of heterogeneity may also underlie the empirical decline in TFP. For example, different types of capital (e.g., structures vs. software) have different depreciation rates and therefore will face different risk premia. Risk-driven changes in capital allocation shares from high-marginal-product to low-marginal-product capital types would lead to falling measured TFP for the same reasons that labor heterogeneity does in our model.

Role of Sticky Wages and Job Duration

In our model, sticky wages play a role that is complementary to, but distinct from, their role in most prior literature. To see this, iterate forward on the vacancy posting condition to show that the value of a match of type i (part-time or full-time) for the firm is given by:

$$J_{i,t} = \underbrace{\sum_{j=0}^{\infty} \frac{(1 - \rho_i)^j \mathbb{E}_t(MPL_{i,t+j} - W_{i,t+j})}{R_{t,t+j}}}_{\text{cash flows}} + \underbrace{\sum_{j=1}^{\infty} (1 - \rho_i)^j Cov_t(M_{t,t+j}, MPL_{i,t+j} - W_{i,t+j})}_{\text{risk premium}}, \quad (41)$$

where we have imposed the transversality condition that $\lim_{j \rightarrow \infty} \mathbb{E}_t[M_{t,t+j} J_{i,t+j}] = 0$.

Equation (41) expresses the value of a match as the sum of two terms. The first is the present value of firms' cash flows, discounted with the risk-free rate $R_{t,t+j} = \mathbb{E}_t[M_{t,t+j}]^{-1}$. The second is a risk premium adjustment. This adjustment depends both on the duration of the firm-worker relationship and the covariance between the stochastic discount factor and the cash flow generated by the worker. The sticky wages of full-time workers tend to increase this covariance, while the lower separation rate (smaller ρ_f) increases the weights on it. In the prior literature focused on the [Shimer \(2005\)](#) puzzle, sticky wages have been used to magnify the fluctuations in the expected cash flows component (the first term above). In our model, however, the main role of sticky wages is rather to amplify the risk premium of full-time labor, by magnifying the negative covariance between the firm's cash flows and the stochastic discount factor (the second term above).

To quantify the importance of both duration differentials and the unique role that sticky wages play in our mechanism, Figure 7 shows the responses to a risk aversion shock in two counterfactual versions of the economy. In the first, we recalibrate $\rho_f = \rho_p = 0.10$, so that the two worker types have the same duration, but continue to differ in terms of the sticky wage. In the second, we keep the calibrated separation rates and wage processes, but counterfactually zero-out the risk premium term from the firm's full-time vacancy posting condition. In this second exercise, we let sticky wages serve the same purpose as in the prior

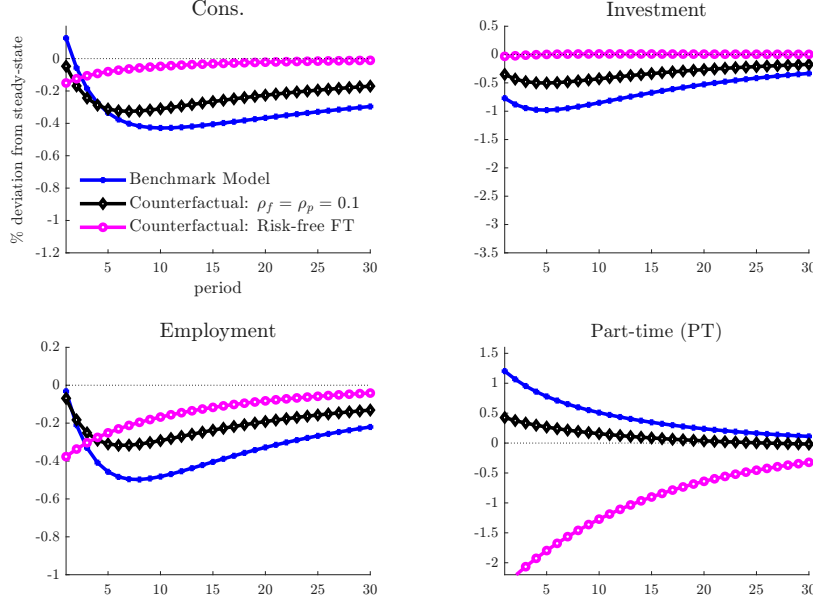


Figure 7: Model responses in counterfactual economies. Consumption, investment, employment, and part-time employment responses to a risk aversion shock in the benchmark model and two counterfactuals: equalized separation rates ($\rho_f = \rho_p = 0.1$) and full-time positions treated as risk-free. Vertical axes are in percentage deviations.

literature (amplified *expected* cash flows variation), but shut down the risk premium channel.

For both counterfactuals, the figure shows a much smaller aggregate effect of risk aversion shocks on real quantities. When separation rates are equalized, the sticky wage of full-time workers means that they are still more risky than part-time workers, only less so. In this case, the model delivers comovement, but the reallocation toward part-time (and corresponding effects on investment and output) is about one half of what they are in our baseline. By contrast, when full-time workers are treated by firms as if they were risk free, the economy responds even less in aggregate and the reallocation to part-time workers is reversed, consistent with the fact that part-time workers are now relatively risky compared to full-time. Most importantly, this counterfactual fails *qualitatively* to generate aggregate comovement, as it implies that consumption and investment move in opposite directions.

Lastly, we also note that while full-time wages in our model adjust slowly to shocks, the *earnings per worker* (i.e., the aggregate) wage can adjust along two additional dimensions. First, the wages of existing part-time workers are flexible and they fall in response to the shock. Second, the share of part-time workers grows in the model as a share of total employment: since those workers have lower earnings, this also reduces earnings per worker. Together, these channels imply that aggregate wages fall in the model when risk aversion rises. Appendix Figure D.1 plots the impulse response of earnings per worker both in the model and the data, and provides some additional discussion. While our assumptions about

wages are a simplification, the overall implications for the aggregate earnings per worker are not strongly counterfactual.

Relationship to the literature on labor risk and macroeconomic fluctuations

We conclude this section with a discussion of the relationship between our findings and [Di Tella and Hall \(2021\)](#) and [Kehoe et al. \(2022\)](#). Like our own, the models in these papers rely on labor returns being subject to risk considerations in order to generate fluctuations in labor demand.²⁴ The key difference across all three papers is the way the different mechanisms can and cannot generate macroeconomic comovement in equilibrium – that is, to obtain not just a fall in N_t , but also a contemporaneous fall in C_t , I_t and Y_t .

[Di Tella and Hall \(2021\)](#) have a model in which higher *idiosyncratic risk* can lead to recessions with comovement between C_t , N_t , and I_t . In their theory, risk-averse entrepreneurs reduce their demand for factors of production and increase their savings demand when faced with higher uninsurable idiosyncratic risk. The first effect reduces demand for both labor and capital services, while the second increases the demand for aggregate capital holdings (as capital is the only store of value in their economy).

The key to the [Di Tella and Hall \(2021\)](#) mechanism is that idiosyncratic risk washes out in the aggregate: investing in the *aggregate* capital stock is safer for entrepreneurs than hiring capital for production. This distinction between aggregate and private capital means that demand for capital as a savings vehicle can increase even as demand for capital as a factor of production falls, generating offsetting effects on I_t in equilibrium. As such, calibrations exist where increases in idiosyncratic risk do not lead to a large fall in investment, even as the labor input falls by a large amount. Because investment does not fall by much, lower output is mirrored by lower consumption, avoiding the typical issue that falling investment pushes up consumption via the resource constraint (as described in our Proposition 1).

Our mechanism differs from that of [Di Tella and Hall \(2021\)](#) because it generates macroeconomic comovement via changes in the desired *composition* of firm investments, rather than fluctuations in the overall demand for savings. The key elements in our composition story are (i) treating hiring as a form of investment, like in [Hall \(2017\)](#), and (ii) assuming two different types of labor positions, one with flexible wages and one with rigid wages. These assumptions introduce multiple durable factors of production with varying risk character-

²⁴This is achieved via search frictions and durable employment relationships in our paper and in [Kehoe et al. \(2022\)](#), while [Di Tella and Hall \(2021\)](#) assume that labor is hired before shocks realize. In order to amplify the impact of risk on labor demand, both [Di Tella and Hall \(2021\)](#) and our paper consider wages that do not respond to the realization of productivity shocks (idiosyncratic productivity in their case, and aggregate productivity in ours). [Kehoe et al. \(2022\)](#) have flexible wages but amplify the effects of risk via human capital accumulation, which increases the effective duration of labor.

istics. Consequently, an increase in risk alters the desired *composition* of investment. In contrast to [Di Tella and Hall \(2021\)](#), this mechanism works with fluctuations in *aggregate* risk or risk appetite and more generally can deliver macroeconomic comovement without requiring a particular response of desired savings.²⁵ Moreover, it can also explain additional facts, such as the patterns of higher aggregate risk premia and the shift toward part-time workers during recessions.

On the other hand, the goal of [Kehoe et al. \(2022\)](#) is to obtain large employment fluctuations and solve the Shimer puzzle rather than to generate macroeconomic comovement *conditional* on risk premium fluctuations. They do not consider the characteristics of business cycles conditional on *only* fluctuations in risk and uncertainty. Rather, they include first-order productivity shocks that also generate risk premium fluctuations, and thus their results reflect a combination of first- and higher-moment shocks, like in [Bloom et al. \(2018\)](#).

5 Conclusions

This paper shows that fluctuations in risk premia can be major drivers of business cycles. Our empirical analysis suggests the possibility of a major causal pathway flowing from risk premia to macroeconomic fluctuations, and our theory embodies one such pathway. In our model, heightened risk premia cause recessions because they drive reallocation of saving towards safer stores of value, which simultaneously have low instantaneous marginal products. Thus, our theory puts risk premia and their effects on precautionary saving at the center of macroeconomic propagation. In this respect, our model bridges a gap between the tradition of risk-driven business cycles à la Keynes and the central lessons of modern macro-finance summarized in [Cochrane \(2017\)](#), all within a real framework.

To focus attention on our novel propagation mechanism, we abstract throughout from many other ingredients that may contribute to risk-driven macroeconomic comovement, including nominal rigidities ([Basu and Bundick, 2017](#)), financial frictions ([Christiano et al., 2014](#)), uninsurable idiosyncratic risk ([Di Tella and Hall, 2021](#)), information frictions ([Ilut and Saijo, 2021](#)), and heterogeneous asset valuations ([Caballero and Simsek, 2020](#)). All of these features likely play a role in the world. Nevertheless, our quantitative analysis demonstrates that the savings reallocation channel is sufficiently powerful to drive a substantial portion of macroeconomic fluctuations on its own.

²⁵The [Di Tella and Hall \(2021\)](#) mechanism does not work with fluctuations in aggregate risk. The reason is that, in response to an increase in aggregate risk no wedge can emerge between the desire to save and the desire to employ risky capital as a factor of production. Without the offsetting increase in desired savings, investment unambiguously falls. This frees up additional resources for consumption, reintroducing the tension described in Proposition 1.

Our theory emphasizes the labor market implications of savings reallocation primarily because our empirical results suggest a flight-to-safety in those markets. Nevertheless, the same patterns likely apply to other forms of saving available in the economy (risky private investments versus safe government bonds, foreign investment for open economies, etc.). Reallocation from new to old capital could also provide a similar amplification mechanism for which there is already intriguing empirical evidence (e.g., [Eisfeldt and Rampini, 2006](#)). Future research should continue to explore the business cycle consequences of these alternative applications of this mechanism.

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Online Appendix

A Empirical Appendix

A.1 Data Construction

Our baseline VAR specification consists of output, consumption, investment, employment, ex-post real stock returns, ex-post real bond returns, and the dividend-price ratio. Our auxiliary series include measures of part-time employment, hours per worker, bond returns, and bond risk premia.

Quantity variables were downloaded from the FRED database of the St. Louis Federal Reserve Bank and are included in seasonally-adjusted, real, per-capita terms. Our population series is the civilian non-institutional population ages 16 and over, produced by the BLS. We convert our population series to quarterly frequency using a three-month average and smooth it using an HP filter with penalty parameter $\lambda = 1600$ to account for occasional jumps in the series that occur after census years and CPS rebasing. Our deflator series is the GDP deflator produced by the BEA national accounts.

For output, we use nominal output produced by the BEA. Our investment measure is inclusive: we take the sum of nominal gross private domestic investment, personal expenditure on durable goods, government gross investment, and the trade balance (i.e. investment abroad). Consumption consists of nominal personal consumption expenditures on non-durables and services.

Our measure of employment is Total Nonfarm Employees (FRED code: PAYEMS) produced by the BLS and divided by population. The measure of part-time employment is the number of people “Employed, usually part-time work” (FRED code: LNS12600000) produced by the BLS and again divided by our population series. This series includes a large discrete jump in the first month of 1994, associated with a reclassification of part-time work. We splice the series by assuming there was no change in employment between 1993M12 and 1994M1. Our measure of hours is “Non-farm Business Sector: Hours of All Persons” (FRED code: HOANBS).

Our quarterly stock market series are based on the S&P 500 monthly data from [Shiller \(2005\)](#). Asset returns are computed inclusive of dividends, and are also deflated by the GDP deflator. Earnings and dividends are from the same source. Our measure of bond risk premia comes from Moody’s corporate bond yield relative to 10-year Treasury bonds (FRED code: BAA10YM). The short-term rate is defined as the real log change in the 3-month Treasury Bills index. The 5-year real bond yield is the real log change in the 5-year Treasury Bond

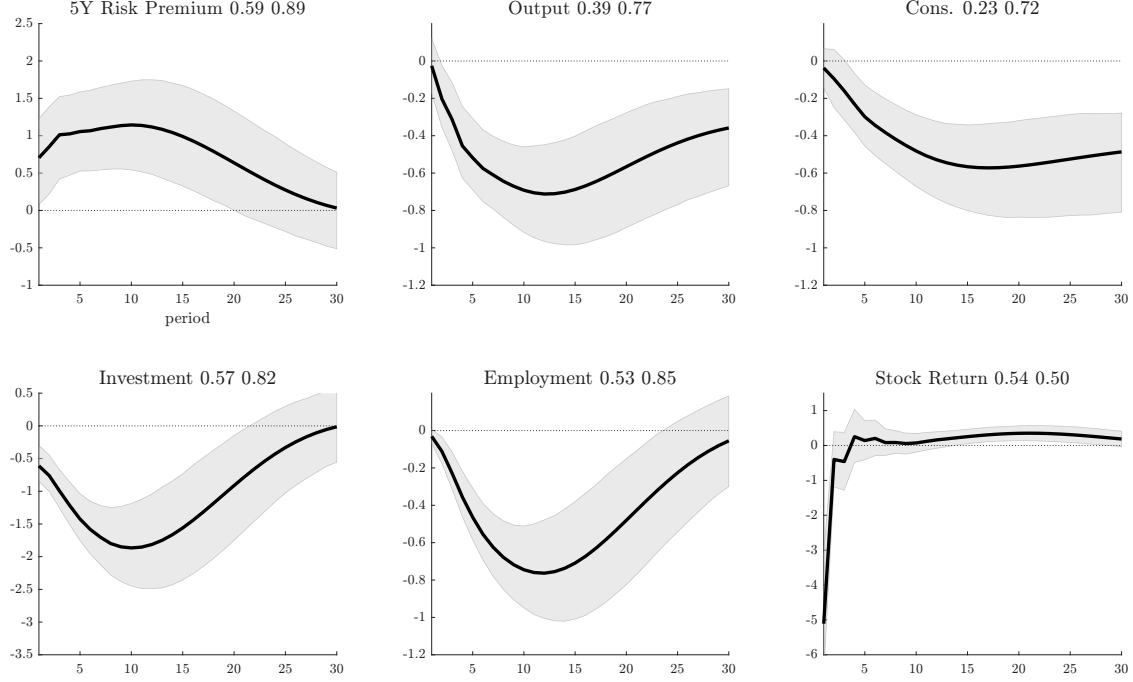


Figure A.1: VAR results based on the 1985-2018 sub-sample.

index.

A.2 Short-sample results

An earlier version of this paper was based on a shorter sample, starting in 1985Q1. Figure A.1 below shows our baseline VAR results, recomputed using that shorter sample. All results are qualitatively the same. The main difference is that the persistence of the increased risk premium is higher.

A.3 Excess Stock Return Predictability

In this section, we examine whether VAR-implied expected returns $\mathbb{E}_t(rp_{t,t+j})$ capture the underlying risk premium effectively, and analyze the specific sources of this predictive power. All of the following forecasting exercises are performed in-sample. We benchmark our results against the comprehensive reviews of Welch and Goyal (2008) and Goyal et al. (2024), who document that return predictability, while being fragile at short horizons, holds some promise at longer horizons (as also argued by Cochrane, 2008) when based on selected macro-based predictors.

The solid black line in Figure A.2 plots the expected excess stock return as estimated by

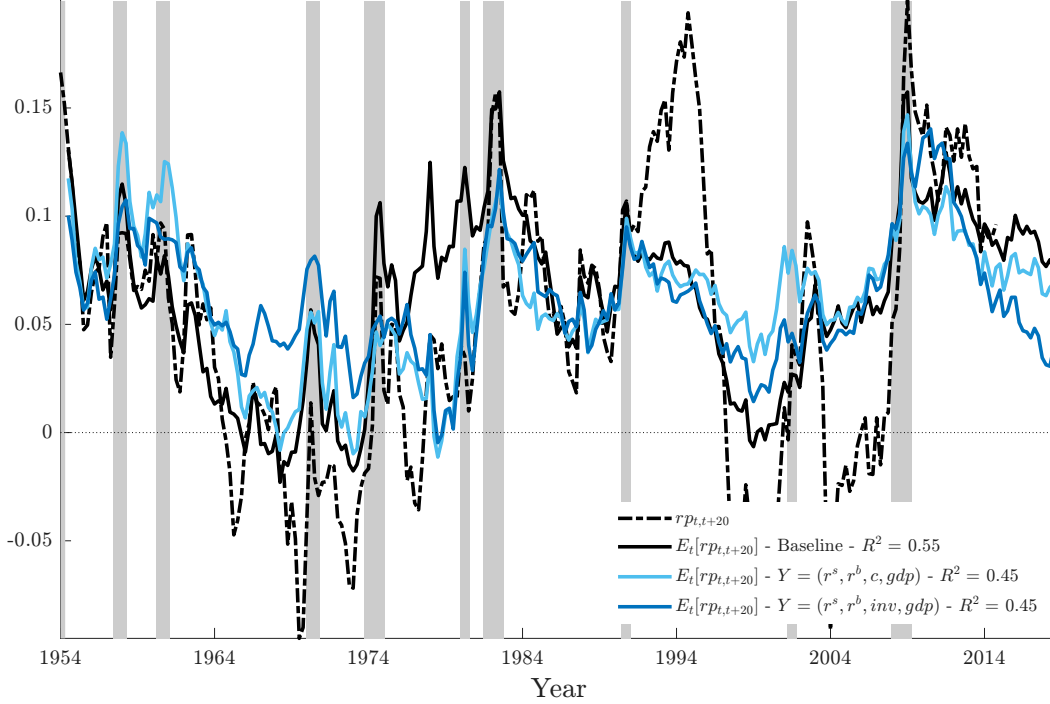


Figure A.2: Ex-post and Ex-ante excess stock returns from alternative VARs.

our VAR, $\mathbb{E}_t(rp_{t,t+20})$, against the realized excess returns over that same forecasting horizon, $rp_{t,t+20}$ (dashed black line). The Figure shows that both series exhibit substantial variation. While the ex-post series is more volatile, the VAR-based expectation of the excess returns tracks it reasonably well.

To understand which specific variables in Y_t are the main predictors our VAR relies on, we investigate the expected excess stock returns $\mathbb{E}_t(rp_{t,t+j})$ as implied by smaller VARs that use a subset of the 7 variables contained in our main specification.

First, we restrict the VAR to only include consumption and GDP in addition to the two return series, so that it spans information similar to the consumption-wealth ratio of [Lettau and Ludvigson \(2001\)](#). As [Figure A.2](#) shows, this restricted VAR already tracks realized 5-year excess returns well, achieving an R^2 of 45%. This is in line with the [Welch and Goyal \(2008\)](#) finding that (ex-post) cay_t is their single strongest in-sample predictor at the 5-year horizon, with an R^2 of 36%.²⁶

Second, we restrict the VAR to instead use information on investment and GDP, so as to span information similar to the investment-based predictors of [Cochrane \(1991\)](#). This

²⁶While we do not include cay_t directly in Y_t , our VAR nevertheless flexibly captures the same information by implicitly estimating the cointegration relationship between consumption and GDP. The flexibility appears to matter: [Goyal et al. \(2024\)](#) find that while the original cay_t variable deteriorates in their updated sample, more flexible consumption-based measures, such as consumption relative to trend and consumption growth, remain among the better performers.

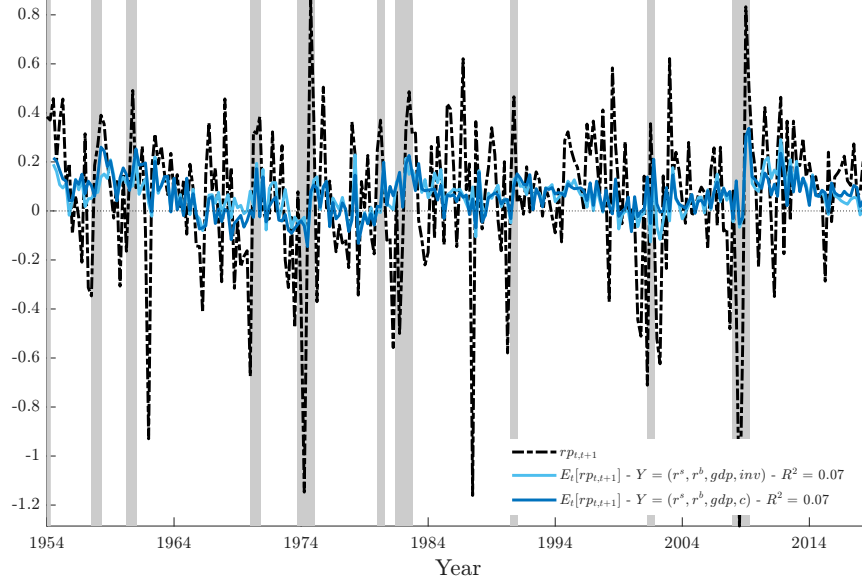


Figure A.3: Predicted vs realized 1-quarter excess returns.

restricted, investment-based VAR delivers a risk premium estimate with an R^2 of 45%, comparable to that of the consumption-based VAR. The similar overall fit, however, masks differences in timing: the consumption-based risk premium tracks realized returns more closely in the late 1960s and early 1970s, whereas the investment-based one does so in the late 1990s and early 2000s. The strong performance of the investment-based VAR is consistent with [Welch and Goyal \(2008\)](#), who find that I/K is a robustly significant predictor at the 5-year horizon with an in-sample R^2 of 34%. Notably, the performance of I/K has in fact improved since 2008, to the extent that [Goyal et al. \(2024\)](#) explicitly recommend considering it as a predictor.

Lastly, we verify that our data and empirical methodology fully agree with the well-documented fact that predictability is weak at *short* horizons ([Welch and Goyal, 2008](#); [Goyal et al., 2024](#)). Specifically, [Goyal et al. \(2024\)](#) find that the only models that deliver significant in-sample predictability when focusing on 1-quarter returns are those based on detrended consumption or on the investment-to-capital ratio, and even these achieve only modest explanatory power, with R^2 values of 5% and 4%, respectively. Our data and methodology paint a very similar picture. Indeed, if we use 4-variable VARs that include consumption and GDP or investment and GDP to predict 1-quarter returns, instead of the 5-year benchmark horizon we use in the paper, we find a much lower R^2 of 7% in both cases. Figure A.3 plots the estimated and the realized risk premia at the one-quarter horizon.

This finding allows us to clarify that we are not claiming short-horizon forecasting power that the rest of the literature has failed to find. Rather, our estimates isolate a small but

persistent component of expected returns that is hard to detect quarter-to-quarter, but compounds into a large share of return variation over 20 quarters, similar to the analysis of [Cochrane \(2008\)](#).

We thus conclude that the joint information in consumption, investment, and GDP plays the most important role in our VAR’s ability to predict excess equity returns at long horizons. At those horizons, our methodology yields comparable results to the estimates of [Welch and Goyal \(2008\)](#) and the predictability loads on the type of variables that they find to be relatively robust as well.

A.4 Relationship with Survey Expectations

One natural question is whether the return expectations implied by our VAR are consistent with those held by investors or firms in the economy. Indeed, a large and very interesting literature has documented that survey of expectations feature numerous biases and potential deviations from rationality (e.g. [Bordalo et al., 2020](#); [Bianchi et al., 2022](#)). In terms of financial data, these expectational deviations are such that reported expectations often do not track well future asset prices and returns ([Greenwood and Shleifer, 2014](#); [Nagel and Xu, 2023](#)), and this has given rise to a large literature that turns its attention away from the classic channel of time-varying risk premia.

However, in a recent NBER working paper, [Thesmar and Verner \(2026\)](#) provide new data on return expectations from Value Line, an independent equity analysis firm, and they find that these forecasts, made by *sophisticated* financial analysts, indeed behave like classic models of time-varying risk premia would suggest. We use data kindly shared with us by David Thesmar. These data come from [Thesmar and Verner \(2026\)](#), which provides a state-of-the-art summary of the finance literature’s findings on expected returns from investor surveys, while providing new data on return expectations from Value Line, an independent equity analysis firm.

The VL data are available quarterly only since 1987. (Before 1987, VL provides forecasts at the annual frequency only.) We use the quarterly sample because it best matches our quarterly VAR. There are a number of differences between the two data sets, so one should not expect the forecasts to match exactly. For example: (i) we forecast returns on the S&P 500 index, while VL forecasts returns for 1,700 individual stocks. Thesmar and Verner aggregate these individual forecasts into a series on “total market” returns using CRSP, and we use this aggregate series to compare with our VAR-forecast; (ii) we forecast excess returns directly, while VL provides forecasts only for total returns. Thesmar and Verner infer an excess return measure using the return on maturity-matched Treasury bonds, which we also

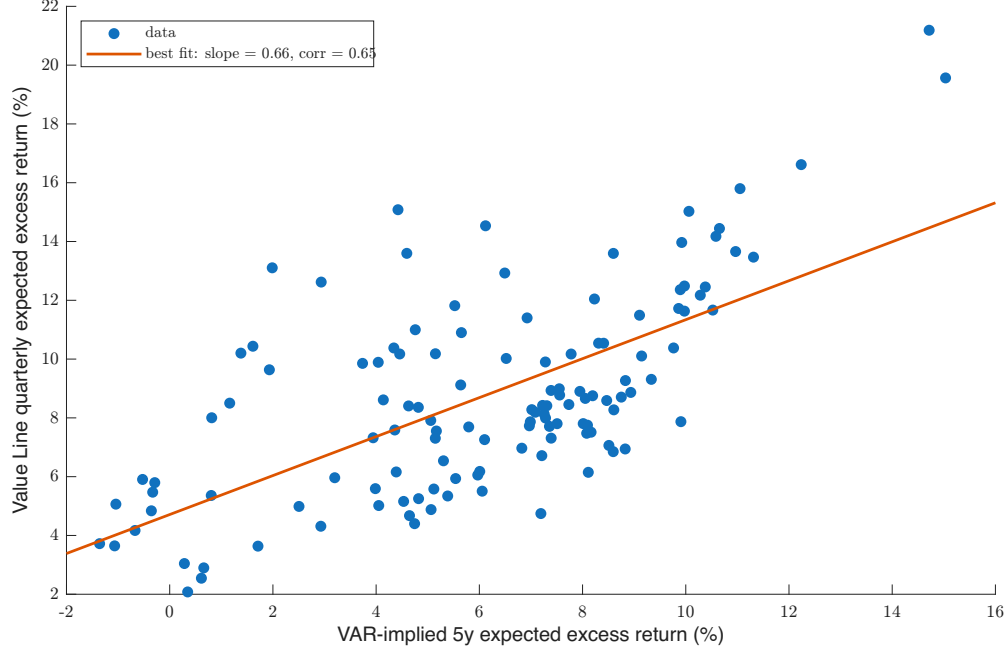


Figure A.4: Value Line versus VAR-implied expectations, 1987Q3-2018Q4.

use; (iii) we forecast the average excess return over exactly 5 years, starting from the next quarter. The long-run VL forecasts are for returns over the next “3 to 5 years.”

Despite these caveats, we see a clear positive relationship between the two sets of quarterly forecasts from 1987Q3 to 2018Q4. The slope coefficient is 0.66 and significant, while the two series exhibit a correlation of 0.65 with one another. See Figure A.4 for a visual representation of this relationship. Thus, our VAR-based forecasts are strongly correlated with the real-time forecasts of actual investors, specifically the expectations of sophisticated investors. Mapping to our theory, the decision-maker is the representative firm, and we think of the real-world firm managers as closer to the “sophisticated” agent behind the VL survey, and not retail investors.

Lastly, the VL forecasts also allow us to address concerns regarding in- vs. out-of-sample forecasts. Using the same quarterly data, a regression of our realized 5-year excess returns on the VL forecasts gives an R^2 of 0.28 (i.e. correlation of 53%). Note that this regression is subject to observations (i)-(iii) above, so the R^2 is a lower bound on the accuracy of the forecasts. Since these subjective forecasts are out-of-sample by definition, this result shows that a good fraction of excess returns can be predicted even out of sample.

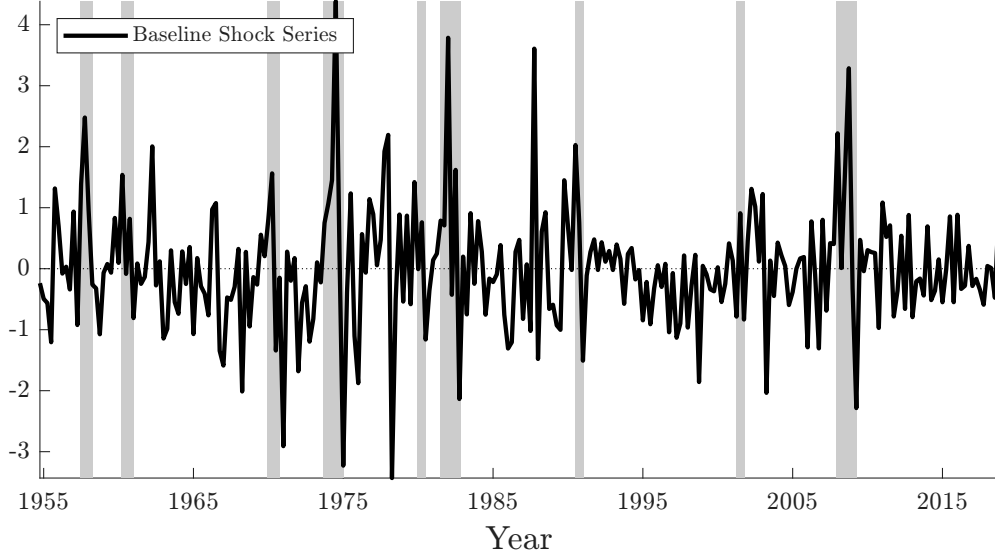


Figure A.5: Implied history of risk premium disturbances from baseline VAR procedure.

A.5 Max Share Identification: Details

Let $\phi(z) \equiv (e_5 - e_6)(\tilde{B} + \tilde{B}^2 + \dots + \tilde{B}^j)(I - \tilde{B}z)^{-1}\tilde{A}$ be the z -transfer function associated with the MA(∞) representation in (3). We can express the variance of $\mathbb{E}_t[rp_{t,t+j}]$ associated with spectra of periodicity $p \equiv [p_1, p_2]$, which we label σ_p^{rp} , as

$$\sigma_p^{rp} = \frac{1}{2\pi} \int_{2\pi/p_2}^{2\pi/p_1} \phi(e^{-i\lambda})\phi(e^{-i\lambda})' d\lambda. \quad (\text{A.1})$$

In turn, we can express the variance of $\mathbb{E}_t[rp_{t,t+j}]$ over those same frequencies, but when only the first element of the shock vector ϵ_t is active, as

$$\sigma_p^{rp} \Big|_{\epsilon_t^{(2)}=\epsilon_t^{(3)}=\dots=0} = \frac{1}{2\pi} \int_{2\pi/p_2}^{2\pi/p_1} \phi(e^{-i\lambda})e_1'e_1\phi(e^{-i\lambda})' d\lambda. \quad (\text{A.2})$$

where again e_1 is a selector vector, this time with 1 in the first position and zeroes everywhere else, and $\epsilon_t^{(k)}$ is the k -th element of the shock vector ϵ_t .

We can then find the matrix A by maximizing (A.2) (recall that $\phi(z)$ is a function of A). This procedure yields a partially identified system, in the sense that the above maximization problem will uniquely determine the first column of A and thus the first element of the shock vector ϵ_t , but not the rest. This is sufficient for our purposes, because we are just interested in $\epsilon_t^{(1)}$, the orthogonal innovation that has the largest possible contribution to fluctuations in the risk premium $\mathbb{E}_t(rp_{t,t+j})$. Figure A.5 shows the time series of our estimated shock $\epsilon_t^{(1)}$.

We have found that our results are robust to estimating the VAR in VECM form so long

as we allow for more than two independent trends in the data. Our results are also robust to increasing the lags in our VAR and changes in the VAR information set, including adding the 5-year Treasury yield or replacing the dividend-price ratio with the earnings-price ratio.

A.6 Response of Employment by Demographic Groups

Differences in part-time employment across demographic groups could, in principle, contribute to the observed increase in aggregate part-time employment if our shock induced a reallocation of employment across these groups.

Among the demographic characteristics recorded in the Current Population Survey, differences in part-time incidence are most pronounced by sex and age. On average, 24% of employed women work part time, compared with 12% of employed men. Similarly, part-time work is much more common among younger and older workers: 42% for individuals aged 16-24 and 21% for those aged 55 and over, compared with only 11% for individuals aged 25-54.²⁷

To address this concern, we have augmented our baseline specification with additional auxiliary variables measuring the employment share of women and the employment shares of the age groups 16-24, 25-54, and 55 and over. The corresponding impulse responses are reported in Figure A.6.

The top-left panel shows that the employment share of women rises modestly following the shock before declining several quarters later, but the response is never statistically significant. Even taking the peak increase of 0.15% at face value, the implied effect on aggregate part-time employment is economically negligible. Indeed, if such part-time employment increase among women were reallocated from men, it would account for an increase in the aggregate part-time share of only 0.01 percentage points.²⁸ The subsequent decline in the female employment share is of similarly negligible magnitude.

The remaining panels show some movement in the age composition of employment. The employment share of prime-age workers (25-54) does not meaningfully move in either direction throughout the response. The employment share of workers aged 55 and over increases modestly, although never significantly, while the employment share of workers aged 16-24 declines significantly. Because young workers have by far the highest part-time incidence, a fall in their employment share mechanically lowers the aggregate part-time share, whichever group absorbs them. Indeed, if the fall in young workers were absorbed by workers aged 55

²⁷These shares are stable over recent decades.

²⁸The increase corresponds to 0.07 percentage points in the employment share of women (relative to its average of 46%), which, multiplied by the 12 percentage point difference in part-time incidence between women and men, implies a rise in the aggregate part-time share of 0.01 percentage points.

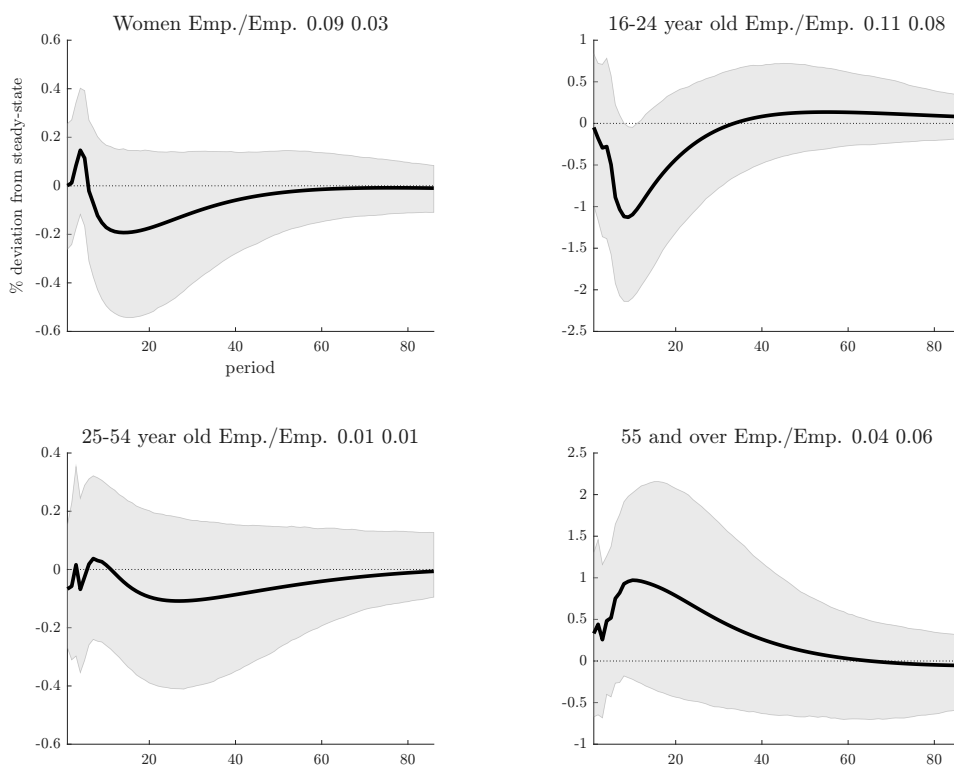


Figure A.6: Estimated response of employment by demographic groups to the risk premium disturbance.

and over, the aggregate part-time share would *decline* by 0.04 percentage points.²⁹

To sum up, along the gender dimension the response is economically negligible; along the age dimension it is both small and attenuating, working against rather than toward the increase we estimate. Both margins thus point to the same conclusion: the adjustment in part-time work occurs primarily within demographic groups rather than through a reallocation of employment across them.

²⁹At its trough, the employment share of young workers falls by 1.13%, corresponding to a decline of 0.19 percentage points relative to its average share of 17%. Multiplying this by the 21 percentage point difference in part-time incidence between young and senior workers yields a reduction in aggregate part-time employment of 0.04 percentage points.

A.7 Controlling for Supply Shocks

One possible concern with the approach in Section 2 is that the disturbances we identify could reflect a combination of underlying structural shocks. For example, the small estimated effects on inflation might result from the commingling of supply and demand disturbances that have a similar effect on real quantities but opposed effects on inflation. Alternatively, our empirical findings could primarily reflect news about future macroeconomic conditions, for which risk premium changes are only a side effect.

While we do not seek to “prove” that a single structural shock drives all risk premium fluctuations, this appendix argues that the patterns we identify in the data are not driven by standard supply-side shocks. We do this by first identifying contemporaneous and/or anticipated supply shocks and controlling for (“cleaning out”) these other shocks before isolating the risk premium disturbance. This ordering is conservative, in the sense that it gives the alternative story the best chance at absorbing the variation explained by our baseline estimate of the risk premium disturbance. If the baseline estimation approach commingles risk with standard supply shocks, we would expect this cleaning step to qualitatively change our results.

Our general approach is to include TFP in our set of auxiliary variables, and then use information about TFP to identify supply shocks using one of several different procedures. In each case, we first identify a set of TFP-related supply shocks and then identify our risk premium disturbance using our max-share approach, after controlling for the identified supply disturbances. Controlling for these supply shocks, we find the same patterns in the data, which suggests to us that our shock is not conflating supply and demand in an important way.

In the first case, labeled “Method 1”, we identify a single supply shock as that which explains the largest possible share of the contemporaneous surprise in productivity. This approach amounts to using a Choleski identification scheme with TFP ordered first, albeit with TFP as an auxiliary variable.

For concreteness, we show the connection assuming only one lag in the VAR. Let

$$\tilde{Y}_t = \Gamma_{tfp} Y_t \tag{A.3}$$

where Γ_{tfp} is an invertible $n_y \times n_y$ matrix whose first row corresponds to the weights from the regression of TFP on the core VAR variables. (The remaining rows can be anything so long as they result in Γ_{tfp} being full rank.) Then, premultiplying by Γ_{tfp} , the VAR in

equation (1) can be transformed into

$$\tilde{Y}_t = \Gamma_{tfp} B \Gamma_{tfp}^{-1} \Gamma_{tfp} Y_{t-1} + \Gamma_{tfp} A \varepsilon_t \equiv \tilde{B} \tilde{Y}_{t-1} + \tilde{A} \varepsilon_t. \quad (\text{A.4})$$

We can then identify $\tilde{A}$ by setting

$$\tilde{A} = \text{chol}(\Gamma_{tfp} \text{cov}(u_t) \Gamma_{tfp}'), \quad (\text{A.5})$$

and find $A = \Gamma_{tfp}^{-1} \tilde{A}$. Since the first row of $\tilde{A}$ has zeros outside of the first entry, this implies that 100% of the surprise in the (fitted) value of TFP is explained by the identified shock.

In the second case, labeled “Method 2” in Figure A.7 below, we identify both an anticipated and a surprise supply shock using the approach of [Chahrour et al. \(2023\)](#), which itself is an extension of the [Kurmman and Sims \(2020\)](#) approach to identifying news shocks. The first stage of this procedure identifies the “news” shock that explains the largest share of *anticipated* fluctuations in TFP at horizons between 20 and 80 quarters into the future. The second stage identifies surprise TFP shocks as those which explain the largest share of the remaining surprises in TFP; this is again equivalent to using a Choleski identification with TFP ordered first, after having controlled for anticipated productivity shocks.

Our third approach, labeled “Method 3” in Figure A.7, is motivated by the concern of [Bouakez and Kemoe \(2023\)](#) that the unconditional correlation between measured TFP and inflation is rather weak, which they argue suggests the possibility of measurement error. We therefore identify as a supply shock the contemporaneous surprise that explains the strongest negative covariance between inflation and TFP. Since this negative relationship is the prototypical signature of supply shocks in most theories, isolating the shock driving this moment before performing our risk premium identification should be particularly effective at reducing the confounding of supply and demand channels in our risk premium disturbance.

The three TFP-based methods described above identify productivity news from VAR-implied expectations. But investor expectations may not correspond to those implied by the VAR and expected cash flows could fluctuate for non-TFP-related reasons. To account for these possibilities, we consider an additional “Method 4.” In this approach, we first extract the shock that explains the largest share of the unconditional variance in firm analysts’ long-term growth expectations, as measured by the LTG series of [Bordalo et al. \(2024\)](#). LTG captures survey-based analysts’ expectations about earnings growth of S&P 500 firms for the next 3-5 years and has been shown to be associated with future forecast errors about stock returns and future fluctuations in investment and activity ([Bordalo et al., 2024](#)). We then isolate our shock as before, imposing that it is orthogonal to this first shock.

Regardless of the method used to control for supply or expectations shocks, Figure A.7

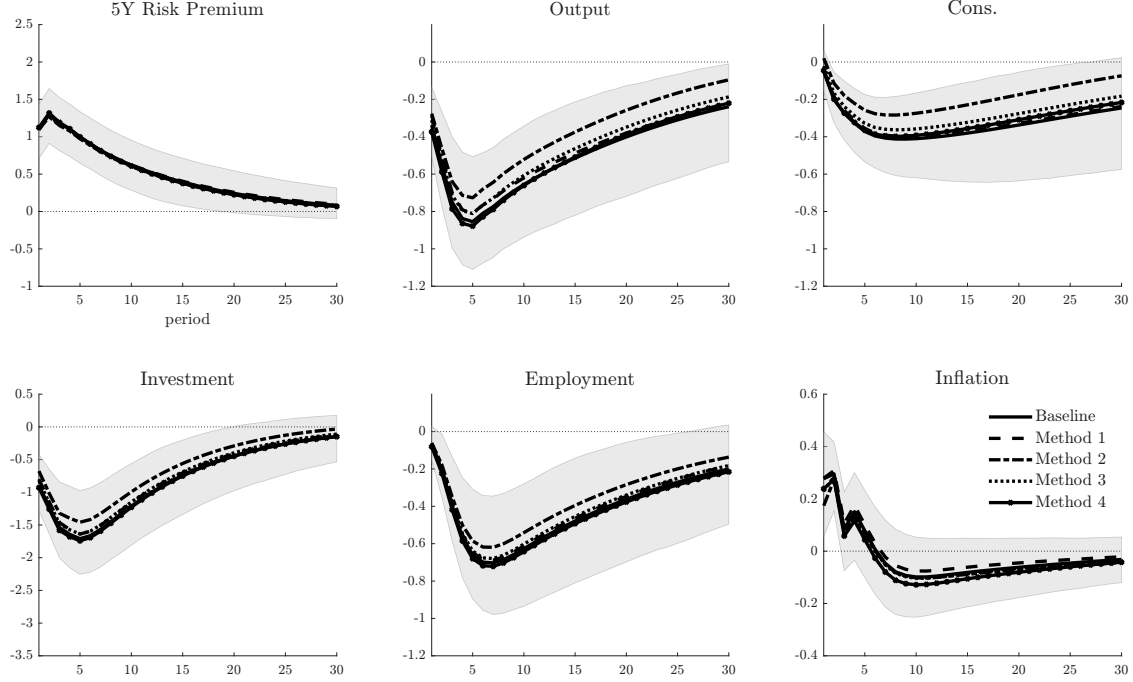


Figure A.7: Different approaches to controlling for alternative shocks.

shows that the properties of the resulting risk premium disturbance $\epsilon_t^{(1)}$ are essentially unchanged. The impulse responses of macro variables to the risk premium disturbance, including the increase in part-time employment and the weak relationship with inflation, are unaffected. Indeed, we find that in all cases the time series of the identified risk premium disturbance is correlated by 97% or more with the shock identified under our baseline procedure.

While our risk premium disturbance could still in principle be associated with more than one deeper structural shock, this evidence indicates that our estimates are unlikely to represent a confounding of conventional supply and demand channels. Moreover, our disturbance appears essentially orthogonal to the main forces driving both expected future productivity and future earnings, implying that changes in expected first moments — whether rational or not — are not likely to account for the patterns we find.

A.8 Evidence regarding over-fitting

One potential concern with targeting in-sample return expectations is that the procedure might overstate the degree of return predictability due to overfitting in finite samples. To address this concern, we performed two exercises using our baseline estimated model as the data generating process. In the first, we estimate our baseline VAR on data simulated by the model, but we replace the actual stock price series with a counterfactual equity price

(and corresponding returns) with a constant risk premium – we do so by counterfactually setting the risk premium component of the Euler equation for this asset to be equal to its unconditional mean, and is thus constant through time. The ex-post returns to this counterfactual asset reflect changes in the risk-free rate and in expected dividends, but not changes in the risk premium. In the second exercise, we replace the counterfactual equity price with the true one implied by the model (which includes time-variation in its risk premium).

In each of these two cases, we simulate 250 samples of artificial data of the same length as our real-world data sample. For each of these artificial samples, we estimate our VAR using our “method one” identification strategy and then perform a bootstrap using 1000 simulations to compute 80% confidence bands, exactly as we do with the real data. We then compute the average coverage probability over the first 16 quarters of the response because this is the horizon over which we find statistical evidence of premium fluctuations.

Using the first DGP, we find that the true response (of zero) falls within our 80% bands around 79% of the time, averaged across horizons, almost identical to the nominal 80%. This suggests that a coverage mismatch cannot account for our findings, all the more so because in the actual data the zero response lies well outside our 80% bands at most horizons.

When we perform the same exercise on the DGP that includes risk premium fluctuations, we find an average coverage probability of around 61% for the risk premium. Meanwhile, coverage rates for output (73%), consumption (75%), investment (70%), and labor (71%) are all close to, but below, their nominal values. The relatively low coverages in this case can be explained by a modest downward bias in the estimates of the risk premium disturbance itself: because every response is traced out from the same estimated disturbance, attenuating it shifts the bands for all of the variables, not only for the risk premium. Although the non-linearity of the DGP and finite lag restrictions of the VAR could contribute to this bias in principle, it almost entirely disappears with longer simulated data samples. This suggests the bias is mostly explained by small-sample issues. If anything, this downward bias makes our estimates conservative for our headline claim, since it understates the true risk premium response.

A.9 Other Risk Measures

The equity risk premium fluctuations associated with our risk premium disturbance are also related to other measures of asset market risk. The first row of panels in Figure A.8 shows that the VIX and Ian Martin’s (2017) more precise proxy for the (1-month) equity premium (SVIX2) react positively but transitorily to our shock. The Pflueger et al. (2020) equity-

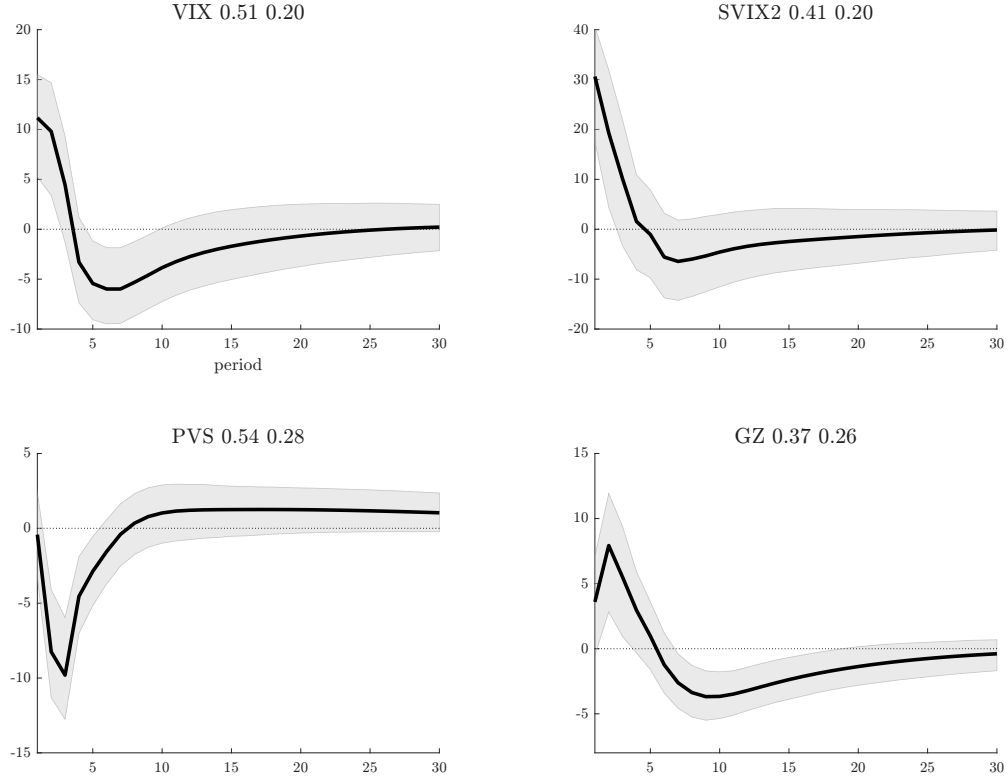


Figure A.8: Impulse responses to the VAR-identified risk premium disturbance for additional variables. Shaded areas cover the 10th- and 90th-percentiles of bootstrapped responses. Vertical axis of the excess bond premium (GZ) is in annualized percentage points, all others in percentage deviations. Numbers in subplot titles correspond to the percent of variance explained at the business cycle (6-32 quarters) and unconditional frequencies (2-500 quarters), respectively.

based perceived risk measure is displayed in the bottom-left of the figure. The series, which is normalized to fall when perceived risk is high, drops on impact but also reverses quickly a couple of quarters after our shock. This shows that our shock, which is identified out of variation in our proxy of the 5-year equity risk premium and thus has more of a medium-run nature, is also related to spikes in a wide range of short-run (1-month) equity risk premium measures.

However, Figure 3 shows that our shock has virtually no effect on safe interest rates, whether short-term or longer-term (five-year) rates. By contrast, the lower-right panel of Figure A.8 shows that the excess bond premium measure of [Gilchrist and Zakrajšek \(2012\)](#) rises for several quarters after the shock. This effect is much shorter-lived than the rise in the 5-year equity risk premium and the impact on the real variables we documented earlier, however. Moreover, we have found that our “risk premium” shock is distinct from the “bond premia” shocks in [Gilchrist and Zakrajšek \(2012\)](#) and [Kurmann and Otrok \(2013\)](#), due in part to having very different effects on inflation and real investment. Thus, our shock seems

to be something different from a direct shock to credit constraints or intertemporal forces.³⁰

B Analytical Model: Proofs of Propositions

B.1 Setup and notation

Before proving the Propositions, we summarize key equations and introduce several definitions to compress the notation required. The labor aggregate $\tilde{N}$ is given by

$$\tilde{N} \equiv \left(N_f^{\frac{\theta-1}{\theta}} + N_p^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}} = Y_0 \quad (\text{B.1})$$

with $\theta > 1$ and $Z_0 = 1$. Absolute and relative marginal products are given by

$$MPN_{i,0} \equiv \frac{\partial Y_0}{\partial N_i} = Y_0^{1/\theta} N_i^{-1/\theta}, \quad \frac{MPN_{p,0}}{MPN_{f,0}} = \left(\frac{N_p}{N_f} \right)^{-1/\theta}. \quad (\text{B.2})$$

Total *employment* (head-count) is given by $N \equiv N_f + N_p$. Note that $N = \tilde{N}$ only as $\theta \rightarrow \infty$. Filling N_i positions requires $N_i/P_i = N_i^{1/\varepsilon}$ vacancies, which cost $\varphi N_i^{1/\varepsilon}$. Period-zero consumption is output net of hiring expenditure,

$$C_0 = Y_0 - \varphi N_f^{1/\varepsilon} - \varphi N_p^{1/\varepsilon}. \quad (\text{B.3})$$

Productivity is the only fundamental shock, with $\log Z_1 \sim N(-\frac{1}{2}\sigma^2, \sigma^2)$, so that $\mathbb{E}[Z_1] = 1$, and $C_1 = Z_1 Y_0$. The household has Epstein–Zin preferences with IES ψ and risk aversion γ ; the stochastic discount factor is

$$M_{0,1} = \beta \left(\frac{C_1}{C_0} \right)^{-1/\psi} \left(\frac{C_1}{(\mathbb{E}_0 C_1^{1-\gamma})^{1/(1-\gamma)}} \right)^{1/\psi-\gamma}. \quad (\text{B.4})$$

Wages are either flexible (renegotiated by Nash bargaining each period, with household weight $\eta = 1 - \varepsilon$) or rigid (set at the expected marginal product at the time of hiring and not adjusted to Z_1). With $\eta = 1 - \varepsilon$ the wage absorbs a fraction $1 - \varepsilon$ of the marginal

³⁰López-Salido et al. (2017), who show that a predictable change in the risk premium on corporate bonds forecasts changes in real economic activity, argue that stock returns predicted using a set of lagged, mostly financial variables do not robustly predict macroeconomic activity. We use a different set of predictive variables to forecast the equity excess return over a longer time horizon and find sharply different results. Our equity risk premium disturbance is in fact a more important business cycle driver (in terms of variance decomposition), as compared to alternative credit market shocks and measures.

product, so the firm's optimality condition for a type- i worker is³¹

$$\varphi N_i^{\frac{1-\varepsilon}{\varepsilon}} = \varepsilon MPN_{i,0} \left[1 + \frac{1}{R} + \lambda_i \text{Cov}(M_{0,1}, Z_1) \right], \quad \lambda_i = \begin{cases} 1, & \text{wage of type } i \text{ flexible,} \\ \frac{1}{\varepsilon}, & \text{wage of type } i \text{ rigid,} \end{cases} \quad (\text{B.5})$$

where $R \equiv \frac{1}{\mathbb{E}_0[M_{0,1}]}$ is the risk-free interest rate. The covariance $\text{Cov}(M_{0,1}, Z_1) < 0$ is multiplied by the larger coefficient $1/\varepsilon$ when the wage is rigid: a rigid wage leaves the firm fully exposed to Z_1 , so a rigid-wage match carries the larger risk premium. We maintain Assumption 1, i.e., that the present value of the firm's second-period surplus is positive, throughout.

Step 1: Simplify Euler equations. Using our log-normality assumptions, we establish the following lemma.

Lemma 1. *The equilibrium risk-free rate and covariance terms are given by*

$$\frac{1}{R} \equiv \mathbb{E}_0[M_{0,1}] = \varrho e^{\gamma\sigma^2}, \quad \text{Cov}(M_{0,1}, Z_1) = \varrho(1 - e^{\gamma\sigma^2}), \quad (\text{B.6})$$

where

$$\varrho \equiv \mathbb{E}_0[M_{0,1}Z_1]. \quad (\text{B.7})$$

Proof. Substituting $C_1 = Z_1 Y_0$ into (B.4), the powers of Z_1 collapse so that

$$\begin{aligned} M_{0,1} &= \beta \left(\frac{Z_1 Y_0}{C_0} \right)^{-1/\psi} \left(\frac{Z_1 Y_0}{(\mathbb{E}_0(Z_1 Y_0)^{1-\gamma})^{1/(1-\gamma)}} \right)^{1/\psi-\gamma} \\ &= \beta (Y_0/C_0)^{-1/\psi} \frac{Z_1^{-1/\psi} (Z_1)^{1/\psi-\gamma}}{(E_0[Z_1^{1-\gamma}])^{\frac{1/\psi-\gamma}{1-\gamma}}} \\ &= c Z_1^{-\gamma} \end{aligned}$$

with $c \equiv \frac{\beta(Y_0/C_0)^{-1/\psi}}{(E_0[Z_1^{1-\gamma}])^{\frac{1/\psi-\gamma}{1-\gamma}}}$. Since Z_1 is log-normal with $\mathbb{E}[Z_1] = 1$, $\mathbb{E}[Z_1^a] = \exp(\frac{\sigma^2}{2}a(a-1))$.

Applying this formula, we have

$$\mathbb{E}[M_{0,1}] = c \exp\left(\frac{\sigma^2}{2}\gamma(\gamma+1)\right) > 0 \quad (\text{B.8})$$

$$\varrho \equiv \mathbb{E}[M_{0,1}Z_1] = c \exp\left(\frac{\sigma^2}{2}\gamma(\gamma-1)\right) > 0 \quad (\text{B.9})$$

³¹These are the part-time and full-time optimality conditions in Section 3.3. The factor ε is the firm's share of the marginal product net of the Nash wage; R is the gross risk-free rate, $1/R = \mathbb{E}_0[M_{0,1}]$.

From these, we can compute $Cov(M_{0,1}, Z_1) = \mathbb{E}[M_{0,1}Z_1] - \mathbb{E}[M_{0,1}]\mathbb{E}[Z_1] = \mathbb{E}[M_{0,1}Z_1](1 - e^{\gamma\sigma^2}) \equiv \varrho(1 - e^{\gamma\sigma^2})$. $\square$

Using the lemma, we can write the bracketed terms in the flexible case in (B.5) as

$$1 + \frac{1}{R} + Cov(M_{0,1}, Z_1) = 1 + \mathbb{E}[M_{0,1}] + Cov(M_{0,1}, Z_1) = 1 + \varrho.$$

Similarly, for the sticky-wage case

$$\begin{aligned} 1 + \frac{1}{R} + \frac{1}{\varepsilon}Cov(M_{0,1}, Z_1) &= 1 + \varrho e^{\gamma\sigma^2} + \frac{1}{\varepsilon}\varrho(1 - e^{\gamma\sigma^2}) \\ &= 1 + \frac{\varrho}{\varepsilon}[1 - (1 - \varepsilon)e^{\gamma\sigma^2}] \\ &= 1 + \frac{\varrho}{\varepsilon}s \end{aligned} \tag{B.10}$$

where $s \equiv 1 - (1 - \varepsilon)e^{\gamma\sigma^2}$.

Using this definition, and noting that s is monotonically decreasing in γ , we can derive feasible ranges for s and γ . Setting γ to its infimum of zero, we immediately have $s < \varepsilon$. Next, note that Assumption 1, applied to a rigid-wage match, requires that $\frac{1}{R} + \frac{1}{\varepsilon}Cov(M_{0,1}, Z_1)$, the discounted future payoff of such a match, is positive. Using (B.10), this implies $\frac{\varrho}{\varepsilon}s > 0$. The largest value for γ consistent with this restriction is given by the solution to $\hat{s} = 1 - (1 - \varepsilon)e^{\hat{\gamma}\sigma^2} = 0$. Hence $s \in (0, \varepsilon)$ and $\gamma \in (0, \hat{\gamma})$.

Step 2: The expenditure decomposition. Define the following

$$H_i \equiv \varphi N_i^{1/\varepsilon}, \quad H \equiv H_f + H_p, \quad \omega_p \equiv \frac{H_p}{H} = \frac{x^{1/\varepsilon}}{1 + x^{1/\varepsilon}}, \quad \omega_f \equiv 1 - \omega_p, \quad x \equiv \frac{N_p}{N_f}. \tag{B.11}$$

In the above, H is total hiring expenditure, the ω_i 's are the shares of hiring expenditure in full- and part-time positions, and x is the ratio of part-time to full-time hiring. Note that ω_p is strictly increasing in x .

We can write the marginal current output per hiring dollar (a.k.a. the dividend-price ratio):

$$r_i \equiv \frac{\partial Y_0}{\partial H_i} = \frac{\varepsilon MPN_{i,0}}{\varphi N_i^{(1-\varepsilon)/\varepsilon}} \tag{B.12}$$

where we have used the production function and $N_i = (H_i/\varphi)^\varepsilon$. Rearranging (B.5), we have

$$r_i^{-1} = 1 + \frac{1}{R} + \text{Cov}(M_{0,1}, Z_1) = 1 + \varrho \quad \text{for flexible wage} \quad (\text{B.13})$$

$$r_i^{-1} = 1 + \frac{1}{R} + \frac{1}{\varepsilon} \text{Cov}(M_{0,1}, Z_1) = 1 + \frac{\varrho}{\varepsilon} s \quad \text{for sticky wages.} \quad (\text{B.14})$$

This immediately implies that $r_i < 1$.

Going forward, for any equilibrium object z we use the notation $\dot{z} \equiv \frac{\partial z}{\partial \gamma}$. Differentiating (B.3) with respect to γ and using $\dot{Y}_0 = r_f \dot{H}_f + r_p \dot{H}_p$ with $H_f = \omega_f H$, $H_p = \omega_p H$, we get

$$\boxed{\frac{\partial C_0}{\partial \gamma} = (\bar{r} - 1) \frac{\partial H}{\partial \gamma} + (r_p - r_f) H \frac{\partial \omega_p}{\partial \gamma}, \quad \bar{r} \equiv \omega_f r_f + \omega_p r_p.} \quad (\text{B.15})$$

This is equation (19). The first term is the *total savings* (or total investment) channel and the second the *reallocation* channel; the proofs below sign each.

B.2 Proof of Proposition 1 (no-comovement benchmark)

Proof. Proposition 1 assumes both sectors share the same wage stickiness: either both flexible, or both rigid with Assumption 1. Then $\lambda_f = \lambda_p$ in (B.5), so that the two inputs share the same return (dividend-price) ratio, i.e., $r_f = r_p$. This equality establishes that the two labor investments carry the same risk premium (Proposition 1, risk premium clause).

Symmetry also implies that $N_f = N_p \equiv \bar{N}$ and $\omega_p = \frac{1}{2}$ for *all* γ . Moreover, since $\bar{r} = r_f = r_p < 1$, equation (B.15) becomes

$$\frac{\partial C_0}{\partial \gamma} = \underbrace{(\bar{r} - 1)}_{<0} \frac{\partial H}{\partial \gamma}. \quad (\text{B.16})$$

Since H is monotonic in $\bar{N}$, C_0 and N cannot move in the same direction (Proposition 1, no-comovement clause). $\square$

Remark. The *direction* in which $\bar{N}$ moves depends on ψ . Under flexible wages, $\psi > 1$ gives $\partial \bar{N} / \partial \gamma < 0$ and $\partial C_0 / \partial \gamma > 0$. These signs are reversed for $\psi < 1$, and $\bar{N}$ is unresponsive at $\psi = 1$. Rigid wages amplify the risk premium and enlarge the region in which $\bar{N}$ falls. Proposition 1 asserts only that C_0 and (Y_0, N) move oppositely, which holds regardless.

B.3 Proof of Proposition 2 (reallocation channel)

Proof of Clause 1. Since $\frac{s}{\varepsilon} < 1$, (B.13)-(B.14) immediately imply that

$$r_f = \frac{1}{1 + \frac{\varrho}{\varepsilon}s} > \frac{1}{1 + \varrho} = r_p. \quad (\text{B.17})$$

A unit invested in full-time hiring yields more current output than a unit in part-time hiring; equivalently, full-time labor carries the higher dividend-price ratio, the equilibrium compensation for its larger risk premium. $\square$

Proof of Clause 2. Because ω_p is strictly increasing in $x = N_p/N_f$ by (B.11), the claim $\partial\omega_p/\partial\gamma > 0$ is equivalent to $\partial x/\partial\gamma > 0$. We reduce the equilibrium to two scalar equations in (x, ϱ) and sign $\partial \log x/\partial\gamma$ by the implicit function theorem.

Divide the part-time labor demand equation in (B.5) by its full-time counterpart, rearrange, and use the marginal-product ratio in (B.1), to get

$$x^q = \Lambda(\varrho, \gamma) \equiv \frac{1 + \varrho}{1 + \frac{\varrho}{\varepsilon}s} \quad (\text{B.18})$$

where $q \equiv \frac{1-\varepsilon}{\varepsilon} + \frac{1}{\theta} > 0$. Since $s < \varepsilon$ (Step 1), we have $1 + \frac{\varrho}{\varepsilon}s < 1 + \varrho$, so $\Lambda > 1$; hence $x^q > 1$, i.e., $x > 1$ and $N_p > N_f$.

Next, multiply the equations in (B.5) by the corresponding N_i and sum together. Use the fact that $\varphi N_p^{\frac{1}{\varepsilon}} + \varphi N_f^{\frac{1}{\varepsilon}} = H$ on the left-hand side. Type by type, this yields the labor-demand identity

$$\frac{H_i}{Y_0} = \varepsilon \mu_i B_i, \quad B_i \equiv r_i^{-1}, \quad (\text{B.19})$$

and summing over i ,

$$\begin{aligned} h \equiv H/Y_0 &= \varepsilon \left[\mu_p(1 + \varrho) + \mu_f(1 + \frac{\varrho}{\varepsilon}s) \right] \\ &= \varepsilon + \varrho\varpi \end{aligned}$$

where $\mu_f = (MPN_{f,0}N_f)/Y_0 = \frac{1}{1+x\zeta}$, $\mu_p = 1 - \mu_f$, with $\zeta \equiv \frac{\theta-1}{\theta} = 1 - \frac{1}{\theta}$ (so $\text{sign } \zeta = \text{sign}(\theta - 1)$), and $\varpi \equiv \varepsilon\mu_p + \mu_f s$. Then, the ratio

$$\Phi \equiv C_0/Y_0 = \frac{Y_0 - H}{Y_0} = 1 - \varepsilon - \varrho\varpi. \quad (\text{B.20})$$

Using the definitions above, we then have

$$\varrho = \beta \Phi^{\frac{1}{\psi}} K(\gamma) \quad (\text{B.21})$$

where $K(\gamma) \equiv \exp\left(-\frac{\sigma^2\gamma(\psi-1)}{2\psi}\right)$.

The system. Equations (B.18) and (B.21) are two equations in (x, ϱ) with γ as a parameter; both are scale-free (the level N_f drops out because μ_p, μ_f and the marginal-product ratio depend only on x). Writing them as

$$F^1(\log x, \varrho; \gamma) \equiv q \log x - \log \Lambda(\varrho, \gamma) = 0, \quad F^2(\log x, \varrho; \gamma) \equiv G \equiv \varrho - \beta \Phi^{1/\psi} K = 0, \quad (\text{B.22})$$

an equilibrium is a root of $F = (F^1, F^2) = 0$. The plan is to apply the implicit function theorem to this system: we compute the partial derivatives of both rows, show that the Jacobian with respect to the unknowns $(\log x, \varrho)$ has strictly positive determinant Δ , so that $\gamma \mapsto (\log x, \varrho)$ is locally a smooth function whose derivatives are delivered by Cramer's rule, and then sign $\log x > 0$ (equivalently $\dot{\omega}_p > 0$), establishing the second clause of the Proposition.

Partials of (B.18). The row F^1 depends on $\log x$ only directly, with $\partial F^1 / \partial \log x = q$, and on (ϱ, γ) only through $\log \Lambda$. Using $\dot{s} = -\sigma^2(1-s)$ since $1-s = (1-\varepsilon)e^{\gamma\sigma^2}$,

$$A \equiv \frac{\partial \log \Lambda}{\partial \gamma} = \frac{\varrho(1-s)\sigma^2}{\varepsilon(1+\frac{\varrho}{\varepsilon}s)} > 0, \quad B \equiv \frac{\partial \log \Lambda}{\partial \varrho} = \frac{1-s/\varepsilon}{(1+\varrho)(1+\frac{\varrho}{\varepsilon}s)} > 0.$$

Partials of (B.21): building blocks. The row $F^2 = G$ has ϱ , $\log x$, and γ as its three arguments: ϱ enters directly and through $\Phi = 1 - \varepsilon - \varrho\varpi$; $\log x$ enters only through $\varpi = \varepsilon\mu_p + \mu_f s$ (via the shares); and γ enters through both K and the risk term s (the latter inside ϖ).

Noting that the shares satisfy $\mu_p/\mu_f = x^\zeta$, we have

$$\frac{d\mu_p}{d\log x} = \zeta\mu_p\mu_f, \quad \frac{d\mu_f}{d\log x} = -\zeta\mu_p\mu_f. \quad (\text{B.23})$$

The factor $K = \exp\left(-\frac{\sigma^2\gamma(\psi-1)}{2\psi}\right)$ has $\log K = \frac{\dot{K}}{K} = -\frac{\sigma^2(\psi-1)}{2\psi}$, and K depends on γ alone (so $K_\varrho = K_{\log x} = 0$).

Partials of Φ . With $\Phi = 1 - \varepsilon - \varrho\varpi$ and $\varpi = \varepsilon\mu_p + \mu_f s$,

$$\begin{aligned} \Phi_\varrho &= -\varpi & (\varpi \text{ depends on } x, \gamma \text{ only, not on } \varrho), \\ \Phi_{\log x} &= -\varrho \frac{\partial \varpi}{\partial \log x} = -\varrho \left(\varepsilon \frac{d\mu_p}{d\log x} + s \frac{d\mu_f}{d\log x} \right) = -\varrho(\varepsilon - s)\zeta\mu_p\mu_f, \quad \text{using } \varepsilon(\zeta\mu_p\mu_f) + s(-\zeta\mu_p\mu_f), \\ \Phi_\gamma &= -\varrho \dot{\varpi} = -\varrho \mu_f \dot{s} = \varrho \mu_f \sigma^2(1-s), & \text{only } s \text{ in } \varpi \text{ depends on } \gamma. \end{aligned}$$

Partials of G . Differentiating $G = \varrho - \beta\Phi^{1/\psi}K$ and using $\frac{\partial}{\partial z}\Phi^{1/\psi} = \frac{1}{\psi}\Phi^{1/\psi}\frac{\Phi_z}{\Phi}$ together with the equilibrium identity $\beta\Phi^{1/\psi}K = \varrho$ (which collapses every $\beta\Phi^{1/\psi}K$ prefactor to ϱ), the generic partial is

$$G_z = \frac{\partial \varrho}{\partial z} - \frac{\varrho}{\psi} \frac{\Phi_z}{\Phi} - \varrho \frac{K_z}{K}. \quad (\text{B.24})$$

Evaluating in turn,

$$\begin{aligned} G_\varrho &= 1 - \frac{\varrho}{\psi} \frac{\Phi_\varrho}{\Phi} = 1 - \frac{\varrho}{\psi} \frac{-\varpi}{\Phi} = 1 + \frac{\varrho\varpi}{\psi\Phi} > 0, \\ G_{\log x} &= -\frac{\varrho}{\psi} \frac{\Phi_{\log x}}{\Phi} = -\frac{\varrho - \varrho(\varepsilon - s)\zeta\mu_p\mu_f}{\psi\Phi} = \frac{\varrho^2(\varepsilon - s)\zeta\mu_p\mu_f}{\psi\Phi} \geq 0, \\ G_\gamma &= -\frac{\varrho}{\psi} \frac{\Phi_\gamma}{\Phi} - \varrho \frac{K_\gamma}{K} = -\frac{\varrho^2\mu_f\sigma^2(1-s)}{\psi\Phi} + \frac{\varrho\sigma^2(\psi-1)}{2\psi} = -\frac{\varrho\sigma^2}{\psi} \left[\frac{\varrho\mu_f(1-s)}{\Phi} - \frac{\psi-1}{2} \right]. \end{aligned}$$

The sign of $G_{\log x}$ follows from $\varepsilon - s = (1 - \varepsilon)(e^{\gamma\sigma^2} - 1) > 0$, giving $\text{sign } G_{\log x} = \text{sign } \zeta = \text{sign}(\theta - 1) \geq 0$.

The Jacobian and the implicit function theorem. Collecting the partials, the Jacobian of F with respect to the unknowns $(\log x, \varrho)$ and its determinant are

$$J \equiv \begin{pmatrix} q & -B \\ G_{\log x} & G_\varrho \end{pmatrix}, \quad \Delta \equiv \det J = qG_\varrho + BG_{\log x} > 0, \quad (\text{B.25})$$

where positivity holds for every $\theta > 1$ because q , G_ϱ , and B are strictly positive and $G_{\log x} \geq 0$. Since F is continuously differentiable and J is nonsingular, the implicit function theorem applies at any equilibrium: $\gamma \mapsto (\log x(\gamma), \varrho(\gamma))$ is a smooth function in a neighborhood, and total differentiation of $F = 0$ along this path gives the linear system $J(\log x, \dot{\varrho})^\top = (A, -G_\gamma)^\top$. Written out, its first row is

$$q \log x = A + B\dot{\varrho} \quad (\text{B.26})$$

and its second row is $G_{\log x} \log x + G_\varrho \dot{\varrho} = -G_\gamma$. By Cramer's rule,

$$\log x = \frac{AG_\varrho - BG_\gamma}{\Delta}, \quad \dot{\varrho} = \frac{-qG_\gamma - AG_{\log x}}{\Delta}. \quad (\text{B.27})$$

Signing $\log x$. It is worth reading the first row (B.26) on its own: since q , A , and B are all positive, $\log x > 0$ *unless* ϱ falls fast enough in γ — precisely, $\log x > 0$ if and only if $\dot{\varrho} > -A/B = -\frac{1-s}{\varepsilon-s}(\varrho(1+\varrho))\sigma^2$. The only term that can make G_γ positive, and thereby push $\dot{\varrho}$ toward this threshold, is the precautionary factor K , which decreases in γ when $\psi > 1$.

We now show this effect is never large enough to change the overall sign.

Consider the numerator $AG_\varrho - BG_\gamma$ in (B.27). Dropping the nonnegative term $\varrho\mu_f(1-s)/\Phi$ from the bracket in G_γ gives, for every $\psi > 0$,

$$G_\gamma \leq \frac{\varrho\sigma^2}{\psi} \cdot \frac{\psi-1}{2} < \frac{\varrho\sigma^2}{2}, \quad (\text{B.28})$$

using $(\psi-1)/\psi < 1$ (when $\psi \leq 1$ the middle expression is already nonpositive). Meanwhile $G_\varrho > 1$, and comparing A with B directly,

$$\frac{A}{B} = \frac{\varrho(1+\varrho)(1-s)\sigma^2}{\varepsilon-s} > \varrho\sigma^2, \quad (\text{B.29})$$

since $1-s > \varepsilon-s$ and $1+\varrho > 1$. Combining the three bounds,

$$AG_\varrho - BG_\gamma > A - B \frac{\varrho\sigma^2}{2} > A - \frac{A}{2} = \frac{A}{2} > 0. \quad (\text{B.30})$$

Hence $\log x > 0$ for every $\psi > 0$, which completes the proof. $\square$

Clause 3: full comovement when $\psi = 1$ and $\theta > \frac{1+\beta}{\beta(1-\varepsilon)}$

To prove this claim, we evaluate the comparative statics at the boundary $\hat{\gamma}$ defined by $s = 0$. We show that the claimed comovements occur at this particular value for γ . Extending by continuity, we know there exists a range of values of γ just below $\hat{\gamma}$ that satisfy the comovement requirements.

Set $\psi = 1$, so $K \equiv 1$ and equation (B.21) implies

$$\varrho = \beta\Phi = \beta(1 - \varepsilon - \varrho\varpi). \quad (\text{B.31})$$

Solving for ϱ yields

$$\varrho = \beta(1 - \varepsilon)/(1 + \beta\varpi) \quad (\text{B.32})$$

which depends only on x and s . In particular $\varrho \leq \beta(1 - \varepsilon)$ for every admissible equilibrium (since $\varpi \geq 0$), a bound we use below.

At $s = 0$ the full-time bracket collapses to $r_f = 1$: the full-time margin returns one unit of current output per hiring dollar. This is the feature that makes the boundary tractable, and it is exactly what decouples the full-time margin in the expenditure decomposition (B.15). Throughout this clause we write $\mu_i = MPN_{i,0}N_i/\tilde{N}$ with $\mu_f + \mu_p = 1$, $k \equiv \frac{1-\varepsilon}{\varepsilon}$ so $q = k + \frac{1}{\theta}$, and we continue to use $\dot{z} \equiv \partial z / \partial \gamma$. Equilibrium objects evaluated at the boundary $\hat{\gamma}$ (where

$\hat{s} = 0$) carry a hat, matching the boundary point $\hat{\gamma}$ itself, e.g., $\hat{x}, \hat{\Phi}, \hat{\mu}_p$.

Step 1: The boundary equilibrium exists and is unique. A boundary equilibrium is a pair $(\hat{x}, \hat{\varrho})$ satisfying the two scale-free equations at $\hat{s} = 0$. There the ratio equation (B.18) gives $\hat{\varrho} = \hat{x}^q - 1$, and (B.32) gives $\hat{\varrho} = \beta(1 - \varepsilon)/(1 + \beta\varepsilon\hat{\mu}_p)$ with $\hat{\mu}_p = \hat{x}^\zeta/(1 + \hat{x}^\zeta)$. Eliminating $\hat{\varrho}$, the boundary ratio is a root of the single *condition*, viewed as an equation in the free variable x ,

$$x^q = 1 + \frac{\beta(1 - \varepsilon)}{1 + \beta\varepsilon \frac{x^\zeta}{1+x^\zeta}}. \quad (\text{B.33})$$

The left-hand side is increasing in x , ranging from 0 to $+\infty$, while the right-hand side is positive, finite, and decreasing; hence (B.33) has a unique interior root, the *solution* $\hat{x}$, with $\hat{\varrho} = \hat{x}^q - 1 > 0$. Evaluating (B.32) at this solution gives the boundary consumption share

$$\hat{\Phi} = \frac{\hat{\varrho}}{\beta} = \frac{1 - \varepsilon}{1 + \varepsilon\beta\hat{\mu}_p}, \quad (\text{B.34})$$

which we use in the consumption step. Finally, $\hat{\varrho} \leq \beta(1 - \varepsilon)$ (since $\hat{\mu}_p \geq 0$) together with $\hat{x}^q = 1 + \hat{\varrho}$ and $q > k = \frac{1-\varepsilon}{\varepsilon}$ bounds the boundary ratio:

$$q \log \hat{x} = \log(1 + \hat{\varrho}) \leq \hat{\varrho} \leq \beta(1 - \varepsilon) \implies \log \hat{x} < \frac{\beta(1 - \varepsilon)}{(1 - \varepsilon)/\varepsilon} = \beta\varepsilon, \quad (\text{B.35})$$

using $\log(1 + u) \leq u$. We use $\log \hat{x} < \beta\varepsilon$ in the head-count step.

Going forward, all equations are evaluated at the boundary equilibrium and we suppress the $\hat{}$ notation on equilibrium objects (retaining it only on the boundary point $\hat{\gamma}$).

Step 2: Reallocation and the boundary partials. At $\psi = 1$ ($K \equiv 1$) and $s = 0$ ($\varpi = \varepsilon\mu_p$, $\dot{s} = -\sigma^2$), the Clause-2 blocks specialize. Crucially, the consistency relation $\varrho = \beta\Phi$ lets us substitute $\Phi = \varrho/\beta$, which cancels the Φ in every G against a power of ϱ :

$$A = \frac{\varrho\sigma^2}{\varepsilon}, \quad B = \frac{1}{1 + \varrho}, \quad G_\varrho = 1 + \beta\varepsilon\mu_p > 0, \quad G_{\log x} = \beta\varepsilon\varrho\zeta\mu_p\mu_f \geq 0, \quad G_\gamma = -\beta\varrho\mu_f\sigma^2 < 0, \quad (\text{B.36})$$

the precautionary term $\frac{\psi-1}{2}$ in G_γ vanishing at $\psi = 1$. With $\Delta = qG_\varrho + BG_{\log x} = q(1 + \beta\varepsilon\mu_p) + \beta\varepsilon b\zeta\mu_p\mu_f > 0$ and $b \equiv \frac{\varrho}{1+\varrho} = \frac{\beta\Phi}{1+\beta\Phi} \in (0, 1)$ (the second form using $\varrho = \beta\Phi$), the

Cramer solution (B.27) collapses to closed forms,

$$\log \dot{x} = \frac{AG_\varrho - BG_\gamma}{\Delta} = \frac{\sigma^2 \varrho \left[\frac{1}{\varepsilon} + \beta \mu_p + \frac{\beta \mu_f}{1+\varrho} \right]}{\Delta}, \quad \dot{\varrho} = \frac{-qG_\gamma - AG_{\log x}}{\Delta} = \frac{\beta \sigma^2 \varrho \mu_f (q - \zeta \varrho \mu_p)}{\Delta}. \quad (\text{B.37})$$

Clearly $\log \dot{x} > 0$, a special case of Clause 2's result at the boundary. Meanwhile, $\dot{\varrho} > 0$. To see this, note that

$$q - \zeta \varrho \mu_p > q - \zeta \beta (1 - \varepsilon) \mu_p > q - (1 - \varepsilon) = \frac{1 - \varepsilon}{\varepsilon} + \frac{1}{\theta} - (1 - \varepsilon) > 0,$$

where the first inequality uses $\varrho < \beta(1 - \varepsilon)$, and the second uses ζ, β, μ_p all less than one.

Step 3: Implications for the level of output and employment. The system (B.18), (B.21) is scale-free, so it pins x, ϱ (hence μ_i, Φ, r_i) but not the level of output, employment or consumption. Using the CES production function, we have

$$\log \dot{Y}_0 = \mu_f \log \dot{N}_f + \mu_p \log \dot{N}_p. \quad (\text{B.38})$$

Taking logs of (B.5) with $MPN_{i,0} = \tilde{N}^{1/\theta} N_i^{-1/\theta}$ and using $\frac{1-\varepsilon}{\varepsilon} + \frac{1}{\theta} \equiv q$ yields

$$q \log \dot{N}_i = \frac{1}{\theta} \log \dot{Y}_0 + \log \dot{B}_i. \quad (\text{B.39})$$

Multiply (B.39) by μ_i and sum over $i \in \{f, p\}$. Then using (B.38) and $\mu_f + \mu_p = 1$, we have:

$$q \log \dot{Y}_0 = \frac{1}{\theta} \log \dot{Y}_0 + \mu_f \log \dot{B}_f + \mu_p \log \dot{B}_p. \quad (\text{B.40})$$

Collecting the $\log \dot{Y}_0$ terms and using $q - \frac{1}{\theta} = k$ yields the single scalar that closes the level block:

$$k \log \dot{Y}_0 = \mu_f \log \dot{B}_f + \mu_p \log \dot{B}_p. \quad (\text{B.41})$$

At the boundary the bracket log-derivatives are

$$\log \dot{B}_p = \frac{\dot{\varrho}}{1 + \varrho}, \quad \log \dot{B}_f \Big|_{s=0} = -\frac{\varrho \sigma^2}{\varepsilon} = -A, \quad (\text{B.42})$$

the first because $B_p = 1 + \varrho$, the second because $B_f = 1 + \frac{\varrho}{\varepsilon} s$ with $B_f = 1$ and $\dot{s} = -\sigma^2$ at $s = 0$ (both brackets from (B.13)–(B.14)). Substituting these values into (B.41) yields

$$k \log \dot{Y}_0 = -\mu_f \frac{\varrho \sigma^2}{\varepsilon} + \mu_p \frac{\dot{\varrho}}{1 + \varrho}. \quad (\text{B.43})$$

Here $-\mu_f A < 0$ while $\dot{\varrho} > 0$ by (B.37), so the two terms compete. To resolve the sign, substitute the closed form $\dot{\varrho} = \beta \varrho \mu_f \sigma^2 (q - \zeta \varrho \mu_p) / \Delta$ from (B.37). The factor ϱ in $\dot{\varrho}$ combines with the $1/(1 + \varrho)$ in (B.43) to produce exactly $b \equiv \varrho/(1 + \varrho)$:

$$\mu_p \frac{\dot{\varrho}}{1 + \varrho} = \underbrace{\frac{\varrho}{1 + \varrho}}_{=b} \cdot \frac{\beta \mu_f \sigma^2 \mu_p (q - \zeta \varrho \mu_p)}{\Delta} = \frac{\beta b \mu_f \mu_p \sigma^2 (q - \zeta \varrho \mu_p)}{\Delta}. \quad (\text{B.44})$$

With $A = \varrho \sigma^2 / \varepsilon$, equation (B.43) becomes

$$k \log \dot{Y}_0 = \mu_f \sigma^2 \left[-\frac{\varrho}{\varepsilon} + \frac{\beta b \mu_p (q - \zeta \varrho \mu_p)}{\Delta} \right]. \quad (\text{B.45})$$

Factoring out the common denominator Δ , we can write

$$k \log \dot{Y}_0 = \frac{\mu_f \sigma^2}{\Delta} \mathcal{N}, \quad (\text{B.46})$$

where $\mathcal{N} \equiv \beta b \mu_p (q - \zeta \varrho \mu_p) - \frac{\varrho}{\varepsilon} \Delta$. Expand each piece, using $\Delta = q(1 + \beta \varepsilon \mu_p) + \beta \varepsilon b \zeta \mu_p \mu_f$:

$$\begin{aligned} \mathcal{N} &= [\beta b q \mu_p - \beta b \zeta \varrho \mu_p^2] - \left[\frac{\varrho q}{\varepsilon} + \beta \varrho q \mu_p + \beta \varrho b \zeta \mu_p \mu_f \right] \\ &= -\frac{\varrho q}{\varepsilon} + \beta q \mu_p (b - \varrho) - \beta b \zeta \varrho \mu_p (\mu_p + \mu_f). \end{aligned}$$

Now apply three facts: $\mu_p + \mu_f = 1$; $b - \varrho = -\frac{\varrho^2}{1 + \varrho} = -\varrho b$ (since $\frac{\varrho}{1 + \varrho} = b$), so the middle term is $-\beta \varrho b q \mu_p$; and the third term is $-\beta \varrho b \zeta \mu_p$. The last two terms then share $-\beta \varrho b \mu_p$:

$$\mathcal{N} = -\frac{\varrho q}{\varepsilon} - \beta \varrho b \mu_p (q + \zeta). \quad (\text{B.47})$$

Finally $q + \zeta = (k + \frac{1}{\theta}) + (1 - \frac{1}{\theta}) = k + 1 = \frac{1}{\varepsilon}$, so $q + \zeta = \frac{1}{\varepsilon}$ and the numerator reduces to a single term:

$$\mathcal{N} = -\frac{\varrho q}{\varepsilon} - \frac{\beta \varrho b \mu_p}{\varepsilon} = -\frac{\varrho}{\varepsilon} (q + \beta b \mu_p) \equiv -\frac{\varrho}{\varepsilon} D, \quad (\text{B.48})$$

where the composite $D \equiv q + \beta b \mu_p > 0$ recurs throughout the remaining steps. Hence $\mathcal{N}$ is clearly negative. Using $k\varepsilon = 1 - \varepsilon$ to solve for $\log \dot{Y}_0$, we have

$$\log \dot{Y}_0 = -\frac{\varrho \mu_f \sigma^2 D}{(1 - \varepsilon) \Delta} < 0, \quad (\text{B.49})$$

implying that output must fall with γ .

Finally, the $i = f$ row of (B.39) gives

$$q \log \dot{N}_f = \frac{1}{\theta} \log \dot{Y}_0 - A < 0, \quad (\text{B.50})$$

so full-time employment also must be falling in γ . The $i = p$ row likewise gives $\log \dot{N}_p$, now using $\log \dot{B}_p = \dot{\varrho}/(1 + \varrho)$ with $\dot{\varrho}$ from (B.37) and (B.49); collecting the $\log \dot{Y}_0$ terms,

$$\log \dot{N}_p = \frac{\frac{1}{\theta} - b}{D} \log \dot{Y}_0. \quad (\text{B.51})$$

Since $\log \dot{Y}_0 < 0$ and $D > 0$, $\log \dot{N}_p > 0$ if and only if

$$\theta > \frac{1}{b}. \quad (\text{B.52})$$

Step 4: Implications for consumption. At the boundary, $r_f = 1$, so that $\bar{r} - 1 = \omega_p(r_p - 1)$ and $r_p - r_f = r_p - 1$. Using these results, (B.15) becomes

$$\dot{C}_0 = (\bar{r} - 1)\dot{H} + (r_p - r_f)H\dot{\omega}_p = (r_p - 1)[\omega_p\dot{H} + H\dot{\omega}_p] = (r_p - 1)\dot{H}_p. \quad (\text{B.53})$$

Because $r_p - 1 = \frac{1}{1+\varrho} - 1 < 0$ and $H_p = \varphi N_p^{1/\varepsilon}$ moves with N_p ,

$$\text{sign } \dot{C}_0 = - \text{sign } \dot{N}_p, \quad (\text{B.54})$$

so *at the boundary consumption falls if and only if part-time employment rises*, that is, if and only if the inequality in (B.52) holds. The full-time margin is consumption-neutral precisely because $r_f = 1$. Using $\frac{1}{b} = 1 + \frac{1}{\beta\Phi}$ together with (B.34), the threshold in primitives is

$$\theta > \frac{1}{b} = 1 + \frac{1}{\beta\Phi} = 1 + \frac{1 + \varepsilon\beta\mu_p}{\beta(1 - \varepsilon)}. \quad (\text{B.55})$$

Since $\mu_p < 1$, the right-hand side is below $1 + \frac{1+\varepsilon\beta}{\beta(1-\varepsilon)} = \frac{1+\beta}{\beta(1-\varepsilon)}$, so the stated condition $\theta > \frac{1+\beta}{\beta(1-\varepsilon)}$ is sufficient (indeed it is the supremum of the right-hand side of (B.55) over $\mu_p \in (\frac{1}{2}, 1)$, hence the tightest sufficient bound expressible in primitives alone).

Step 5: Implication for total employment. Suppose that inequality (B.52) holds, so that $\dot{N}_p > 0$. From $\dot{\tilde{N}} = MPN_{f,0}\dot{N}_f + MPN_{p,0}\dot{N}_p$, solve for $\dot{N}_f$ and add $\dot{N}_p$, to get

$$\begin{aligned}\dot{N} &= \frac{1}{MPN_{f,0}}\dot{\tilde{N}} + \left(1 - \frac{MPN_{p,0}}{MPN_{f,0}}\right)\dot{N}_p \\ &= \frac{1}{MPN_{f,0}}\dot{\tilde{N}} + (1 - x^{-1/\theta})\dot{N}_p,\end{aligned}\tag{B.56}$$

where the second equality uses $MPN_{p,0}/MPN_{f,0} = x^{-1/\theta}$.

In (B.56) the two terms have opposite signs: the first is negative, since $\dot{\tilde{N}} = \dot{Y}_0 < 0$ by (B.49) and $MPN_{f,0} > 0$; the second is positive, since $x > 1$ gives $1 - x^{-1/\theta} > 0$ and $\dot{N}_p > 0$ in the case considered. Hence $\dot{N} < 0$ holds if and only if the negative first term outweighs the positive second:

$$(1 - x^{-1/\theta})\dot{N}_p < -\frac{1}{MPN_{f,0}}\dot{\tilde{N}}.\tag{B.57}$$

Multiply both sides by $MPN_{f,0}/\tilde{N} > 0$. On the right,

$$-\frac{1}{MPN_{f,0}}\dot{\tilde{N}} \cdot \frac{MPN_{f,0}}{\tilde{N}} = -\frac{\dot{\tilde{N}}}{\tilde{N}} = -\log \dot{Y}_0,\tag{B.58}$$

since $\tilde{N} = Y_0$. On the left, write $\dot{N}_p = N_p \log \dot{N}_p$ and use $MPN_{f,0}N_p/\tilde{N} = \mu_f x$ (because $\mu_f = MPN_{f,0}N_f/\tilde{N}$ and $N_p/N_f = x$):

$$(1 - x^{-1/\theta}) \frac{MPN_{f,0}N_p}{\tilde{N}} \log \dot{N}_p = (1 - x^{-1/\theta}) \mu_f x \log \dot{N}_p < -\log \dot{Y}_0.\tag{B.59}$$

Next substitute $\log \dot{N}_p = \frac{b-1/\theta}{D} (-\log \dot{Y}_0)$ from (B.51), to get

$$(1 - x^{-1/\theta}) \mu_f x \frac{b-1/\theta}{D} (-\log \dot{Y}_0) < -\log \dot{Y}_0.\tag{B.60}$$

Finally divide both sides by $-\log \dot{Y}_0 > 0$ and multiply by $D > 0$:

$$(1 - x^{-1/\theta}) \mu_f x \left(b - \frac{1}{\theta}\right) < D.\tag{B.61}$$

We next bound each term. First, $1 - x^{-1/\theta} < \frac{\log x}{\theta} \leq \frac{\beta\varepsilon}{\theta}$ by (B.35). Second, $\mu_f < \frac{1}{2}$ (since $x > 1 \Rightarrow \mu_p > \frac{1}{2}$). Third, inverting the log function in (B.35), we have $x < e^{\beta\varepsilon} \leq e$. Finally,

$b - \frac{1}{\theta} < b \leq \beta\Phi \leq \beta(1 - \varepsilon)$. Hence

$$\text{LHS of (B.61)} < \frac{\beta\varepsilon}{\theta} \cdot \frac{1}{2} \cdot e \cdot \beta(1 - \varepsilon) = \frac{e}{2} \beta^2 \varepsilon (1 - \varepsilon) \frac{1}{\theta} \leq \frac{e}{8} \frac{1}{\theta}, \quad (\text{B.62})$$

using $\beta^2 \leq 1$ and $\varepsilon(1 - \varepsilon) \leq \frac{1}{4}$, while the right side satisfies $D \geq q \geq \frac{1}{\theta}$. Since $\frac{e}{8} < 1$, (B.61) holds strictly for all $\beta, \varepsilon \in (0, 1)$ and $\theta > 1$, so $\dot{N} < 0$.

Step 6: Conclusion of Clause 3. The determinant $\Delta = q(1 + \beta\varepsilon\mu_p) + \beta\varepsilon b \zeta\mu_p\mu_f$ remains strictly positive at the boundary equilibrium (Step 2), so the implicit function theorem argument of Clause 2 applies at $\hat{\gamma}$: the equilibrium $\gamma \mapsto (x, \varrho)$, and hence (C_0, Y_0, N) , are smooth functions in a neighborhood of $\hat{\gamma}$. Thus, existence and uniqueness extend from $s = 0$ to a left-neighborhood $(\hat{\gamma} - \delta, \hat{\gamma}]$.

We have shown that $\dot{Y}_0 < 0$ and $\dot{N} < 0$ unconditionally, and that $\dot{C}_0 < 0$ under (B.55), for which $\theta > \frac{1+\beta}{\beta(1-\varepsilon)}$ is sufficient. Therefore, all three derivatives are strictly negative in a left-neighborhood of $\hat{\gamma}$. $\square$

C Model

This section contains a detailed derivation of the real business cycle model that we use in our quantitative analysis.

C.1 Households

The economy is populated by a representative household with a continuum of members of unit measure. In period t , the household chooses aggregate consumption (C_t), government bond holdings (B_{t+1}), corporate bond holdings (B_{t+1}^c), and firm share holdings (X_{t+1}), to maximize lifetime utility

$$\mathbb{V}_t = \max \left[(1 - \beta) C_t^{1-1/\psi} + \beta (\mathbb{E}_t \mathbb{V}_{t+1}^{1-\gamma_t})^{\frac{1-1/\psi}{1-\gamma_t}} \right]^{\frac{1}{1-1/\psi}} \quad (\text{C.1})$$

subject to the period budget constraint, denoted in terms of the consumption numeraire,

$$C_t + P_t^e X_{t+1} + Q_t^c (B_{t+1}^c - dB_t^c) + \frac{1}{R_t} B_{t+1} \leq (D_t^e + P_t^e) X_t + B_t^c + B_t + E_t^l + T_t. \quad (\text{C.2})$$

In the above, Q_t^c is the price of a multi-period corporate bond with average duration $(1-d)^{-1}$, R_t is the one-period safe real interest rate, P_t^e is the price of a share of the representative

firm that pays a real dividend D_t^e , and E_t^l is the household's total labor earnings (detailed below). Risk aversion is denoted by γ_t , while ψ is the intertemporal elasticity of substitution.

Epstein–Zin preferences imply the following stochastic discount factor:

$$M_{t,t+1} = \left(\frac{\partial \mathbb{V}_t / \partial C_{t+1}}{\partial \mathbb{V}_t / \partial C_t} \right) = \beta \left(\frac{C_{t+1}}{C_t} \right)^{1-1/\psi} \left(\frac{C_t}{C_{t+1}} \right) \left(\frac{\mathbb{V}_{t+1}}{(\mathbb{E}_t \mathbb{V}_{t+1}^{1-\gamma_t})^{\frac{1}{1-\gamma_t}}} \right)^{1/\psi-\gamma_t}. \quad (\text{C.3})$$

The first-order conditions for the households yield

$$\begin{aligned} 1 &= R_t \mathbb{E}_t M_{t,t+1}, \\ P_t^E &= \mathbb{E}_t [M_{t,t+1} (D_{t+1}^E + P_{t+1}^E)], \\ Q_t^c &= \mathbb{E}_t [M_{t,t+1} (dQ_{t+1}^c + 1)]. \end{aligned}$$

C.2 Firms

The representative firm chooses $N_{f,t}$, $N_{p,t}$, $V_{f,t}$, $V_{p,t}$, K_{t+1} , and I_t to maximize its discounted cash flow:

$$\max \mathbb{E}_t \sum_{s=0}^{\infty} \left(\frac{\partial \mathbb{V}_t / \partial C_{t+s}}{\partial \mathbb{V}_t / \partial C_t} \right) D_{t+s}, \quad (\text{C.4})$$

subject to the production function:

$$Y_t \leq (K_t)^\alpha (Z_t \tilde{N}_t)^{1-\alpha}, \quad (\text{C.5})$$

and the labor aggregator (with $N_t \equiv N_{f,t} + N_{p,t}$ denoting total employment):

$$\tilde{N}_t = \left((1 - \Omega) N_{f,t}^{\frac{\theta-1}{\theta}} + \Omega N_{p,t}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}. \quad (\text{C.6})$$

The capital accumulation equation is

$$K_{t+1} = \left(1 - \delta - \frac{\phi_K}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 \right) K_t + I_t, \quad (\text{C.7})$$

and the laws of motion for employment in the full-time and part-time sectors are given by

$$N_{f,t} = (1 - \rho_f) N_{f,t-1} + \Theta_{f,t} V_{f,t}, \quad (\text{C.8})$$

$$N_{p,t} = (1 - \rho_{p,t}) N_{p,t-1} + \Theta_{p,t} V_{p,t}, \quad (\text{C.9})$$

where ρ_f and ρ_p are the exogenous components of separation rates and $\rho_{p,t} \equiv \rho_p + (1 - \rho_p)P_{f,t}$ takes into account part-timers that endogenously switch to a full-time job. The cash flows of the firm are given by

$$D_t = Y_t - W_{f,t}N_{f,t} - W_{p,t}N_{p,t} - I_t - \varphi_{f,t}V_{f,t} - \varphi_{p,t}V_{p,t}. \quad (\text{C.10})$$

The problem of the firms yields the following equilibrium conditions:

$$q_t = \mathbb{E}_t \left[M_{t,t+1} \left(R_{t+1}^K + q_{t+1} \left(1 - \delta - \frac{\phi_K}{2} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right)^2 + \phi_K \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right) \frac{I_{t+1}}{K_{t+1}} \right) \right) \right] \quad (\text{C.11})$$

$$\frac{1}{q_t} = 1 - \phi_K \left(\frac{I_t}{K_t} - \delta \right), \quad (\text{C.12})$$

$$\tilde{R}_t^K K_t = \alpha(K_t)^\alpha (Z_t \tilde{N}_t)^{1-\alpha}, \quad (\text{C.13})$$

and finally

$$\frac{\varphi_{f,t}}{\Theta_{f,t}} = (1 - \Omega)(1 - \alpha)Z_t \left(\frac{K_t}{Z_t \tilde{N}_t} \right)^\alpha \left(\frac{\tilde{N}_t}{N_{f,t}} \right)^{\frac{1}{\theta}} - W_{f,t} + \mathbb{E}_t \left\{ M_{t,t+1} \frac{(1 - \rho_f)\varphi_{f,t+1}}{\Theta_{f,t+1}} \right\}, \quad (\text{C.14})$$

$$\frac{\varphi_{p,t}}{\Theta_{p,t}} = \Omega(1 - \alpha)Z_t \left(\frac{K_t}{Z_t \tilde{N}_t} \right)^\alpha \left(\frac{\tilde{N}_t}{N_{p,t}} \right)^{\frac{1}{\theta}} - W_{p,t} + \mathbb{E}_t \left\{ M_{t,t+1} \frac{(1 - \rho_{p,t+1})\varphi_{p,t+1}}{\Theta_{p,t+1}} \right\}. \quad (\text{C.15})$$

In equilibrium, $\Theta_{i,t} = \frac{\mathcal{M}_i(S_{i,t}, v_{i,t})}{v_{i,t}}$, where $\mathcal{M}_i$ is the Cobb-Douglas matching function for sector i given by (33).

C.3 Labor Market Search and Nash Bargaining

Our analysis of the labor market assumes that the household reallocates search effort across markets until it is indifferent between searching in the two markets. The value of search in a given market depends on the likelihood of finding a job in each market, the wages offered conditional on employment and the continuation value in each sector.

From the household perspective, the stock of workers evolves according to

$$N_{f,t} = (1 - \rho_f)N_{f,t-1} + P_{f,t}S_{f,t} \quad (\text{C.16})$$

$$N_{p,t} = (1 - \rho_{p,t})N_{p,t-1} + P_{p,t}S_{p,t} \quad (\text{C.17})$$

where $(1 - \rho_{p,t}) \equiv (1 - \rho_p)(1 - P_{f,t})$ is the part-time retention rate, and $P_{f,t}$ and $P_{p,t}$ capture the employment returns to search effort in each market, which the household takes as given.

Define $\mathbb{W}_{f,t}$ as the value of being a matched worker in the full-time sector, $\mathbb{W}_{p,t}$ as the value of being a matched worker in the part-time sector, $\mathbb{S}_{f,t}$ as the value of search effort in the full-time market, $\mathbb{S}_{p,t}$ as the value of search effort in the part-time market, and $\mathbb{U}_t$ as the value of being currently unemployed. The surplus terms are determined by the following equations

$$\mathbb{W}_{f,t} = W_{f,t} + \mathbb{E}_t \{M_{t,t+1} [(1 - \rho_f)\mathbb{W}_{f,t+1} + \rho_f \mathbb{S}_{f,t+1}]\}, \quad (\text{C.18})$$

$$\mathbb{W}_{p,t} = (W_{p,t} + \kappa_t) + \mathbb{E}_t \{M_{t,t+1} [(1 - \rho_p)[(1 - P_{f,t+1})\mathbb{W}_{p,t+1} + P_{f,t+1}\mathbb{W}_{f,t+1}] + \rho_p \mathbb{S}_{f,t+1}]\} \quad (\text{C.19})$$

$$\mathbb{S}_{f,t} = P_{f,t}\mathbb{W}_{f,t} + (1 - P_{f,t})\mathbb{U}_t \quad (\text{C.20})$$

$$\mathbb{S}_{p,t} = P_{p,t}\mathbb{W}_{p,t} + (1 - P_{p,t})\mathbb{U}_t \quad (\text{C.21})$$

$$\mathbb{U}_t = b_t + \mathbb{E}_t \{M_{t,t+1}\mathbb{S}_{f,t+1}\}. \quad (\text{C.22})$$

In equation (C.19) above, the term $P_{f,t+1}\mathbb{W}_{f,t+1}$ reflects the possibility that a worker with an ongoing part-time job finds a full-time job in the following period. As noted in the main text, in equilibrium $\mathbb{S}_{f,t} = \mathbb{S}_{p,t}$.

To implement Nash bargaining, we need to compute firm surpluses as well. The value of a matched worker is given by:

$$J_{i,t} = MPL_{i,t} - W_{i,t} + \mathbb{E}_t \{(1 - \rho_{i,t+1})M_{t,t+1}J_{i,t+1}\} \quad (\text{C.23})$$

Under free-entry, $J_{i,t} = \frac{\varphi_{i,t}}{\Theta_{i,t}}$. We assume that if workers walk away from a match at the bargaining stage, then they remain unemployed for the period. Using the above surplus definitions, the Nash-bargaining surplus-sharing rules are given by:

$$\mathbb{W}_{f,t} - \mathbb{U}_t = \frac{\eta_f}{1 - \eta_f} \frac{\varphi_{f,t}}{\Theta_{f,t}} \quad (\text{C.24})$$

$$\mathbb{W}_{p,t} - \mathbb{U}_t = \frac{\eta_p}{1 - \eta_p} \frac{\varphi_{p,t}}{\Theta_{p,t}}. \quad (\text{C.25})$$

We use (C.24) only to determine the steady-state wage of full-time workers, but impose (C.25) period-by-period to model flexible part-time wages.

C.4 Stationary Equilibrium

The model economy follows a balanced-growth path driven by the technology process, Z_t , which we assume is a random walk:

$$\log(Z_t) = \log(Z_{t-1}) + \sigma_z \epsilon_t^z. \quad (\text{C.26})$$

To describe the dynamics of the model in terms of stationary variables, we stationarize any of the trending variables, X_t , by defining their stationary counterpart, $\hat{X}_t \equiv \frac{X_t}{Z_{t-1}}$. The equilibrium of the economy in terms of these stationary variables is a sequence for $\{\hat{Y}_t, \hat{C}_t, \hat{I}_t, \hat{G}_t, \hat{K}_t, V_{f,t}, V_{p,t}, \tilde{N}_t, N_{f,t}, N_{p,t}, S_{f,t}, S_{p,t}, U_t, \tilde{R}_t^K, q_t, R_t, M_t, \hat{V}_t, \hat{W}_{f,t}, \hat{W}_{p,t}, \hat{\mathbb{W}}_{f,t}, \hat{\mathbb{W}}_{p,t},$

$\hat{\mathbb{S}}_{f,t}, \hat{\mathbb{S}}_{p,t}, \hat{\mathbb{U}}_t, \hat{P}_t^E, \hat{D}_t^E, \hat{B}_t^c, Q_t^c, \hat{\Gamma}_t\}$ that satisfies the following conditions:

$$\hat{Y}_t = (\hat{K}_t)^\alpha (\Delta Z_t \tilde{N}_t)^{1-\alpha}, \quad (\text{C.27})$$

$$\tilde{N}_t = \left((1 - \Omega) N_{f,t}^{\frac{\theta-1}{\theta}} + \Omega N_{p,t}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}, \quad (\text{C.28})$$

$$N_{f,t} = (1 - \rho_f) N_{f,t-1} + \mathcal{M}_1(S_{f,t}, V_{f,t}), \quad (\text{C.29})$$

$$N_{p,t} = (1 - \rho_{p,t}) N_{p,t-1} + \mathcal{M}_2(S_{p,t}, V_{p,t}), \quad (\text{C.30})$$

$$S_{f,t} + S_{p,t} = 1 - (1 - \rho_f) N_{f,t-1}, \quad (\text{C.31})$$

$$\hat{\mathbb{S}}_{f,t} = \hat{\mathbb{S}}_{p,t} \quad (\text{C.32})$$

$$U_t = 1 - N_{f,t} - N_{p,t}, \quad (\text{C.33})$$

$$\begin{aligned} \frac{\hat{\Gamma}_t \varphi_{f,t} V_{f,t}}{\mathcal{M}_1(S_{f,t}, V_{f,t})} = & (1 - \Omega)(1 - \alpha) \Delta Z_t \left(\frac{\hat{K}_t}{\Delta Z_t \tilde{N}_t} \right)^\alpha \left(\frac{\tilde{N}_t}{N_{f,t}} \right)^{\frac{1}{\theta}} - \hat{W}_{f,t} + \\ & + \mathbb{E}_t \left\{ M_{t,t+1} \Delta Z_t \frac{(1 - \rho_f) \hat{\Gamma}_{t+1} \varphi_1 V_{f,t+1}}{\mathcal{M}_1(S_{f,t+1}, V_{f,t+1})} \right\}, \end{aligned} \quad (\text{C.34})$$

$$\begin{aligned} \frac{\hat{\Gamma}_t \varphi_{p,t} V_{p,t}}{\mathcal{M}_2(S_{p,t}, V_{p,t})} = & \Omega(1 - \alpha) \Delta Z_t \left(\frac{\hat{K}_t}{\Delta Z_t \tilde{N}_t} \right)^\alpha \left(\frac{\tilde{N}_t}{N_{p,t}} \right)^{\frac{1}{\theta}} - \hat{W}_{p,t} + \\ & + \mathbb{E}_t \left\{ M_{t,t+1} \Delta Z_t \frac{(1 - \rho_{p,t+1}) \hat{\Gamma}_{t+1} \varphi_2 V_{p,t+1}}{\mathcal{M}_2(S_{p,t+1}, V_{p,t+1})} \right\}, \end{aligned} \quad (\text{C.35})$$

$$\hat{\mathbb{W}}_{f,t} = W_{f,t} + \mathbb{E}_t \left\{ M_{t,t+1} \Delta Z_t \left[(1 - \rho_f) \hat{\mathbb{W}}_{f,t+1} + \rho_f \hat{\mathbb{S}}_{f,t+1} \right] \right\}, \quad (\text{C.36})$$

$$\begin{aligned} \hat{\mathbb{W}}_{p,t} = & (\hat{W}_{p,t} + \hat{\Gamma}_t \kappa) \\ & + \mathbb{E}_t \{ M_{t,t+1} \Delta Z_t [(1 - \rho_p) [(1 - P_{f,t+1}) \hat{\mathbb{W}}_{p,t+1} + P_{f,t+1} \hat{\mathbb{W}}_{f,t+1}] + \rho_p \hat{\mathbb{S}}_{f,t+1}] \} \end{aligned} \quad (\text{C.37})$$

$$\hat{\mathbb{S}}_{f,t} = P_{f,t} \hat{\mathbb{W}}_{f,t} + (1 - P_{f,t}) \hat{\mathbb{U}}_t, \quad (\text{C.38})$$

$$\hat{\mathbb{S}}_{p,t} = P_{p,t} \hat{\mathbb{W}}_{p,t} + (1 - P_{p,t}) \hat{\mathbb{U}}_t, \quad (\text{C.39})$$

$$\hat{\mathbb{U}}_t = \hat{\Gamma}_t b + \mathbb{E}_t \left\{ M_{t,t+1} \Delta Z_t \hat{\mathbb{S}}_{f,t+1} \right\}, \quad (\text{C.40})$$

$$\hat{\mathbb{W}}_{p,t} - \hat{\mathbb{U}}_t = \frac{\eta_p}{1 - \eta_p} \frac{\hat{\Gamma}_t \varphi_2}{\Theta_{p,t}}, \quad (\text{C.41})$$

$$\hat{W}_{f,t} = \hat{\Gamma}_t \hat{W}_1^{ss}, \quad (\text{C.42})$$

$$M_{t,t+1} = \beta \left(\frac{\hat{C}_{t+1} \Delta Z_t}{\hat{C}_t} \right)^{1-1/\psi} \left(\frac{\hat{C}_t}{\hat{C}_{t+1} \Delta Z_t} \right) \left(\frac{\hat{\mathbb{V}}_{t+1}}{(\mathbb{E}_t \hat{\mathbb{V}}_{t+1}^{1-\gamma_t})^{\frac{1}{1-\gamma_t}}} \right)^{1/\psi-\gamma_t}, \quad (\text{C.43})$$

$$\hat{P}_t^E = \mathbb{E}_t \left[M_{t,t+1} \Delta Z_t \left(\hat{D}_{t+1}^E + \hat{P}_{t+1}^E \right) \right], \quad (\text{C.44})$$

$$Q_t^c = \mathbb{E}_t \left[M_{t,t+1} (dQ_{t+1}^c + 1) \right], \quad (\text{C.45})$$

$$1 = R_t \mathbb{E}_t M_{t,t+1}, \quad (\text{C.46})$$

$$\tilde{R}_t^K = \alpha \left(\frac{\hat{K}_t}{\Delta Z_t \tilde{N}_t} \right)^{\alpha-1}, \quad 78 \quad (\text{C.47})$$

$$q_t = \mathbb{E}_t \left[M_{t,t+1} \left(\tilde{R}_{t+1}^K + \right. \right. \quad (\text{C.48})$$

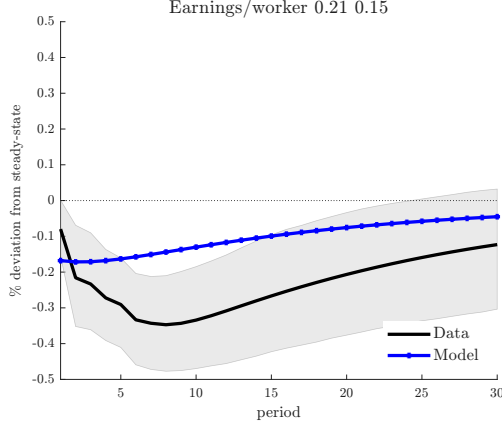


Figure D.1: Response of earnings per worker in model and data.

D Earnings per worker

Our assumptions about sticky wages in the full-time sector are stark for simplicity. Of course, the literature has come to contrasting and often opposed conclusions about the cyclicalty of wages. One straightforward and common measure of wages is earnings per worker. Figure D.1 plots the empirical and model-implied response of earnings per worker, under our baseline empirical and model estimates.

The figure shows that earnings per worker fall modestly in response to the shock both in the data and in the model. While the model fall is somewhat less than in the data, it falls in or near the empirical confidence band. In the model, there are two reasons this measure of wages shows non-trivial adjustment in response to risk shocks, even though the wages of full-time workers do not respond. First, the wages of existing part-time workers are flexible and they fall in response to the shock. Second, the number of part-time workers grows in the model as a share of total employment: since those workers have lower earnings, this also reduces earnings per worker. We conclude from this figure that, while our assumptions about wages are probably an oversimplification, the overall implications for earnings per worker are not strongly counterfactual.

E TFP calculation

Taking log changes of the production function (25), we can write the change in output following a risk aversion shock as:

$$\Delta \log(Y_t) = (1 - \alpha) \Delta \log \tilde{N}_t + \alpha \Delta \log(K_t), \quad (\text{E.1})$$

$\Delta \log Z_t$ does not appear in (E.1) since changes in true TFP are zero in response to the risk aversion shock.

To define measured TFP we need to specify what data the econometrician observes and what he knows about the model that generated the data. The quarterly measure of TFP typically used for business cycle analysis, and the one we confront our model with, builds on Basu et al. (2006) and is implemented in quarterly data by Fernald (2014). Aggregate TFP is constructed using data on output and production inputs such as capital and labor. The methodology takes into account certain dimensions of input heterogeneity by using actual or estimated relative factor prices to control for differences in implied marginal products. The dimensions of labor heterogeneity considered are education, age, sex, race/ethnicity, industry, and occupation, but do not include part-time status. Thus, the effect of part-time status on wages is controlled for only to the extent that it is explained by these six included variables.

It follows that in our model, where there is no such heterogeneity along those six dimensions, the “naive” TFP econometrician would not distinguish between N_p and N_f , and, given data on output, capital and employment, would construct the change in measured TFP, $\Delta \log \tilde{Z}_t^e$, as:

$$\Delta \log \tilde{Z}_t^e = \Delta \log(Y_t) - \omega_N \Delta \log(N_t) - (1 - \omega_N) \Delta \log(K_t), \quad (\text{E.2})$$

where $\omega_N = \frac{W_f N_f + W_p N_p}{Y}$ is the labor share. Since there are no distortions in physical capital investment, capital gets paid its true output elasticity, which implies $\alpha = 1 - \omega_N$. Thus, plugging (E.1) into (E.2), we can rewrite the baseline measured TFP in response to a risk aversion shock as:

$$\Delta \log \tilde{Z}_t^e \equiv (1 - \alpha) \Delta \log(\tilde{N}_t) - \omega_N \Delta \log(N_t). \quad (\text{E.3})$$

Figure E.1 shows the response of such TFP measures in our model alongside the empirical response of Fernald’s TFP. We discuss the finding in the main text.

In some applications, the TFP econometrician may have data on hours in addition to employment. In this case, the measure of TFP would be:

$$\Delta \log \tilde{Z}_t^h \equiv (1 - \alpha) \Delta \log(\tilde{N}_t) - \omega_N \Delta \log(N_{f,t} + x N_{p,t}). \quad (\text{E.4})$$

Above, x represents the ratio of hours worked by a part-time worker relative to the hours of a full-time worker, which in Section 4.2 we calibrated to be 0.5. The labor share ω_N remains unchanged because it depends on the total labor income of each group.

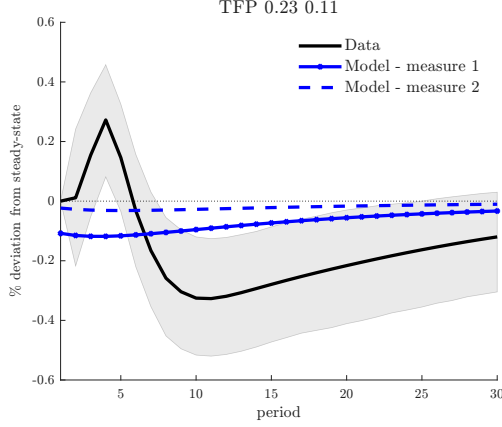


Figure E.1: Empirical and alternative model responses of measured TFP.

F Robustness

F.1 Alternative Estimation Procedure

Our baseline estimation procedure compares theoretical impulse responses to those estimated from a particular empirical procedure on real data. In general, however, applying our empirical procedure to data generated by the model only imperfectly identifies the theoretical response to risk aversion disturbances. To alleviate concern that our results could be driven by potential misspecification of the identification procedure, we reestimated the model using an alternative procedure that aligns analogous objects in the model and data. In particular, define $\hat{\Psi}^m(\Pi)$ to be the vector of impulse responses generated by applying our “method 1” empirical procedure to a sample of 5,000 periods of data generated by the model under parameter vector Π (stacked along with the same unconditional moments as $\Psi(\Pi)$). Then, the parameter vector is estimated to minimize the loss function

$$\mathcal{L}^m(\Pi) \equiv (\hat{\Psi} - \hat{\Psi}^m(\Pi))'W(\hat{\Psi} - \hat{\Psi}^m(\Pi)). \quad (\text{F.1})$$

The blue line in Figure F.1 presents the impulse responses for the model when reestimated in this way. The figure shows that the model impulse responses are very similar to those from our baseline procedure. The cyan line depicts the impulse responses that are generated by applying the empirical procedure to the model data. The close alignment of the blue and cyan lines demonstrates that our empirical procedure does a very good job of identifying the true effects of risk shocks on model-simulated data. The second column of Table F.1 shows the parameters of the reestimated model. Most parameters remain close to our baseline estimates.

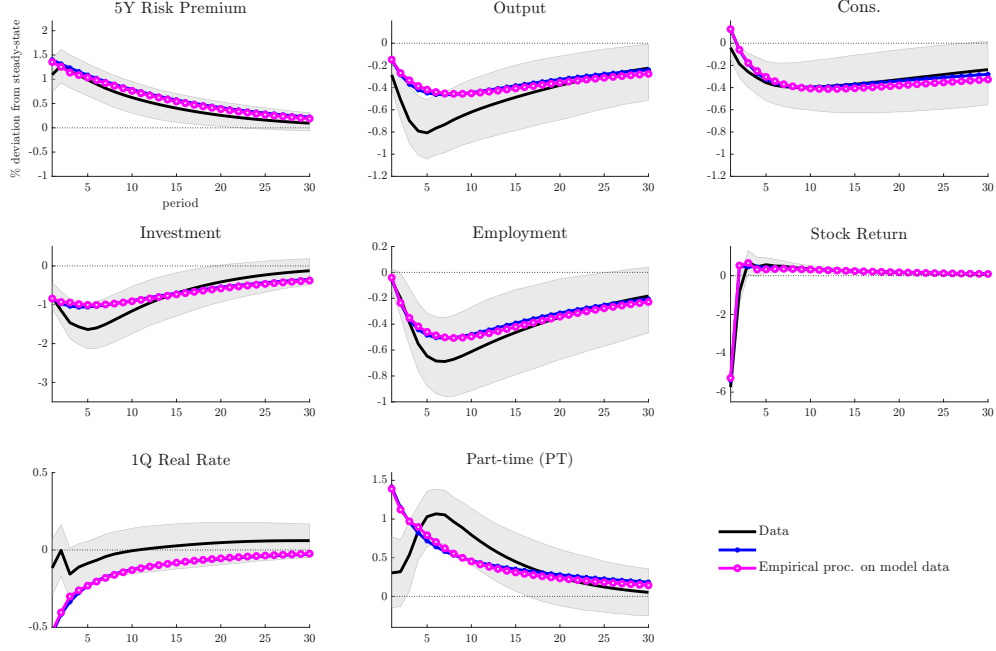


Figure F.1: Model responses when estimated using alternative procedure.

F.2 Low IES results

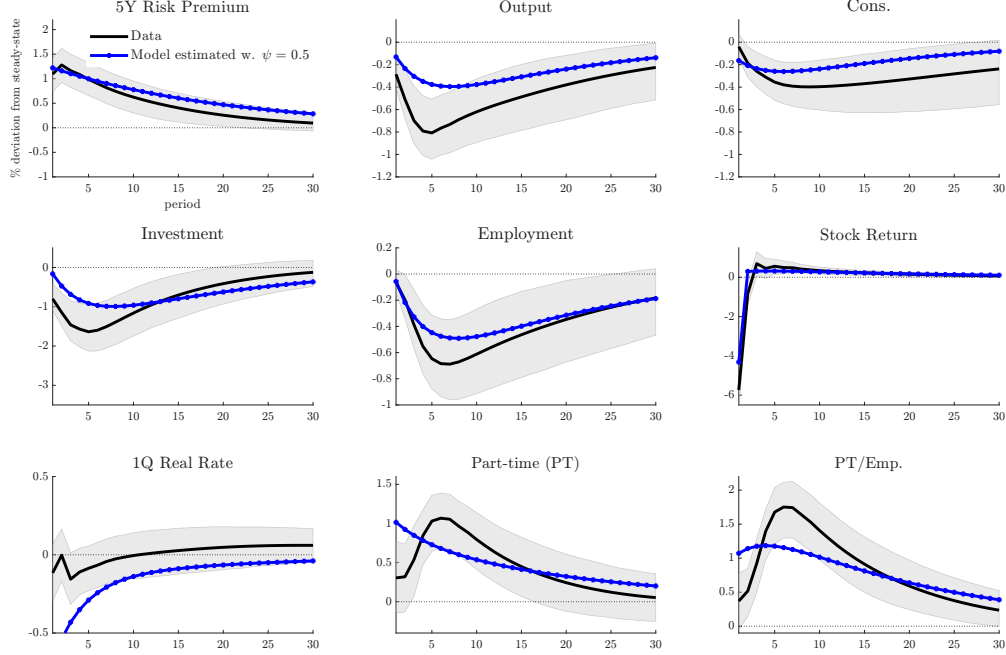


Figure F.2: Model responses with low intertemporal elasticity of substitution.

To emphasize that a high intertemporal elasticity is not essential for our results, Figure F.2 plots the implied impulse response for the model, estimated with $\psi = 0.5$. The model

Table F.1: Estimated Parameters - Robustness

Name	Description	Base Est.	Alt. Procedure	Low IES	Leverage < 0.65
γ_{ss}	Steady-state risk aversion	36.236	47.970	101.159	66.729
ν	Leverage Ratio	0.767	0.765	0.554	0.650
Labor Markets					
φ_f	Vacancy posting cost - full-time	0.955	0.810	2.202	1.008
φ_p	Vacancy posting cost - part-time	0.234	0.332	0.325	0.188
κ	Value if no perm posit.	0.725	0.764	0.796	0.652
b	Value if unemployed	1.194	1.221	1.244	1.126
Ω	Labor contrib. of part-time	0.228	0.201	0.186	0.221
θ	Elas. between full- & part-time	5.079	2.943	2.573	4.790
ϵ_f	Matching elasticity - full-time	0.392	0.373	0.138	0.467
ϵ_p	Matching elasticity - part-time	0.025	0.025	0.025	0.025
χ_f	Matching technology - full-time	0.449	0.503	0.316	0.499
χ_p	Matching technology - part-time	1.594	1.001	0.875	2.701
ω	Gradual wage adj.	0.976	0.970	0.986	0.970
Risk Aversion Process					
ρ_γ	AR(1) risk av. shock	0.934	0.938	0.951	0.936
σ_γ	Std. dev. of risk av. shock	0.438	0.430	0.336	0.289

matches the data nearly as well as our baseline, but with a somewhat larger miss on the interest rate response. The third column of Table F.1 presents the parameters estimated in this case.

F.3 Low leverage results

To assess the robustness of our quantitative results with respect to leverage, we reestimate the model imposing $\nu \leq 0.65$, which corresponds to a liability-to-equity ratio of approximately 2. This value is conservative relative to our baseline estimate and is close to the average ratio of total liabilities to net worth for the nonfinancial corporate sector in the Financial Accounts of the Flow of Funds over our sample. Figure F.3 reports the impulse responses from this restricted estimation, alongside the baseline responses and the empirical targets.

The restricted estimation produces two main changes. First, steady-state risk aversion, γ , increases relative to the baseline, as shown in the last column of Table F.1. This adjustment is natural: the lower leverage reduces the model-implied equity premium, so higher risk aversion is needed to continue matching the unconditional equity premium (which remains one of our estimation targets). Second, the responses of real variables are essentially unchanged while the response of risky asset prices is muted. Output, investment, and other real aggregates move similarly to the baseline, indicating that higher risk aversion partly substitutes for lower leverage in the transmission of risk aversion shocks to real variables. However, the

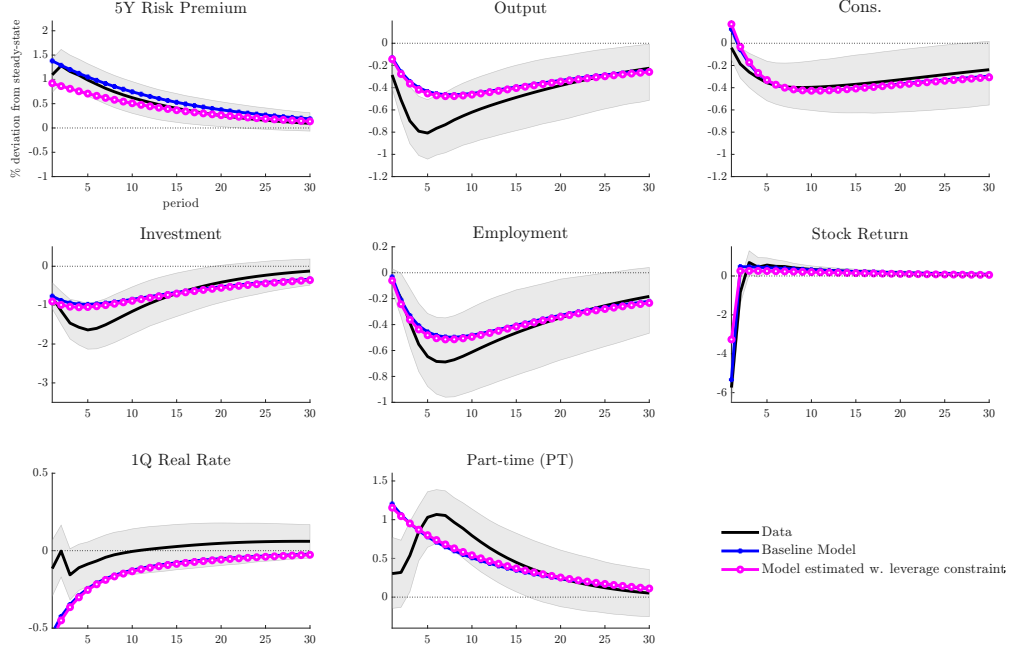


Figure F.3: Model responses with low leverage.

restricted model generates weaker responses of the equity premium and stock returns, even though it continues to deliver the same response of the risk-free rate as in the baseline. This result reflects the classic asset-pricing tension between generating large movements in risk premia and maintaining a smooth risk-free rate: raising γ further would improve the fit of equity returns but would generate excessive movements in the real rate. The estimation instead prefers to contain the response of the risk-free rate and undershoots stock-return and equity premium responses (while staying within the confidence interval of the empirical responses for the latter). Thus, while lower leverage reduces the model's ability to match the full magnitude of asset price responses, it leaves the paper's central real transmission results largely unchanged.

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